Keep, Measure, Erase

What a Quantum Loop Does with Its History

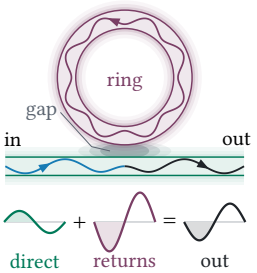
ANONYMOUS AUTHOR(S)

Quantum loops have a reputation for paradox. We trace the paradoxes to a single fact: the history of a quantum loop, the record of how many times its body ran, is itself quantum information. A loop can erase its history, keep it, or measure it, and these choices give three different programs. Existing accounts make this choice implicitly. We make the choice explicit in iso, a reversible language in the Π family, and the paradoxes dissolve. The coherent loop, which erases its history, is the trace of its body: for every finite body it is total and unitary and compiles exactly into loop-free code, so the question of when it halts disappears. The other two loops come from the coherent one through two universal constructions on quantum processes: R, which adds allocation, yields the loop that keeps its history, and L, which then adds hiding, yields the loop that measures it. Measuring the guard on every round gives the same data as keeping the exact number of returns and then forgetting it. Forgetting gives the coherent data exactly when history is independent of data. With decidable coefficient equality, a compiler can test this. Finally, an ideal ring resonator computes the scalar trace. For any given finite unitary body, a lossless optical multiport with its return ports fed back computes the trace at a chosen frequency. This gives the trace a physical meaning that recent work had left open, and it opens a new front for quantum programming: loops that run as optical feedback.

1 Introduction

Can a quantum computer run a loop? The question sounds settled, yet the answers on record contradict one another. Let the loop's guard be in superposition, so that one branch exits after three rounds while another keeps going. One answer says the computer then cannot halt without spoiling its computation [24]. A second says the loop must measure its guard on every round, which leaves a classical loop running over quantum data [36]. A third keeps the guard coherent but bounds the rounds and pads every branch until all of them finish together [37, 38]. A fourth defines the loop as a trace, an infinite sum over the paths of its control flow [3, 5]. We claim that all four answers are right about the loops they describe. We compare them by following the history of a loop, the record of how many times its body ran, through to the output.

Physics has a loop that nobody counts. An ideal lossless ring resonator has a straight waveguide beside a ring. Their fields overlap across the narrow gap. This is the coupler. Part of the incoming light goes straight through. The rest goes around the ring and splits again each time it comes back. Some light leaves after one trip, some after two, some after a hundred. Under steady illumination at a single frequency, all these contributions reach the same outgoing mode, so their amplitudes add [2, Eq. (6)]. In this plane-wave reading of feedback [3, § 5.3], the execution formula of geometry of interaction adds the exits that reach this mode [1, 5].



In Q# and OpenQASM 3, a `while` loop tests a classical condition [26, 27]. Measuring the guard on each round tells an early exit from a late one. Arrival time can do the same in the ring. Send in a pulse shorter than one round trip. The pulse that goes straight through has passed the output before the first return arrives. Each further trip gives a separate pulse [28, Eqs. (30)–(33)] [3, § 3.4].

A quantum language has three choices for what to do with a loop's history:

- **Keep it.** Return a count register with the data.
- **Measure it.** Measure the guard after each call to decide whether to continue.
- **Erase it.** Add the amplitudes from different rounds at the same output, as the ring does.

For classical programs the three coincide on the returned data when the count is ignored, which is why classical languages never had to choose. For quantum programs they give three different loop constructs, and a language that provides only one has chosen for its programmers.

We make this choice explicit in iso, a reversible language in the Π family [8]. We write the body once using symmetric pattern-matching clauses [30, §§ 3–5], with an exit and a back edge, and close it in three ways. The coherent loop `trace` connects the back edge, count_K counts returns modulo K (zero for an immediate exit), and `while` measures the guard after each call. Each choice comes with something a compiler can use:

- **Coherent loops compile away.** Eliminating one returning basis value at a time gives a finite clause table with exact coefficients and no back edge, even when infinitely many rounds contribute [3, Thm. 3.3.15].
- **Forgetting can be certified.** Each exit map gives the part of the output that leaves after a given number of returns. With m returning basis values and a nonempty input type, we check whether the first $m+1$ maps are multiples of one map. With decidable equality of exact coefficients, this decides when the measured loop can replace the coherent data channel, even for a caller who keeps a reference entangled with the input.
- **We can probe the coherent loop without a counter.** Each trip picks up a phase on the back edge. One control qubit lets us compare two nearby phases and estimate how much weight circulates. Averaging over the phase gives the measured mean number of returns. For the rotation with $\cos\theta = 3/5$ and $\sin\theta = 4/5$, that mean is one. With no added phase, four times the input weight circulates and the input comes back with its sign flipped.
- **An ideal ring realizes the scalar loop.** Fairbanks and Selinger ask whether a device can take the trace of a unitary supplied as a black box [13, App. B]. For a given matrix, we can build its optical network. The ring is a familiar model of feedback [12]. Each trip gives the corresponding exit amplitude of our scalar `trace`. For a frequency-independent coupler and phase proportional to frequency, the circulating weight for unit input times the round-trip time gives the Wigner–Smith delay [31].

2 Finite programs in iso

An iso $\omega : a \leftrightarrow b$ is a reversible program from data type a to data type b. We use the finite types of Π: 0 has no values, 1 has the value $*$, $a + b$ holds a tagged value, and $a \otimes b$ holds a pair [18, § 3.1]. Write $\mathscr{B}(a)$ for the basis values of a and $|s\rangle$ for the vector of $s \in \mathscr{B}(a)$. The two values of $1 + 1$ carry the same payload $*$. Their tags distinguish left $*$ from right $*$.

2.1 Syntax

An iso can be given by a table of clauses. To read a clause $v \leftrightarrow e$, match the input against v and use the variables bound by the match to evaluate e. For example, the pattern left x matches left s and binds x to the payload s. A pattern may be unit $*$, a variable x, a tagged pattern left v or right v, or a pair $\langle v_1, v_2 \rangle$. Each variable occurs at most once in the pattern. The grammar also allows an expression to form a finite superposition with coefficients $\alpha_i \in \mathbb{C}$, or to apply an iso locally:

$$
\begin{aligned}
e &::= v \mid \textstyle\sum_i \alpha_i e_i \mid \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e, \\
\mathscr{T} &::= \{v_i \leftrightarrow e_i\}_{i \in I}, \qquad t ::= e \mid \omega\, t, \\
\omega &::= f \mid \mathscr{T} \mid \omega; \omega \mid \omega \oplus \omega \mid \omega \otimes \omega \mid \omega^{\dagger}.
\end{aligned}
$$

Here I is finite. In a local application, q assembles the argument to ω, and p binds the result for the rest of the expression. Both p and q use only unit, variables and pairs. A variable can bind a value

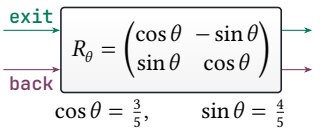
of any type, including a whole tagged value. A pair pattern takes apart a result of product type. In this way the same binding works for every basis value that the local iso can return.

An iso may be a clause table, a name f for an earlier checked definition, or a combination of isos. Sequence $f;g$ runs f first and then g. A sum acts on the alternative selected by the tag, a tensor acts on the two factors of a pair, and dagger reverses the iso. A closed term t has no free value variables. We write application $\omega\, t$ as `omega on t` in the executable syntax. Thus `f; g on t` applies the whole sequence to t: it feeds the result of f directly into g.

In general, `round` $f : a + c = \mathcal{T}$ names a table of clauses $\mathcal{T}$ and gives the open body f type $(a + c) \leftrightarrow (a + c)$. This abbreviation uses the same external type a on both sides. The outer tags `exit` and `back` stand for `left` and `right` and carry values of type a and c, respectively. The payload can itself be tagged. For example, `Qubit` is the executable name for $1 + 1$, with `0` for left $*$ and `1` for right $*$. Thus `exit 0` expands to left (left $*$): the outer tag selects the port and the inner tag gives the qubit value. Similarly, `Unit` names the type 1 and `unit` its value $*$.

For the loop from the introduction, take both payload types to be `Unit`. The guard then has just two basis values, `exit unit` and `back unit`. We want one call to rotate this guard through $\theta = \arccos(3/5)$. With the exit basis value first and the return basis value second, the two clauses give the two columns of the rotation matrix shown beside them.

```
round guardRound : Unit + Unit = {
  exit unit ↔ [ 3/5 . exit unit + 4/5 . back unit];
  back unit ↔ [-4/5 . exit unit + 3/5 . back unit]
}
```



On input `exit unit`, the first clause gives $(3/5)|\text{exit}\ *\rangle + (4/5)|\text{back}\ *\rangle$. The brackets group the superposition, and the dot scales each tagged value by its amplitude. These two amplitudes form the first column of R_θ. The `back unit` clause gives the second column in the same order.

The closing brace ends the definition of the body. In Section 3 we connect the lower output to the lower input to run the body again on the returning amplitude.

To evaluate a source iso ω, we unfold its names and replace each checked clause table by an atom $[M]$ containing that table's matrix. The result is a program κ built from *matrix combinators*:

$$\kappa ::= [M] \mid \kappa;\kappa \mid \kappa \oplus \kappa \mid \kappa \otimes \kappa \mid \kappa^\dagger.$$

For `guardRound` the whole table becomes the single atom $\left[\begin{smallmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{smallmatrix}\right]$.

2.2 Statics

To check a table $\omega : a \leftrightarrow b$, we first check that each basis input of a matches exactly one clause and that each output expression has type b under the variables its pattern binds. For a basis input s, its clause gives column s of $M = [\![\omega]\!]$. Its entry in row t is the amplitude for output t. For `guardRound`, the marked entry $M_{ts} = -4/5$ is the exit amplitude on a returning input.

	exit	s=back
t=exit	3/5	−4/5
back	4/5	3/5

Both columns of `guardRound` have squared norm one, and their inner product is zero: the exit row contributes $-12/25$ and the return row contributes $12/25$ (Figure 1). In general, we check that a and b have equally many basis values and that $M^\dagger M = I$, where dagger takes the conjugate transpose. The diagonal entries of this product are the squared column norms and the off-diagonal entries are the inner products of distinct columns. A square matrix that passes this check is unitary.

The judgment $\Sigma; \Delta \vdash e : b$ gives e output type b in the earlier iso definitions Σ and the context of value variables Δ. These variables are used linearly. In a pair, the two components use disjoint parts of Δ. In a superposition, each alternative uses the same variables once. A local application splits the variables between its argument and its continuation, where the result pattern binds fresh variables.

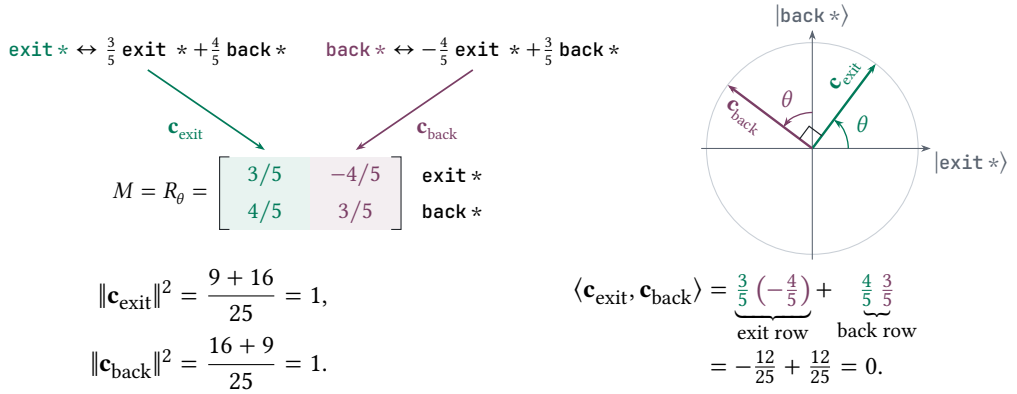


Fig. 1. Each clause gives one column of R_θ. The rotation turns each basis vector through the same angle θ, so the columns lie on the unit circle and remain perpendicular. Each product in their inner product uses entries from the same output row. For complex columns we conjugate the first entry of each product.

A checked iso has judgment $\Sigma \vdash \omega : a \leftrightarrow b$. Matrix atoms pass the same column test as tables, and the combinators must agree on the types they connect. A named declaration enters Σ only after it passes these checks, so evaluating a clause during checking can use only earlier definitions. Within a definition, checking and evaluation recurse on proper subprograms. The clauses and linear contexts adapt symmetric pattern matching [30, §§ 3–5] [23, Ch. 3]. Appendix A gives the full rules.

2.3 Dynamics

In Figure 1, we obtained each column by evaluating the right-hand side of a clause. Write $\gamma \vdash e \Downarrow \psi$ when expression e evaluates to the vector ψ under the basis values assigned by γ. A value v gives $|v[\gamma]\rangle$, and a sum adds its weighted outputs. A local application passes each basis result, with its amplitude, to the continuation and adds the resulting vectors.

Write $\mathscr{E}_\Sigma(\omega) = \kappa$ for elaboration and $\Sigma \vdash \omega \Downarrow M$ when κ evaluates to M. For `guardRound`, the atom rule $[R_\theta] \Downarrow R_\theta$ gives the matrix whose columns we just checked. A closed term then applies the matrix to its input: if $t \Downarrow \psi$, then $\omega\, t \Downarrow M\psi$.

Suppose we apply `guardRound` twice in sequence, starting with `exit unit`. After the first call the amplitudes are $3/5$ on **exit** and $4/5$ on **back**. The second call therefore uses both clauses. Its output is the first column weighted by $3/5$ plus the second column weighted by $4/5$:

$$\frac{3}{5}\begin{pmatrix}3/5\\4/5\end{pmatrix} + \frac{4}{5}\begin{pmatrix}-4/5\\3/5\end{pmatrix} = \begin{pmatrix}-7/25\\24/25\end{pmatrix}.$$

This is ordinary sequence: the whole output of the first call becomes the input to the second. In general, if $\kappa \Downarrow M$ and $\lambda \Downarrow N$, then $\kappa;\lambda \Downarrow NM$, since M acts first on the input column. Elaboration preserves sequence, sum, tensor and dagger; the last three evaluate by taking the direct sum, tensor product and conjugate transpose of the corresponding matrices.

2.4 Denotation

A type a denotes the space $[\![a]\!] = \mathbb{C}^{\mathscr{B}(a)}$, with one orthonormal basis vector for each basis value of a. Thus 0 and 1 denote the zero space and $\mathbb{C}$, and sum and product types denote direct sums and tensor products, respectively.

The evaluated columns of a checked iso $\omega : a \leftrightarrow b$ give its linear map $[\![a]\!] \to [\![b]\!]$.

For `guardRound`, an input with amplitudes u on `exit` and v on `back` gives the first column weighted by u plus the second weighted by v:

$$[\![\texttt{guardRound}]\!]\begin{pmatrix}u\\v\end{pmatrix} = u\begin{pmatrix}3/5\\4/5\end{pmatrix} + v\begin{pmatrix}-4/5\\3/5\end{pmatrix} = \begin{pmatrix}(3u-4v)/5\\(4u+3v)/5\end{pmatrix}.$$

Thus R_θ gives the output of the clauses on every input. The column check makes it unitary. Appendix A proves the correspondence with evaluation for all checked isos.

In the sequence above, we passed both output coordinates to the next call. To form a loop, we let the `exit` amplitude leave and feed only the `back` amplitude into the next call.

3 What the coherent loop computes

Closing `guardRound` leaves the caller with a single output of type `Unit`. Every amplitude that leaves through `exit` contributes to that output, however many trips it makes around the loop.

3.1 From a round to its exits

Follow the lower wire in Figure 2. On the second call, only the `back` clause contributes: it takes the returning amplitude $4/5$ and sends $(4/5)(-4/5) = -16/25$ to the external output. It sends $(4/5)(3/5) = 12/25$ back for the third call.

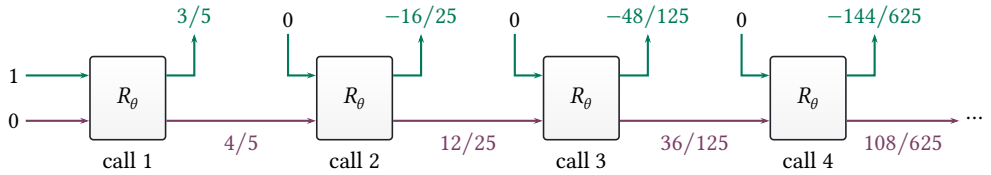


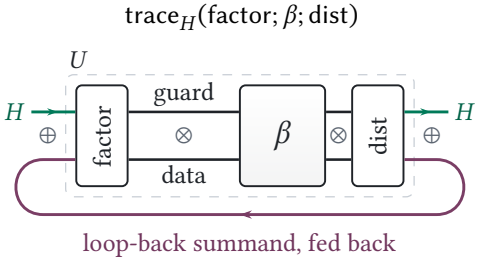
Fig. 2. Four calls to `guardRound`. The upper outputs contribute to the caller's result; the first two add to $-1/25$. The lower wire carries the returning amplitude from one call to the next. A zero at an input means no amplitude enters that port.

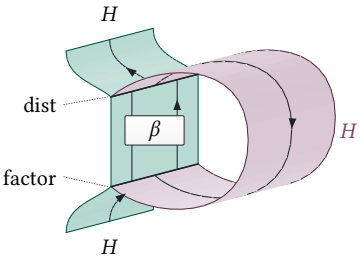
Syntax and statics. The programmer writes `trace guardRound`. Unfolding the name and replacing its two clauses by their checked matrix atom gives the evaluator $\mathrm{trace}_1[R_\theta]$. Both programs connect the same return wire. The two typing rules describe this connection before and after elaboration:

$$\underbrace{\frac{\Sigma \vdash \omega : a + c \leftrightarrow b + c}{\Sigma \vdash \mathrm{trace}_c\, \omega : a \leftrightarrow b}}_{\text{source iso}} \qquad \underbrace{\frac{\kappa : a + c \leftrightarrow b + c}{\mathrm{trace}_c\, \kappa : a \leftrightarrow b}}_{\text{matrix combinator}}.$$

In the source rule, Σ supplies earlier definitions. Their names have already been unfolded in the matrix rule. In both rules closing c leaves a and b at the interface. Thus `trace guardRound` has type $1 \leftrightarrow 1$. The `round` declaration supplies the return type. Elaboration carries the new constructor from one language to the other: $\mathscr{E}_\Sigma(\mathrm{trace}_c\, \omega) = \mathrm{trace}_c\, \mathscr{E}_\Sigma(\omega)$. In `trace f; g` we close `f` first and then apply `g`. QWhile uses this interface for its do-while combinator [3, § 5.1].

We can evaluate the combinator to a matrix or draw its connections. Each box denotes a morphism in Unitary, the category of finite-dimensional Hilbert spaces and unitary maps. Joining boxes composes their maps. Here both tags carry the same payload H, and closing the return wire takes the direct-sum trace over H [5, 21].





Read the circuit from left to right and the sheets from bottom to top. At factor, `exit` x becomes $|0\rangle \otimes x$ and `back` x becomes $|1\rangle \otimes x$. This is the unitary $H \oplus H \to G \otimes H$ with $G = \mathbb{C}^2$, including the swap that puts the guard first. Each alternative has its own sheet. Between factor and dist, guard and data travel through β as two wires on one sheet [11, Defs. 16, 18–20]. The dist box applies the inverse of factor and separates the alternatives again. The purple sheet carries a return back to factor, leaving $H \to H$ for the caller. The general rule also allows different payload types.

The exit maps. To follow the same calculation for any body $\beta : a + c \leftrightarrow b + c$, evaluate it to a matrix U and group its columns by the input tag and its rows by the output tag. Write $X = [\![a]\!]$, $X' = [\![b]\!]$ and $Y = [\![c]\!]$. On external input $u \in X$ and returning input $y \in Y$, one call gives

$$U\begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} Au + By \\ Cu + Dy \end{pmatrix}. \tag{3.1}$$

The top row gives the external output in X' and the bottom row gives the returning output in Y. For `guardRound`, the first clause gives $A = 3/5$ and $C = 4/5$, and the second gives $B = -4/5$ and $D = 3/5$. Reading the same rows and columns gives $A : X \to X'$, $B : Y \to X'$, $C : X \to Y$ and $D : Y \to Y$. These are morphisms in **FdHilb**, where all linear maps are allowed. Since U itself is unitary and the same finite space Y occurs on both sides, $\dim X = \dim X'$.

The first call receives $(u, 0)$, so Au leaves immediately and Cu goes back. The second call receives $(0, Cu)$, so BCu leaves and DCu goes back. These are components of the original input amplitude, with no normalization between calls. The zero denotes a port with no incoming amplitude. Taking either component of the output is a projection in **FdHilb**.

Write $x_k : X \to X'$ for the exit map after k returns. The immediate exit has made zero returns. To exit after $k+1$ returns, a contribution first enters Y through C, makes k further passages through D, and finally leaves through B. Thus

$$x_0 = A, \qquad x_{k+1} = BD^kC. \tag{3.2}$$

After $n + 1$ calls the contributions to the external output are $x_0u, \dots, x_nu$, with D^nCu still on the return wire. Coherent feedback adds all the exits at their common output:

$$[\![\mathrm{trace}_c\, \beta]\!] = L = \sum_{k \geq 0} x_k.$$

This is the execution formula from the introduction [1]. In the ring, x_0 gives the light that goes straight through and x_k the contribution after k trips. A finite unitary body can have infinitely many nonzero exits. We find their sum from the amplitude that builds up on the return wire.

3.2 Computing the sum without a last round

On input `unit`, the first call to `guardRound` sends $4/5$ back. Each later call returns $3/5$ of what it receives (Figure 2). Write y for the sum on the return wire. After the first term, the sum is $(3/5)y$.

$$y = \frac{4}{5} + \underbrace{\frac{12}{25} + \frac{36}{125} + \frac{108}{625} + \frac{324}{3125} + \cdots}_{(3/5)y} = \frac{4}{5} + \frac{3}{5}y.$$

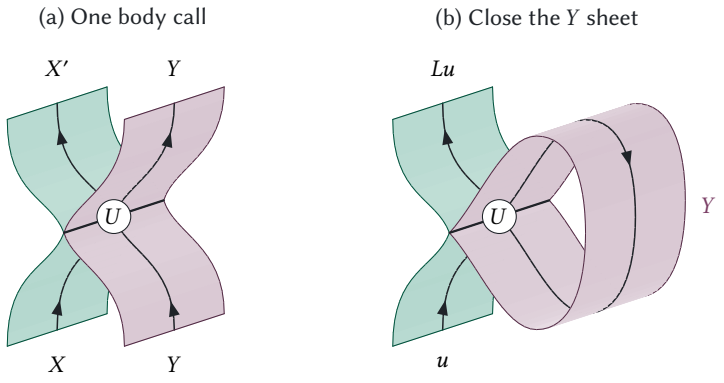
This is the *wire equation*. Subtracting $(3/5)y$ gives $(2/5)y = 4/5$, so the returns sum to $y = 2$. The first call sends $3/5$ straight to the output, and the `back` clause sends $-4/5$ of each return there. Their sum is $3/5 - (4/5)y = -1$, so `trace guardRound on unit` $\Downarrow -|*\rangle$.

For a finite unitary body the returns reached from an external input decay geometrically (Theorem B.1). The first is Cu and each later return comes from applying D once more. Adding them gives

$$y = Cu + DCu + D^2Cu + \cdots = Cu + Dy,$$
$$Lu = Au + B(Cu + DCu + D^2Cu + \cdots) = Au + By.$$

For finite unitary bodies the exits are absolutely summable, L is unitary, and $\sum_k x_k^\dagger x_k = I_X$ [5, Thms. 4, 7, 8] [3, Thm. 3.3.15, Cor. 3.3.16].

To realize Figure 3 with light, we build a multiport for the given finite unitary U, with one optical mode per basis value. Beam splitters and phase shifters implement its matrix [29]. We connect each return output to its matching input by a lossless delay line, choosing its length so that the field comes back in phase at the working frequency. The field supplied on those inputs must then satisfy $y = Cu + Dy$. The steady field on the external outputs is $Au + By = Lu$, just what the loop returns on input u (Appendix H).



For `guardRound` on input `unit`, the wire equation gives

$$R_\theta \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

The purple output matches the purple input. The caller receives the green output, -1.

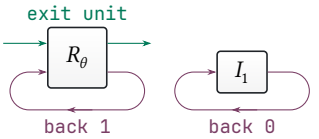
Fig. 3. In Unitary, the open body at left is $U : X \oplus Y \to X' \oplus Y$. Joining the two ends of the purple Y sheet at right gives its direct-sum trace $L = \mathrm{Tr}^Y(U) : X \to X'$.

To compute the whole gate, we solve the wire equation for each basis input $|s\rangle$. Column s of F holds its summed returns, so $F|s\rangle = C|s\rangle + DF|s\rangle$. Collecting the columns gives $(I_Y - D)F = C$. We evaluate the body to U, find a solution F, and send the returns to the external output through B. Adding the immediate exits gives $A + BF$.

$$\frac{\kappa \Downarrow U \quad U = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \quad F : X \to Y \quad (I_Y - D)F = C}{\mathrm{trace}_c\, \kappa \Downarrow A + BF}.$$

An undetermined return. For `guardRound` we found the return by dividing by $1-3/5$. To see what happens when $I_Y - D$ has no inverse, give the body two return values. Use `back 1` for the original return and let the new value, `back 0`, return to itself.

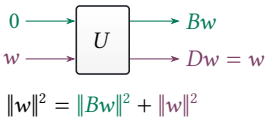
```
round extraPort : Unit + Qubit = {
  exit unit ↔ [ 3/5 . exit unit + 4/5 . back 1];
  back 1    ↔ [-4/5 . exit unit + 3/5 . back 1];
  back 0    ↔ back 0
}
```



In the order `exit unit`, `back 1`, `back 0`, the first two clauses give R_θ and the last gives I_1. The body is $R_\theta \oplus I_1$. On input u, the amplitude on `back 1` satisfies $y_1 = \frac{4}{5}u + \frac{3}{5}y_1$, so $y_1 = 2u$. On `back 0` we get $y_0 = y_0$. This leaves y_0 free and makes $I_Y - D = \mathrm{diag}(2/5, 0)$ *singular*, meaning it has no inverse. Starting from u gives $y_0 = 0$ because no call reaches `back 0`. Every choice of y_0 gives the same external output, $(3/5)u - (4/5)y_1 = -u$.

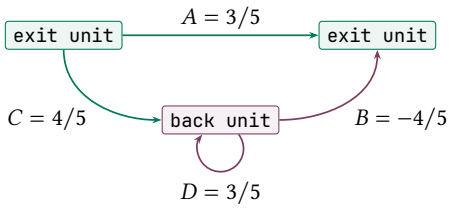
The ring brings light back to the coupler one trip later. Czachor uses the same connection to model a wormhole. Part of a particle's later output feeds its earlier input. The grandfather story follows a traveler who goes back and prevents their own departure. That story gives the traveler a definite history. In the model, amplitudes for different trips add at the external port. The particle is reflected at the mouth with probability one [12, Thm. 1 and § IV].

Czachor asks whether the external output is still fixed when the inverse fails, as he finds in the scalar case [12, after Eq. (45)]. For finite unitary bodies, Bartha's trace theorem says it is [5, Thm. 8]. Summing the returns gives one solution. Suppose we find another on the same input u. Call the difference between the two returns w. When we subtract their equations $y = Cu + Dy$, the two copies of Cu cancel, leaving $w = Dw$. In the picture, all the squared norm that enters on the lower wire leaves on that wire. Since U preserves norm, $Bw = 0$. The caller gets the same output from either solution (Appendix B, Figure 14).



3.3 Compiling away the loop

The wire equation gave us the gate $L = A + BF$. We can also write its clauses directly. In `guardRound`, the first clause sends $3/5$ to `exit unit` and $4/5$ to `back unit`. Follow the latter in the picture. One call sends $B = -4/5$ to the exit and $D = 3/5$ back to the same clause.



From unit amplitude on `back unit`, each further trip contributes one more factor of D. The exits sum to

$$B + BD + BD^2 + \cdots = \frac{B}{1-D} = -2.$$

The first clause sends $C = 4/5$ there, giving $(4/5)(-2)$. The wire equation groups the same factors as $BF = (-4/5)\cdot 2$, with $F = C/(1-D)$.

In the first clause we replace `back unit` by $-2\,(\texttt{exit unit})$. Both terms now leave by the same port, so we can add their coefficients and drop the external tag.

$$\texttt{exit unit} \leftrightarrow \tfrac{3}{5}(\texttt{exit unit}) + \tfrac{4}{5}\big(-2\,(\texttt{exit unit})\big) = -\texttt{exit unit} \quad \rightsquigarrow \quad \{* \leftrightarrow -1 \cdot *\}.$$

With two return patterns, removing the first can change the second. In `flipRound` each call rotates the guard and flips a data bit (Appendix F.1). Figure 4 replaces `back 0` by its two outputs

wherever it occurs. The path from `back 1` through `back 0` and back to `back 1` now gives a self-loop with coefficient $(3/5)^2 = 9/25$. We close it before substituting in the external clauses.

Remove `back 0`. Replace it by $-\frac{4}{5}\texttt{exit}\,1 + \frac{3}{5}\texttt{back}\,1$.

$$\texttt{exit}\,1 \mapsto \tfrac{3}{5}\texttt{exit}\,0 + \tfrac{4}{5}\texttt{back}\,0 \qquad \texttt{back}\,1 \mapsto -\tfrac{4}{5}\texttt{exit}\,0 + \tfrac{3}{5}\texttt{back}\,0$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$\tfrac{3}{5}\texttt{exit}\,0 + \tfrac{4}{5}\big(-\tfrac{4}{5}\texttt{exit}\,1 + \tfrac{3}{5}\texttt{back}\,1\big) \qquad -\tfrac{4}{5}\texttt{exit}\,0 + \tfrac{3}{5}\big(-\tfrac{4}{5}\texttt{exit}\,1 + \tfrac{3}{5}\texttt{back}\,1\big)$$

$$= \tfrac{3}{5}\texttt{exit}\,0 - \tfrac{16}{25}\texttt{exit}\,1 + \tfrac{12}{25}\texttt{back}\,1 \qquad = -\tfrac{4}{5}\texttt{exit}\,0 - \tfrac{12}{25}\texttt{exit}\,1 + \tfrac{9}{25}\texttt{back}\,1$$

Delete the `back 0` clause. The third clause is $\texttt{exit}\,0 \mapsto \frac{3}{5}\texttt{exit}\,1 + \frac{4}{5}\texttt{back}\,1$.

Close `back 1`. Sum its exits with factor $1/(1 - 9/25) = 25/16$.

$$\texttt{back}\,1 \;\longmapsto\; \tfrac{25}{16}\big(-\tfrac{4}{5}\texttt{exit}\,0 - \tfrac{12}{25}\texttt{exit}\,1\big) = -\tfrac{5}{4}\texttt{exit}\,0 - \tfrac{3}{4}\texttt{exit}\,1.$$

Substitute for `back 1` **in both exit clauses.**

$$\texttt{exit}\,0 \mapsto \tfrac{3}{5}\texttt{exit}\,1 + \tfrac{4}{5}\big(-\tfrac{5}{4}\texttt{exit}\,0 - \tfrac{3}{4}\texttt{exit}\,1\big) = -\texttt{exit}\,0,$$

$$\texttt{exit}\,1 \mapsto \tfrac{3}{5}\texttt{exit}\,0 - \tfrac{16}{25}\texttt{exit}\,1 + \tfrac{12}{25}\big(-\tfrac{5}{4}\texttt{exit}\,0 - \tfrac{3}{4}\texttt{exit}\,1\big) = -\texttt{exit}\,1.$$

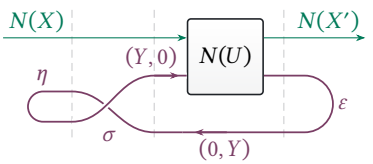
Fig. 4. Compiling `trace flipRound`. The arrows substitute for `back 0`. Closing `back 1` and substituting its exits leaves each input bit with amplitude -1.

For a general body, we evaluate each basis input as in Section 2 and choose a return $\texttt{back}\,q$. Write its clause as $\texttt{back}\,q \mapsto d\,(\texttt{back}\,q) + e$, where $d = D_{qq}$ is the amplitude to return to q and e contains the other outputs. If $d \neq 1$, we replace each occurrence of $\alpha\,(\texttt{back}\,q)$ in the other clauses by $\alpha e/(1-d)$, collect equal patterns and delete the clause for q. If $d = 1$, unitarity makes the other entries in its row and column zero, so we can delete it directly, as in `extraPort`.

We repeat with the smaller unitary table. After $|\mathcal{B}(c)|$ steps we drop the external tags and obtain an iso $a \leftrightarrow b$ with matrix L. This is elimination for finite contractions [3, Thm. 3.3.15] applied to clauses. Proposition B.2 shows that every order gives the same map and keeps the coefficients in their original subfield.

In Unitary we closed U with one operation, $L = \mathrm{Tr}^Y(U)$. To build the same loop out of bends and crossings, we need a type for a wire that runs back. In Int(Unitary) [21, § 4] an object $P = (P^+, P^-)$ gives the spaces for its rightward and leftward ports. A morphism $P \to Q$ takes input from both ends and is a unitary $P^+ \oplus Q^- \to Q^+ \oplus P^-$. The embedding N sends Y to $(Y, 0)$ and keeps each gate's matrix. Here 0 is the zero-dimensional space. A wire of type $(Y, 0)$ carries amplitude only to the right, and a wire of type $(0, Y)$ only to the left.

Follow the return output of $N(U)$ around the cap ε. It travels left to the cup η and turns right. The crossing σ swaps the port order so this leg enters the gate and the other passes below it. Both bends carry the amplitude unchanged, so the return still satisfies $y = Cu + Dy$. The composite in Figure 39 is therefore $N(L)$. Its matrix is the gate L we compiled in Unitary.



3.4 Making the sign observable

The clause for `trace guardRound` sends $|*\rangle$ to $-|*\rangle$. A caller who measures the result still sees $*$ with certainty, just as for the identity. To see the sign, the caller needs a second amplitude to compare

with the one that went through the loop. In `bypassRound`, the two amplitudes belong to one qubit. Figure 5 draws them as two paths. The upper path carries $|0\rangle$ straight through. The lower path carries $|1\rangle$ through `guardRound`. The return still carries `Unit`.

```
round bypassRound : Qubit + Unit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ [ 3/5 . exit 1 + 4/5 . back unit];
  back unit ↔ [-4/5 . exit 1 + 3/5 . back unit]
}
```

$$U_{\text{bypass}} = \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & t & -s \\ \hline 0 & s & t \end{array}\right)$$

The columns follow the clauses in order, with $t = 3/5$ and $s = 4/5$. Only the first clause carries the bypass, so the immediate exit is $x_0 = \mathrm{diag}(1, t)$. Every later exit comes from `back unit` and gives $x_{k+1} = \mathrm{diag}(0, -s^2 t^k)$ for $k \geq 0$. On the lower path these are the exits of `guardRound`. Closing the loop leaves $|0\rangle$ alone and sends $|1\rangle$ to $-|1\rangle$.

The caller in Figure 5 runs the Hadamard test on the scalar loop [9, p. 1, Eqs. (1)–(3)]. Its program is `H; trace bypassRound; H; measure on 0`. The first Hadamard prepares $(|0\rangle + |1\rangle)/\sqrt{2}$. The loop changes this to $(|0\rangle - |1\rangle)/\sqrt{2}$. Measuring either state in the computational basis would give 0 and 1 with equal probability. The second Hadamard brings the two paths together at each output.

$$H\frac{|0\rangle - |1\rangle}{\sqrt{2}} = \tfrac{1}{2}(|0\rangle + |1\rangle) - \tfrac{1}{2}(|0\rangle - |1\rangle) = (\tfrac{1}{2} - \tfrac{1}{2})|0\rangle + (\tfrac{1}{2} + \tfrac{1}{2})|1\rangle = |1\rangle.$$

The contributions cancel at output 0 and add at output 1, so the caller reads 1 with certainty. Had the loop preserved the sign, the two Hadamards would have returned the caller to 0.

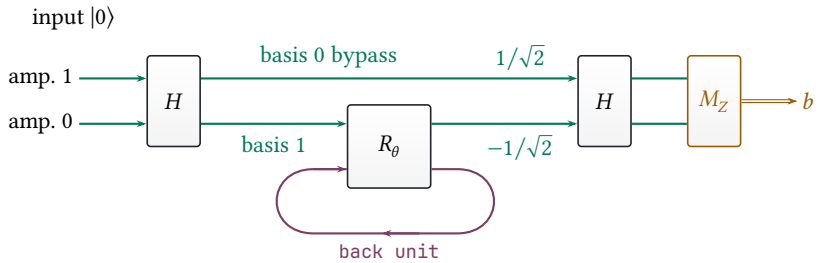


Fig. 5. The two rails carry the basis alternatives of one qubit $G = \mathbb{C}^2$. Each $H : G \to G$ and the closed loop $\mathrm{diag}(1, -1) : G \to G$ are in Unitary. The final measurement $M_Z : G \rightsquigarrow G$ is a channel in CPTP. Its output is diagonal in the computational basis and gives bit 1 with certainty.

Fairbanks and Selinger observe that the trace of this rotation jumps from $+1$ at $\theta = 0$ to -1 for $0 < \theta < 2\pi$ [13, Remark B.1]. At $\theta = 0$ all the light goes straight through. To see what happens near zero, we switch on a steady input of amplitude one with the ring empty and $z = 1$. At first only the direct field $t = \cos\theta$ reaches the output. After n trips the first n delayed exits have arrived, giving $t - (1 - t^2)(1 + t + \cdots + t^{n-1}) = -1 + (1 + t)t^n$ for $n \geq 1$. For any fixed n this approaches $+1$ as θ tends to zero. For any fixed small positive θ, the factor t^n decays and the output approaches -1. The weaker the coupling, the longer it takes for the returning light to reverse the sign.

We can use this sign to separate light by wavelength. The splitters in Figure 6 act like the two Hadamards in Figure 5. The first sends amplitude $1/\sqrt{2}$ down each path. The ring changes the sign on the lower path. The second splitter takes their sum at the upper port and their difference at the lower one, each divided by $\sqrt{2}$. All the light leaves by the lower port. Wang et al. used a ring on the shorter of two paths to separate evenly spaced optical frequencies. The odd-numbered frequencies went mostly to one output and the even-numbered ones to the other [32, Figs. 1 and 6].

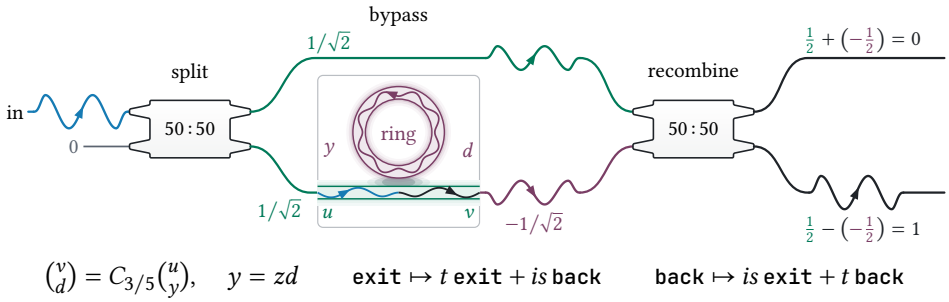
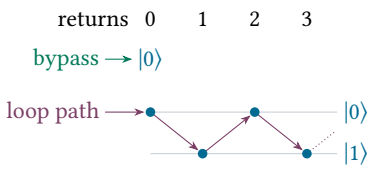


Fig. 6. The lower path contains the ring from the introduction. At the gap, light stays in its guide with amplitude $t = 3/5$ or crosses with amplitude $is = 4i/5$. The straight guide carries u to v. The ring feeds d back as $y = zd$ [2, 28]. At $z = 1$ the two crossings give $(is)^2 = -s^2$ and each further trip gives t, exactly the exits of `guardRound`.

The bypass lets the caller read the sign of the closed rotation. An exit after one return and an exit after two both reach the second splitter on the same path. To tell them apart there, we can carry a record around the loop.

4 What allocation and hiding do to the loop

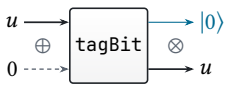
We run the same three clauses of `bypassRound`, this time with a counter bit. We start it at 0 and flip it on each return. The bypass exits at once with 0. On the lower path, one return takes the bit to 1 and two bring it back to 0. At the second Hadamard, exits with an even count can still interfere with the bypass. Odd counts carry a different bit.



The caller did not supply a count register. We need the ability to allocate that register.

Adding allocation to iso. Our caller supplies one qubit, with two basis values. Returning it alongside a counter bit gives four output pairs. An iso needs a square table, so we tag the input to give it four basis values as well.

```
tagBit :: (Qubit + Qubit) ↔ (Qubit x Qubit)
tagBit = {left x  ↔ ⟨0, x⟩;
          right x ↔ ⟨1, x⟩}
lift tagBit on 1
```



The bit records the input tag, so either clause can run backwards. The last line uses `lift` to supply the data through the left input. On `1` it gives `<0, 1>`. On the caller's superposition $u = (|0\rangle + |1\rangle)/\sqrt{2}$, it gives $|0\rangle \otimes u$. The 0 on the unused input means zero amplitude.

Following Heunen and Kaarsgaard, we call the extended language iso_a [18, § 3.3]. It embeds each iso with arr and adds lift:

$$\frac{\omega : a \leftrightarrow b}{\mathsf{arr}\,\omega : a \rightarrowtail b} \qquad \frac{\omega : a + d \leftrightarrow b}{\mathsf{lift}\,\omega : a \rightarrowtail b}.$$

If ω evaluates to U, then $\mathsf{arr}\,\omega$ uses the whole matrix. On input u, $\mathsf{lift}\,\omega$ gives $U(u, 0)$: only the columns for a are used. Those columns are still orthonormal, so the map preserves inner products. This is an *isometry*, written $\rightarrowtail$. These maps form the category $\mathsf{Isometry} \simeq R[\mathsf{Unitary}]$, where programs in iso_a compose in sequence, by direct sum and by tensor [18, Thm. 22].

For $[K]$, the sum of $K \geq 1$ copies of 1, the same construction gives alloc_j, which prepares j alongside the data. Starting at zero and advancing on each return gives $\mathsf{count}_K\,\beta : a \rightarrowtail [K] \otimes b$

for a body $\beta : a + c \leftrightarrow b + c$. We write `count K f on t`, where the numeral `K` gives the number of counter values, `f` is the round, and `t` is its external input.

Adding hiding to iso_a. The counting loop returns the counter and the data. Suppose we want to return just the data to the caller for the second Hadamard. We need the ability to hide the counter. To see what this does, group the exits of `bypassRound` by the bit they carry. Write u_0 for the sum of the data vectors at even counts, including the bypass, and u_1 for their sum at odd counts. The joint output is $|0\rangle \otimes u_0 + |1\rangle \otimes u_1$. If the caller keeps only the data, we take the partial trace over the bit. A unit vector w has density matrix $ww^\dagger$. For our joint output, grouping its rows and columns by the counter gives

$$\begin{pmatrix} u_0 u_0^\dagger & u_0 u_1^\dagger \\ u_1 u_0^\dagger & u_1 u_1^\dagger \end{pmatrix} \xrightarrow{\text{hide the bit}} u_0 u_0^\dagger + u_1 u_1^\dagger .$$

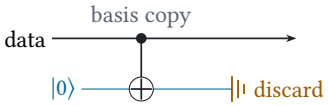
The partial trace adds the two diagonal blocks. The other two drop out because $\langle 0 \mid 1 \rangle = 0$. Exits carrying the same bit have already added their amplitudes in u_0 or u_1.

This second extension, iso_a^χ, embeds an isometry with arr and adds hide [18, § 3.4]:

$$\frac{v : a \rightarrowtail b}{\mathsf{arr}\, v : a \rightsquigarrow b} \qquad \frac{v : a \rightarrowtail b \otimes g \qquad \mathscr{B}(g) \neq \emptyset}{\mathsf{hide}\, v : a \rightsquigarrow b}.$$

The rule hides the last factor. The executable `count` returns the data first, so `count 2 bypassRound; discard` keeps the data. On a density matrix ρ (a positive operator of trace one), $\mathsf{arr}\, v$ gives $V \rho V^\dagger$ and $\mathsf{hide}\, v$ gives $\mathrm{tr}_g(V \rho V^\dagger)$, where V is the matrix of v. These channels compose in sequence and by tensor in the category of completely positive trace-preserving maps $\mathbf{CPTP} \simeq L[\mathbf{Isometry}]$ [19, Cor. 7]. To discard an inhabited b as in Figure 12, we hide b in $b \leftrightarrow 1 \otimes b$. This sends ρ to $\mathrm{tr}\, \rho$.

From a fresh bit to a measurement. Start a fresh bit at 0 and flip it when the data bit is 1. On basis input s this writes a copy of s. On a superposition $\alpha|0\rangle + \beta|1\rangle$, the two wires carry $\alpha|00\rangle + \beta|11\rangle$. Discarding the copy destroys the coherence between 0 and 1. Reading the data now gives 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.



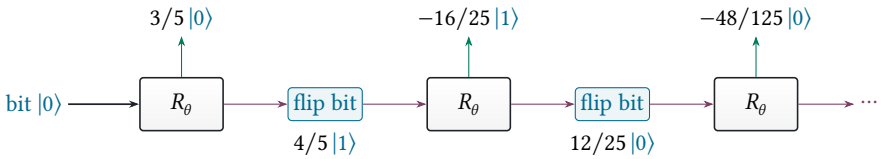
For any inhabited b, we lift a permutation that sends each selected input s to $\langle s, s \rangle$. This gives the basis copier $\mathsf{clone}_b : b \rightarrowtail b \otimes b$. Hiding its copy measures b in the computational basis [18, § 3.4.1].

$$\mathsf{measure}_b = \mathsf{hide}\, \mathsf{clone}_b, \qquad [\![\mathsf{measure}_b]\!](\rho) = \sum_{s \in \mathscr{B}(b)} P_s \rho P_s, \qquad P_s = |s\rangle\langle s|.$$

In $\mathsf{while}_c\, \beta : a \rightsquigarrow b$ we copy and discard only the tag after each call. This preserves coherence within each payload. Appendix A.5 gives the rules for empty types.

4.1 Keep a count

In `count 2 guardRound on unit`, the 2 tells us to keep the count modulo two. The first call sends $3/5\,|0\rangle$ to the exit with the bit still at 0. The other $4/5$ goes around with the bit flipped to 1. After two returns the bit is 0 again, so the first and third exits carry the same bit and payload.

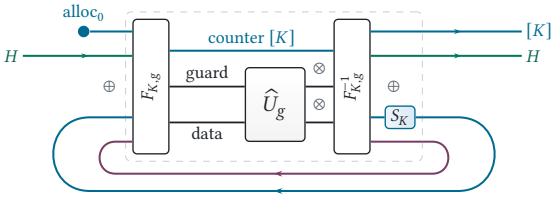


Every two further returns multiply a delayed exit by $(3/5)^2$. Summing the even and odd exits separately gives 0 and -1.

$$\underbrace{\frac{3}{5} - \frac{48/125}{1-(3/5)^2}}_{\text{even returns}} = 0, \qquad \underbrace{\frac{-16/25}{1-(3/5)^2}}_{\text{odd returns}} = -1.$$

The loop path of `bypassRound` thus leaves with counter state $-|1\rangle$, and the bypass with $|0\rangle$. On input $(|0\rangle + |1\rangle)/\sqrt{2}$, the joint output is $(|0\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle)/\sqrt{2}$, with the counter first. Hiding it leaves $I/2$. The second Hadamard preserves this equal mixture, so the caller reads 1 half the time.

With K values, the bit flip becomes $S_K : j \mapsto j+1 \bmod K$. For a common payload H, $F_{K,g}$ separates counter, guard and payload. The counter passes beside the body. After $F_{K,g}^{-1}$ restores the ports, S_K advances the counter on the return wire.



We allocate the counter at zero and close the return wire of the enlarged body β_K. After k returns the exit $x_k u$ carries $k \bmod K$ (Figure 7):

$$\mathrm{count}_K\, \beta = \mathrm{alloc}_0\, ;\, \mathrm{arr}(\mathrm{trace}_{[K]\otimes c}\, \beta_K), \qquad R_K u = \sum_{k\geq 0} |k \bmod K\rangle \otimes x_k u. \tag{4.1}$$

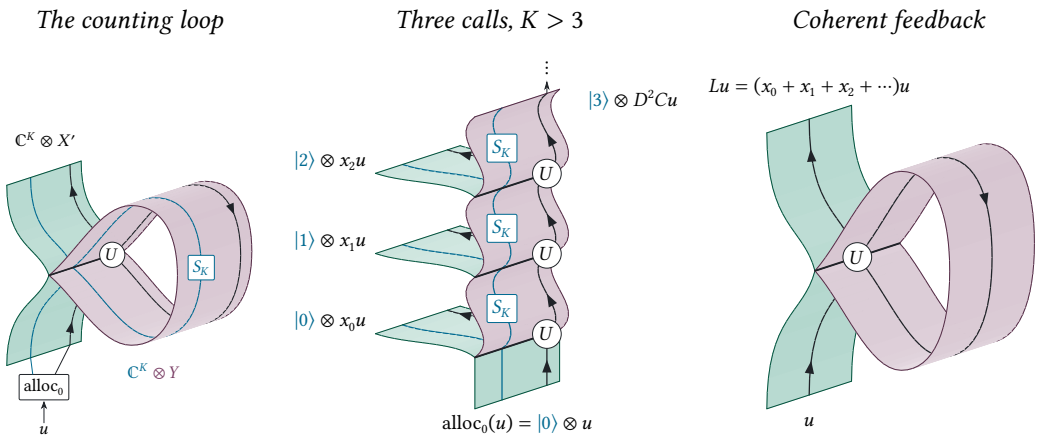


Fig. 7. Read upwards. Each trip around the purple sheet advances the blue counter. After three calls, $|3\rangle \otimes D^2 Cu$ continues around the loop. Without the counter, all exits reach one output and their amplitudes add.

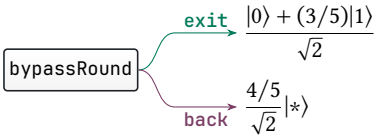
To hide the counter we swap it to the last position and apply hide. Exits with the same label have already added. With an exact count we get $Ru = \sum_k |k\rangle \otimes x_k u$ with $H_{\mathrm{rec}} = \ell^2(\mathbb{N})$. Both R_K and R are isometries. Embedding the first K labels gives $R_K \to R$ in operator norm (Appendix C). Ying's coin loop records a one for each return and a zero for the exit. The word $1^k 0$ thus records k returns [35, Eq. (34)]. Assolini and Di Pierro copy the guard into a fresh qubit at each test [4, § 4].

In the bounded loop of Yuan et al., an input register says how many times to apply an operation. A branch that finishes early executes no-op instructions until all branches reach the same control state at the same machine step [37, § 6.1, Fig. 7]. The original count remains in the input register. Register-machine compilation likewise computes wait counts within a chosen time bound [38, § 5] to address the common-clock halting question [24]. In our loop, amplitude from a single input can leave after different numbers of returns. The counter labels those exits as the body runs, even when there is no last round.

4.2 Measure the returning tag

We now run `H; while bypassRound; H; measure on 0`. The counter gave the first and third exits the same bit. Measuring the tag after every call distinguishes the first exit from every later one.

The first Hadamard sends $1/\sqrt{2}$ down each path. The bypass and $3/5$ of the lower path's amplitude both leave on the first `exit`. At output 1 the second Hadamard takes their difference. The amplitude is $(1-3/5)/2 = 1/5$, so this exit contributes probability $1/25$.



Only the lower path returns, with probability $|(4/5)/\sqrt{2}|^2 = 8/25$. Each further call returns with probability $9/25$, so this part eventually exits. It carries $|1\rangle$, which the second Hadamard sends to 1 half the time. Adding the first and later exits gives $1/25 + (8/25)(1/2) = 1/5$.

We match the bypass to the direct exit in phase and arrival time. Of the pulses leaving the ring in Figure 8, only the first meets the bypass at the second splitter. Half of each later pulse goes to the difference port, giving $1/5$ there when we include the first pulse. Under steady illumination at the frequency of Figure 6, that port gets all the light [28, Eqs. (30)–(33)].

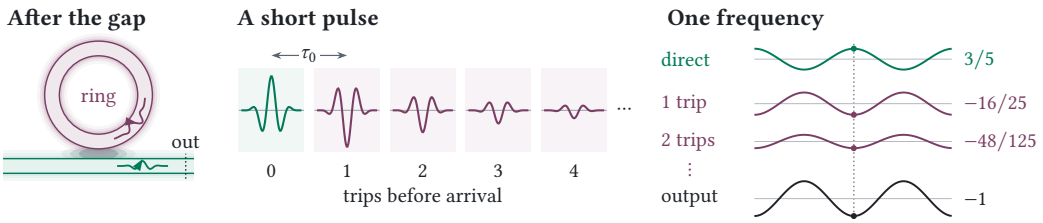


Fig. 8. A pulse shorter than one round-trip time τ_0 gives a separate pulse for each exit. With steady light at round-trip phase $z = 1$, the dots mark the field at the same instant. The output sums all the exits.

What one call measures. In `while f`, the body supplies the return type c, as in `trace f`. After each call we copy its tag into a fresh bit and hide the bit. On the first exit of `bypassRound` both paths carry the same tag, so copying it preserves their interference.

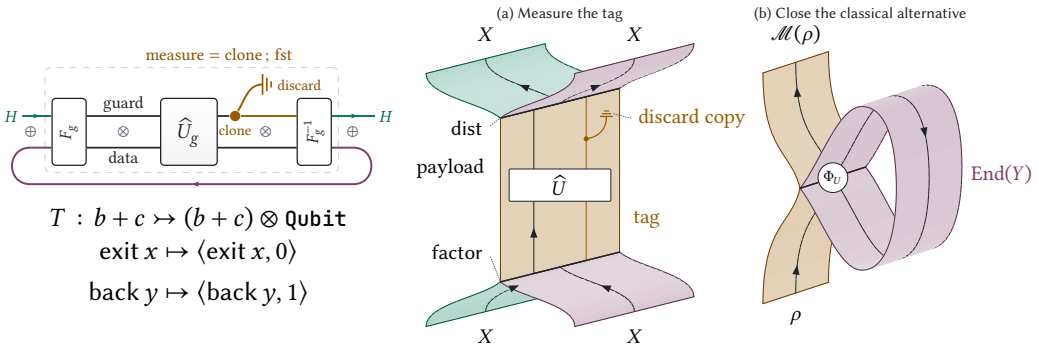


Fig. 9. With a common payload, F_g factors the ports into guard and data. One call is hide(arr β; T). We discard the copied guard, then F_g^{-1} restores the ports. The sheets show this same call from bottom to top and the return block that passes to the next call.

For external input ρ and returning input σ, one call gives

$$\Phi_U(\rho,\sigma) = \big(\underbrace{A\rho A^\dagger + B\sigma B^\dagger}_{\text{exit}},\ \underbrace{C\rho C^\dagger + D\sigma D^\dagger}_{\text{return}}\big).$$

Their traces give the probabilities on normalized input. The first call on $(\rho, 0)$ emits $A\rho A^\dagger$. Its return $C\rho C^\dagger$ gives $(BC)\rho(BC)^\dagger$ on the next exit. Ying and Feng test the guard first [36]. When the external types agree, we run U once and then enter their loop. Each unfolding adds one more measured exit. Their sum gives Ying's least fixed point [34, Prop. 5.1] (Appendix C).

Adding the measured exits. After k returns, one exit gives `exit` on call $k+1$ while every later exit still gives `back`. The tag measurement therefore distinguishes every pair of counts. The first two exits of `guardRound` add to $3/5 - 16/25 = -1/25$ in the coherent loop. Here their probabilities add to $(3/5)^2 + (-16/25)^2$ (Table 1).

On input ρ, we add the exits' density matrices, just as when we hide the exact count:

$$\mathscr{M}(\rho) = \sum_{k\ge 0} x_k \rho x_k^\dagger = \mathrm{tr}_{H_{\mathrm{rec}}}(R\rho R^\dagger), \qquad \mathscr{M} = [\![\mathsf{while}_c\ \beta]\!]. \tag{4.2}$$

The partial trace drops the cross terms between different counts. Appendix C derives the same sum by closing the return block of Φ_U. Since $\sum_k x_k^\dagger x_k = I$ (Theorem B.1), a finite unitary body exits with probability one on every normalized input, even when there is no last possible call.

Now consider the same body $\beta : a + c \leftrightarrow b + c$ in all three loops. In `trace` all exits meet. In count_K only exits with the same label meet. In `while` we add their density matrices. For `bypassRound` on $|+\rangle$, we hide any counter and apply H to the data ($t = 3/5$, $K = 2$). The same caller reads 1 with certainty for coherent feedback, with probability $1/2$ when we keep parity, and with probability $1/5$ when we measure on every round.

Loop and type	*General output*	`bypassRound`	Pr(1)
$\mathsf{trace}_c\ \beta : a \leftrightarrow b$	$Lu = \sum_{k\ge 0} x_k u$	$\lvert - \rangle$	1
$\mathsf{count}_K\ \beta : a \rightarrowtail [K] \otimes b$	$R_K u = \sum_{k\ge 0} \lvert k \bmod K\rangle \otimes x_k u$	$\dfrac{\lvert 0\rangle\otimes\lvert 0\rangle - \lvert 1\rangle\otimes\lvert 1\rangle}{\sqrt{2}}$	$1/2$
$\mathsf{while}_c\ \beta : a \rightsquigarrow b$	$\mathscr{M}(\rho) = \sum_{k\ge 0} x_k \rho x_k^\dagger$	$\frac{1}{2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}$	$1/5$

5 When can a caller forget the difference?

Suppose the caller needs only the data. Hiding the exact count adds one density matrix for each exit. Closing the wire adds the exit amplitudes. We want a test on the body that tells us whether the two loops give the caller the same state on every input.

In `guardRound`, every exit carries `unit`. The first two exits multiply it by $3/5$ and $-16/25$, and each later exit picks up another factor of $3/5$. Their sum is $3/5 - (16/25)/(1 - 3/5) = -1$. The caller therefore receives `unit` with certainty whether we close the wire or measure the tag. Adding the bypass changes what an exit does to the input. The first exit carries amplitude from both paths, but only the rotation path contributes to the later exits. Recording even one return therefore singles out the rotation path.

For a general body, write $x_k u$ for the output after exactly k returns on input u. The coherent loop gives $Lu = \sum_k x_k u$. Suppose every x_k applies this same gate to the data, up to a scalar b_k that depends only on the count. Keeping the exact count then gives

$$Ru = \sum_{k\ge 0} |k\rangle \otimes x_k u = \Big(\sum_{k\ge 0} b_k |k\rangle\Big) \otimes Lu.$$

For `guardRound`, $L = -1$, so factoring out Lu leaves $-\frac{3}{5}|0\rangle + \frac{16}{25}|1\rangle + \frac{48}{125}|2\rangle + \cdots$ in the counter. The count can vary, but this state is the same for every input. When hiding it gives the caller the coherent output, we call the agreement on the data *free erasure*.

Theorem 5.1 (Agreement on the returned data). *Let $\beta : a + c \leftrightarrow b + c$ be a finite iso with nonempty external input type a. Let $m = |\mathscr{B}(c)|$ be the number of basis values on the return wire. The following are equivalent.*
(1) The loops agree on the returned data, $\mathscr{M}(\rho) = L\rho L^\dagger$ for every input state ρ.
(2) One unit count vector $\xi \in \ell^2(\mathbb{N})$ satisfies $Ru = \xi \otimes Lu$ for every input u.
(3) One of $x_0, \ldots, x_m$ is nonzero and every map in this list is a scalar multiple of it.
In this case $x_k = b_k L$ for every k, where $\xi = \sum_k b_k|k\rangle$, $\sum_k b_k = 1$ and $\sum_k |b_k|^2 = 1$.

To see why agreement forces this factorization, give the coherent loop a unit input u. A measurement along any direction orthogonal to Lu has probability zero. In the measured loop each exit contributes its own nonnegative probability to that same outcome. For their sum to be zero, each $x_k u$ must lie along Lu. If an exit multiplied two basis outputs by different scalars, it would change their relative amplitudes on a superposition of those inputs. Thus at each count one scalar works for every input. For finitely many exits this also follows from the Kraus criterion [33, Cor. 2.23]. Appendix D gives the proof for the full sequence.

To check all the exits, we need only the four maps of the body. Write A for the immediate exit and C for the part of the first call that enters the return wire. On that wire B leads out and D goes around again. The exits in order are therefore

$$x_0 = A, \qquad x_1 = BC, \qquad x_2 = BDC, \qquad x_{j+1} = BD^jC \quad (j \geq 0).$$

If there are $m > 0$ return values, D is an m-by-m matrix. By Cayley–Hamilton, every power from D^m onward is a linear combination of $I, D, \ldots, D^{m-1}$. Multiplying on the left by B and on the right by C expresses every later exit as a linear combination of $x_1, \ldots, x_m$. When $m = 0$ there is only A.

We can therefore stop at x_m, even when there is no last round. At least one of these maps is nonzero, since their span contains the unitary L. We choose a nonzero entry in one map. For each other map, we divide its corresponding entry by the chosen one and check whether this ratio works for every entry. With decidable equality of exact coefficients, this gives a finite test.

Suppose $|0\rangle$ exits at once and $|1\rangle$ exits after $m > 0$ returns, both unchanged (Appendix D). A counter modulo m labels both exits zero, so $|+\rangle$ survives. Measurement leaves $I/2$. The first m exit maps are multiples of x_0. Only x_m reveals the difference.

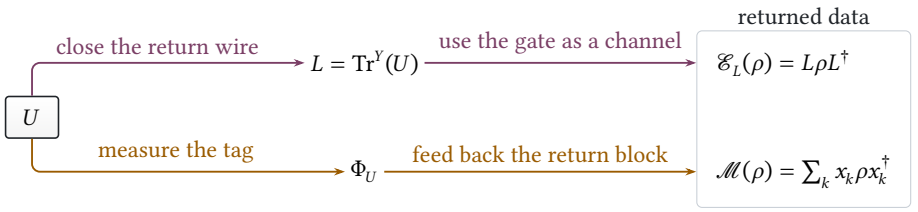


Fig. 10. Closing $U : X \oplus Y \to X' \oplus Y$ in Unitary gives $L : X \to X'$. Measuring the tag gives $\Phi_U : \mathrm{End}(X) \oplus \mathrm{End}(Y) \to \mathrm{End}(X') \oplus \mathrm{End}(Y)$, whose return we close by the classical-sum trace. Both routes give the caller a channel from X to X' in CPTP.

Replacing the loop in a program. Andrés-Martínez gives an example where the two routes differ [3, § 3.3.3]. Theorem 5.1 tells us exactly when they agree. When $x_k = b_k L$, the upper path in Figure 10

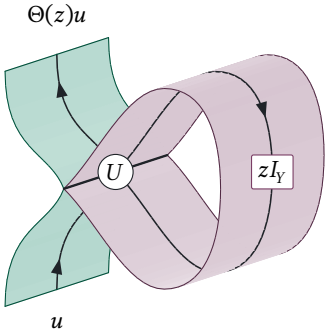
adds the exits to give $(\sum_k b_k)L = L$. On the lower path, exit k contributes $|b_k|^2 L\rho L^\dagger$. These weights sum to one, so the caller receives $L\rho L^\dagger$ along either path.

Leaving a hole in the caller where it uses the loop's data channel gives a *channel context*. Equal channels give the same output probabilities even if the caller keeps a reference entangled with the input or runs the loop on one branch of a measurement. If they differ, some input and an output measurement distinguish them. A coherent bypass belongs to the body we test before we give its channel to the caller.

Corollary 5.2 (Replacement in a caller). *Let β be a finite iso with m basis values on its return wire. Hiding the exact count always gives the measured loop. On a nonempty external input type, the measured loop may replace the coherent loop's data channel in every channel context if and only if $x_0, \dots, x_m$ span a line. In that case, hiding the counter in $\mathrm{count}_K\,\beta$ gives the same channel for every $K \geq 1$. On an empty external input type all these channels agree.*

6 A count without a counter

The caller who receives only the data from `guardRound` gets `unit` whether we close the wire or measure the tag. Even the coherent gate alone tells us little about the returns. A clause that exits at once with sign -1 gives the same gate. Only the rotation sends amplitude back. To see the difference, we put a phase on the return wire before we close it.



Follow the arrow around the purple sheet. Each time the body sends amplitude back, the gate on that sheet multiplies it by $z = e^{i\varphi}$. The immediate exit still gives $3/5$. The next exit crosses the gate once and gives $-(16/25)z$. The one after it crosses twice and gives $-(48/125)z^2$. Write $t = 3/5$ and $s = 4/5$. Each further trip contributes another factor tz, so the output amplitude is

$$\Theta_{\mathrm{rot}}(z) = t - s^2 z(1 + tz + t^2 z^2 + \cdots) = \frac{t - z}{1 - tz}.$$

Suppose we measure the tag after each call. On `exit unit` the first clause sends amplitude $s = 4/5$ to `back unit`. The probability of reading **back** is the squared magnitude of that amplitude, $|s|^2 = 16/25$. If we read **back**, the next call starts in `back unit`. Its clause sends amplitude $t = 3/5$ back to that same port, so the conditional probability of another return is $|t|^2 = 9/25$. The probabilities of making at least one return, at least two, and at least three are therefore s^2, s^2t^2, s^2t^4. Adding all these probabilities counts a run with k returns k times, giving the mean $s^2/(1 - t^2) = 1$.

Now close the wire coherently and look just before the phase gate. For input amplitude 1, successive returns contribute $s, stz, s(tz)^2, \dots$ to the same wire. At $z = 1$ they add to $s/(1 - t) = 2$, as in the wire equation of Section 3. Its squared magnitude for unit input is the *circulating weight*, here $|2|^2 = 4$. At $z = -1$ the returns alternate in sign. They add to $s/(1 + t) = 1/2$, giving a circulating weight of $1/4$ for the same input.

The bypass lets the caller see the change in the output. At $z = 1$ the loop gives -1, so the second Hadamard sends the caller to output 1. At $z = -1$ the loop gives $+1$ and the caller reads 0. To read the weight itself, we compare two nearby settings. For a small angle δ, $\Theta_{\mathrm{rot}}(e^{i\delta}) = -1 - 4i\delta + O(\delta^2)$. The output phase changes by 4δ to first order. The immediate clause gives -1 at either setting.

For a ring with round-trip time τ_0, changing the frequency by $\Delta\omega$ adds $\tau_0\Delta\omega$ to the return phase. With a frequency-independent coupler, the output phase changes by $4\tau_0\Delta\omega$ to first order, giving

a Wigner–Smith delay of $4\tau_0$ relative to the direct path [31]. For a weaker coupling the weight $\sin^2\theta/(1-\cos\theta)^2 = \cot^2(\theta/2)$ grows without bound as $\theta \to 0$ (Appendix H).

From one return wire to a finite body. On input u, the successive returns just before the phase gate are $Cu, zDCu, (zD)^2Cu, \ldots$. Write Y_1 for the orthogonal complement of the return states that never emit. Restricting B, D to this space gives B_1, D_1, with decaying powers of D_1 as in Section 3. The sum $F_z u$ solves $F_z u = Cu + zD_1F_z u$. The return crosses the phase gate before B_1 takes it to the output. Adding the immediate exit gives

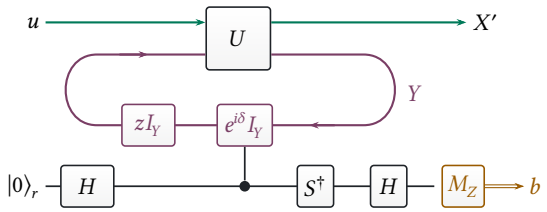
$$F_z = (I - zD_1)^{-1}C, \qquad \Theta(z) = A + zB_1F_z = \sum_{k\ge 0} z^k x_k.$$

Theorem 6.1 (The measured mean and the phase derivative). *For a finite unitary body, let* $N = \sum_{k\ge 0} k\, x_k^\dagger x_k$. *For* $z = e^{i\varphi}$,

$$Q(z) = -i\Theta(z)^\dagger \frac{d}{d\varphi}\Theta(e^{i\varphi}) = F_z^\dagger F_z, \qquad \frac{1}{2\pi}\int_0^{2\pi} Q(e^{i\varphi})\,d\varphi = N, \qquad \operatorname{tr} N = \dim Y_1.$$

If every exit makes k returns, then $\Theta(e^{i\varphi}) = e^{ik\varphi}L$ and $Q(z) = kI$. When exits from different rounds interfere, the derivative instead gives $\langle u, Q(z)u\rangle = \|F_z u\|^2$ for unit input u. Appendix E derives this identity by comparing the wire equation at two phases [6, Eq. (4.10)].

Reading the weight with one qubit. To compare two nearby phases in one run, we prepare the reference in $(|0\rangle + |1\rangle)/\sqrt{2}$. Both branches put z on every return. Only the $|1\rangle$ branch adds the extra angle δ. By the time the data exits, the two branches have passed through different closed gates.



Before the final two gates

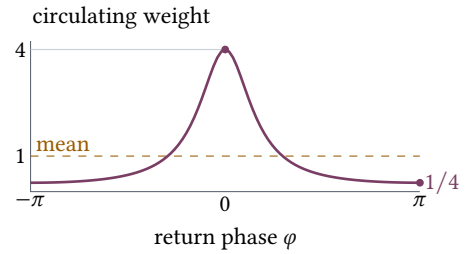
$$\frac{|0\rangle \otimes \Theta(z)u + |1\rangle \otimes \Theta(ze^{i\delta})u}{\sqrt{2}}$$

The last two gates, $S^\dagger$ and H, turn the imaginary part of the overlap into a difference of probabilities. If b is the bit we read in this Hadamard test [9, p. 1, Eqs. (1)–(3)], then

$$\Pr(b=0) - \Pr(b=1) = \operatorname{Im}\langle \Theta(z)u, \Theta(ze^{i\delta})u\rangle = \delta\,\|F_z u\|^2 + O(\delta^2).$$

For `guardRound` at $z = 1$, the caller therefore reads 0 with probability $1/2 + 2\delta + O(\delta^2)$. A clause that exits immediately still gives 0 half the time. Over repeated runs, we subtract the fraction of ones from the fraction of zeros and divide by δ. For small δ this estimates the weight 4 (Appendix I).

Averaging over the phase. Now choose the angle uniformly over a full turn and keep it fixed on every return in that run. The curve shows the circulating weight of `guardRound` at each setting. Its height varies from 1/4 to 4, yet its average is the measured mean of one return.



In $|s + stz + st^2z^2 + \cdots|^2$, each term multiplies the conjugate of every term in the sum. A pair with exponents k, j carries the phase $e^{i(k-j)\varphi}$. If $k \neq j$ this factor averages to zero. What remains is $s^2 + s^2t^2 + s^2t^4 + \cdots$, the sum of the probabilities of reaching each successive return.

For a general body on unit input u, the measured mean is $\sum_k k\|x_k u\|^2$ [7, § 3, Eq. (8)]. Appendix E sums these means over an orthonormal input basis. The sum is $\dim Y_1$, the dimension of the part of the return space that can emit.

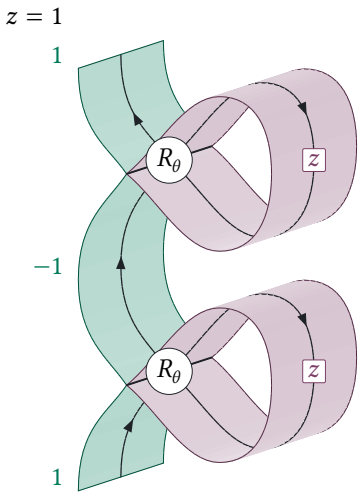
If we forget which phase we chose, terms from different counts cancel at the output too. The caller gets the data it would get by measuring the guard on every round.

$$\frac{1}{2\pi}\int_0^{2\pi} \Theta(e^{i\varphi})\rho\Theta(e^{i\varphi})^\dagger \, d\varphi = \mathscr{M}(\rho). \tag{6.1}$$

With the bypass the caller reads 1 with certainty at $z = 1$, with probability $1/2$ for an equal choice of $+1$ or -1 (hidden parity), and with probability $1/5$ over the full circle. For `guardRound` alone the weight varies, yet the caller always gets `unit`. By Section 5, when all exits are multiples of one gate, z changes only its overall phase. Conversely, a channel that is the same at every setting agrees at $z = 1$ with its average, $\mathscr{M}$.

7 What the caller can tell

We close two copies of `guardRound` and feed the output of the first into the second. With the return phase fixed at $z = 1$, the first loop changes the sign and the second changes it back. Together they give the caller the same gate as a straight wire.



Follow the green sheet upwards. At $z = 1$ the input amplitude 1 leaves the first loop as -1 and the second as 1 again. Each purple sheet goes back to its own rotation.

Now let us vary z, keeping it the same on both return wires. A path that goes around the lower loop once and the upper loop twice picks up z in the first and z^2 in the second. A path that makes all three returns in the lower loop also carries z^3. Their amplitudes add in the coefficient of z^3.

For bodies U and V, the first loop sends $x_i^U u$ into the second after i returns. An exit after j more returns therefore contributes $x_j^V x_i^U u$ and carries the phase z^{i+j}. All the ways of making k returns reach the same output, so $x_k = \sum_{i+j=k} x_j^V x_i^U$. For two copies of `guardRound`, each loop multiplies its input by $\Theta_{\mathrm{rot}}(z)$ and the pair gives $\Theta_{\mathrm{rot}}(z)^2$.

If we measure the tag after every call, each loop makes one return on average, giving two in all. With coherent feedback we can compare two nearby settings using the reference qubit. Starting at $z = 1$, a change δ in the return angle shifts each loop's output phase by 4δ to first order. The two shifts add, so the caller in Section 6 reads 0 with probability $1/2 + 4\delta + O(\delta^2)$. On a straight wire the probability stays at $1/2$.

We can see what it means to count these returns by going back to the ring from the introduction. Put two identical rings in sequence and send in a pulse shorter than one round trip. The first pulse to arrive went straight through. The next made one trip, and the one after that made two. Since a trip takes the same time in either ring, the arrival time tells us how many trips the light made in all.

Now illuminate the same rings steadily at a single frequency. Light that went straight through and light that went around the rings reach the same outgoing mode, and their amplitudes add. At the frequency where each ring changes the sign, the two signs cancel. To see the effect of the returns, we change the frequency slightly. Each trip picks up the same small change in phase, so a contribution that made three trips picks up that change three times. Near our starting frequency, the output phase changes eight times as fast as the phase on one trip. For unit input, that eight is the total circulating weight in the two rings.

If we repeat the measurement over a full turn of the return phase and average the weights, we get two, the mean number of returns we found by measuring the guard. We can learn that mean while the exits still interfere, without assigning a number of trips to any one detection.

AI Use. The author used ChatGPT Codex (GPT-6.1 Sol) for formatting, TikZ, code, literature searches and brainstorming, verified the results and takes responsibility for the revised paper.

Data Availability. We plan to submit iso and its examples for Artifact Evaluation.

References

[1] Samson Abramsky, Esfandiar Haghverdi, and Philip Scott. 2002. Geometry of Interaction and linear combinatory algebras. *Mathematical Structures in Computer Science* 12, 5 (2002), 625–665. doi:10.1017/S0960129502003730

[2] Paul M. Alsing, Edwin E. Hach, III, Christopher C. Tison, and A. Matthew Smith. 2017. Quantum-optical description of losses in ring resonators based on field-operator transformations. *Physical Review A* 95 (2017), 053828. arXiv:1608.01280 [quant-ph] doi:10.1103/PhysRevA.95.053828

[3] Pablo Andrés-Martínez. 2022. *Unbounded loops in quantum programs: categories and weak while loops.* Ph. D. Dissertation. University of Edinburgh. arXiv:2212.05371 doi:10.48550/arXiv.2212.05371

[4] Nicola Assolini and Alessandra Di Pierro. 2025. A Denotational Semantics for Quantum Loops. arXiv:2506.23320 [cs.PL]. doi:10.48550/arXiv.2506.23320

[5] Miklós Bartha. 2014. Quantum Turing automata. In *Proceedings of DCM 2012 (EPTCS, Vol. 143)*. Open Publishing Association, Waterloo, NSW, Australia, 17–31. arXiv:1404.0074 doi:10.4204/EPTCS.143.2

[6] Jussi Behrndt, Seppo Hassi, and Henk de Snoo. 2009. Boundary Relations, Unitary Colligations, and Functional Models. *Complex Analysis and Operator Theory* 3, 1 (2009), 57–98. doi:10.1007/s11785-008-0064-z

[7] J. Bourgain, F. A. Grünbaum, L. Velázquez, and J. Wilkening. 2014. Quantum Recurrence of a Subspace and Operator-Valued Schur Functions. *Communications in Mathematical Physics* 329, 3 (2014), 1031–1067. arXiv:1302.7286 doi:10.1007/s00220-014-1929-9

[8] Jacques Carette, Chris Heunen, Robin Kaarsgaard, and Amr Sabry. 2024. How to Bake a Quantum Π. *Proceedings of the ACM on Programming Languages* 8, ICFP, Article 236 (2024), 29 pages. doi:10.1145/3674625

[9] Andrew M. Childs. 2013. Quantum Algorithms, Lecture 18: Approximating the Jones Polynomial. University of Waterloo lecture notes. https://www.cs.umd.edu/~amchilds/teaching/w13/l18.pdf

[10] Alexandre Clément and Simon Perdrix. 2020. PBS-Calculus: A Graphical Language for Coherent Control of Quantum Computations. In *45th International Symposium on Mathematical Foundations of Computer Science (LIPIcs, Vol. 170)*. Schloss Dagstuhl, Dagstuhl, Germany, 24:1–24:14. doi:10.4230/LIPIcs.MFCS.2020.24

[11] Cole Comfort, Antonin Delpeuch, and Jules Hedges. 2020. Sheet diagrams for bimonoidal categories. arXiv:2010.13361 doi:10.48550/arXiv.2010.13361

[12] Marek Czachor. 2019. Time travel without paradoxes: Ring resonator as a universal paradigm for looped quantum evolutions. *Physics Letters A* 383 (2019), 2704–2712. doi:10.1016/j.physleta.2019.05.043

[13] Aaron David Fairbanks and Peter Selinger. 2025. On traces in categories of contractions. In *Proceedings of the 22nd International Conference on Quantum Physics and Logic (QPL 2025) (EPTCS, Vol. 426)*, Alejandro Díaz-Caro, Ognyan Oreshkov, and Ana Belén Sainz (Eds.). Open Publishing Association, Waterloo, NSW, Australia, 79–99. arXiv:2502.14993 doi:10.4204/EPTCS.426.3

[14] Robert Glück and Robin Kaarsgaard. 2018. A Categorical Foundation for Structured Reversible Flowchart Languages: Soundness and Adequacy. *Logical Methods in Computer Science* 14, 3, Article 16 (2018), 38 pages. doi:10.23638/LMCS-14(3:16)2018 (arXiv:1710.03666).

[15] John E. Gough. 2014. Feedback network models for quantum transport. *Physical Review E* 90 (2014), Article 062109. doi:10.1103/PhysRevE.90.062109

[16] John Gough and Matthew R. James. 2009. Quantum Feedback Networks: Hamiltonian Formulation. *Communications in Mathematical Physics* 287 (2009), 1109–1132. doi:10.1007/s00220-008-0698-8

[17] Masahito Hasegawa. 1997. Recursion from Cyclic Sharing: Traced Monoidal Categories and Models of Cyclic Lambda Calculi. In *Typed Lambda Calculi and Applications (Lecture Notes in Computer Science, Vol. 1210)*. Springer, 196–213. doi:10.1007/3-540-62688-3_37

[18] Chris Heunen and Robin Kaarsgaard. 2022. Quantum information effects. *Proc. ACM Program. Lang.* 6, POPL (2022), 2:1–2:27. arXiv:2107.12144 doi:10.1145/3498663

[19] Mathieu Huot and Sam Staton. 2019. Universal Properties in Quantum Theory. In *Proceedings of the 15th International Conference on Quantum Physics and Logic (QPL 2018)*, EPTCS 287, 213–223. doi:10.4204/EPTCS.287.12

[20] Roshan P. James and Amr Sabry. 2013. Isomorphic Interpreters from Logically Reversible Abstract Machines. In *Reversible Computation: 4th International Workshop, RC 2012, Revised Papers (Lecture Notes in Computer Science, Vol. 7581)*. Springer, Berlin, Heidelberg, 57–71. doi:10.1007/978-3-642-36315-3_5

[21] André Joyal, Ross Street, and Dominic Verity. 1996. Traced monoidal categories. *Mathematical Proceedings of the Cambridge Philosophical Society* 119, 3 (1996), 447–468. doi:10.1017/S0305004100074338

[22] Robin Kaarsgaard and Mathys Rennela. 2021. Join Inverse Rig Categories for Reversible Functional Programming, and Beyond. In *Proceedings of the 37th Conference on the Mathematical Foundations of Programming Semantics (MFPS 2021) (EPTCS, Vol. 351)*. Open Publishing Association, Waterloo, NSW, Australia, 152–167. doi:10.4204/EPTCS.351.10 (arXiv:2105.09929).

[23] Louis Lemonnier. 2024. *The Semantics of Effects: Centrality, Quantum Control and Reversible Recursion.* Ph. D. Dissertation. Université Paris-Saclay. arXiv:2406.07216 doi:10.48550/arXiv.2406.07216

[24] Noah Linden and Sandu Popescu. 1998. The halting problem for quantum computers. arXiv:quant-ph/9806054 doi:10.48550/arXiv.quant-ph/9806054

[25] Octavio Malherbe, Philip Scott, and Peter Selinger. 2012. Partially traced categories. *Journal of Pure and Applied Algebra* 216, 12 (2012), 2563–2585. arXiv:1107.3608 doi:10.1016/j.jpaa.2012.03.026

[26] Microsoft. 2025. Conditional loops in Q#. *Microsoft Learn.* https://learn.microsoft.com/en-us/azure/quantum/user-guide/language/expressions/conditionalloops.

[27] OpenQASM. Classical instructions. *OpenQASM Language Specification*, section on while loops. https://openqasm.com/language/classical.html#while-loops.

[28] M. G. Raymer and C. J. McKinstrie. 2013. Quantum Input-Output Theory for Optical Cavities with Arbitrary Coupling Strength: Application to Two-Photon Wave-Packet Shaping. *Physical Review A* 88 (2013), 043819. arXiv:1309.5915 [quant-ph] doi:10.1103/PhysRevA.88.043819

[29] Michael Reck, Anton Zeilinger, Herbert J. Bernstein, and Philip Bertani. 1994. Experimental realization of any discrete unitary operator. *Physical Review Letters* 73 (1994), 58–61. doi:10.1103/PhysRevLett.73.58

[30] Amr Sabry, Benoît Valiron, and Juliana Kaizer Vizzotto. 2018. From symmetric pattern-matching to quantum control. In *FoSSaCS 2018*. Springer, Cham, 348–364. arXiv:1804.00952 doi:10.1007/978-3-319-89366-2_19

[31] Felix T. Smith. 1960. Lifetime Matrix in Collision Theory. *Physical Review* 118, 1 (1960), 349–356. doi:10.1103/PhysRev.118.349

[32] Songli Wang, Yuyang Wang, Swarnava Sanyal, Robert Parsons, Xingchen Ji, Yoshitomo Okawachi, Xiang Meng, Michal Lipson, Alexander Gaeta, and Keren Bergman. 2025. Automated tuning of a ring-assisted MZI-based interleaver for Kerr frequency combs. *Optics Letters* 50, 2 (2025), 698–701. doi:10.1364/OL.546772

[33] John Watrous. 2018. *The Theory of Quantum Information.* Cambridge University Press, Cambridge, UK. doi:10.1017/9781316848142

[34] Mingsheng Ying. 2011. Floyd–Hoare Logic for Quantum Programs. *ACM Transactions on Programming Languages and Systems* 33, 6, Article 19 (2011), 49 pages. doi:10.1145/2049706.2049708

[35] Mingsheng Ying. 2014. Quantum Recursion and Second Quantisation. arXiv:1405.4443 [quant-ph]. doi:10.48550/arXiv.1405.4443

[36] Mingsheng Ying and Yuan Feng. 2010. Quantum loop programs. *Acta Informatica* 47, 4 (2010), 221–250. arXiv:quant-ph/0605218 doi:10.1007/s00236-010-0117-4

[37] Charles Yuan, Agnes Villanyi, and Michael Carbin. 2024. Quantum control machine: the limits of control flow in quantum programming. *Proc. ACM Program. Lang.* 8, OOPSLA1 (2024), 1–28. arXiv:2304.15000 doi:10.1145/3649811

[38] Zhicheng Zhang and Mingsheng Ying. 2025. Quantum register machine: efficient implementation of quantum recursive programs. *Proc. ACM Program. Lang.* 9, PLDI, Article 180 (2025), 26 pages. arXiv:2408.10054 doi:10.1145/3729283

A Typing and interpreting the clauses

A.1 Syntax

$$
\begin{aligned}
a, b &::= 0 \mid 1 \mid a + b \mid a \otimes b \\
v &::= * \mid x \mid \mathsf{left}\ v \mid \mathsf{right}\ v \mid \langle v_1, v_2 \rangle \mid \mathsf{exit}\ v \mid \mathsf{back}\ v \\
p, q &::= * \mid x \mid \langle p, q \rangle \\
e &::= v \mid \textstyle\sum_i \alpha_i e_i \mid \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e \\
\omega &::= f \mid \{v_i \leftrightarrow e_i\}_{i \in I} \mid \omega\,;\, \omega \mid \omega \oplus \omega \mid \omega \otimes \omega \mid \omega^\dagger \mid \mathsf{trace}_c\ \omega \\
t &::= e \mid \omega\, t \\
decl &::= f :: a \leftrightarrow b\,;\ f = \omega \mid \mathsf{round}\ f : a + c = \{v_i \leftrightarrow e_i\}_{i \in I}
\end{aligned}
$$

Fig. 11. Finite iso definitions. A name f refers to an earlier closed definition. The coefficient α_i is a complex scalar. A signature uses $::$, while the `round` abbreviation uses $:$. In trace_c, the return has type c.

The surface patterns exit v and back v elaborate recursively to left v and right v. This rewrite preserves payloads and nesting, and is performed before matching or typing by the core rules below. The remaining executable abbreviations are those of Section 2: `Unit`, `unit`, `Qubit` and its basis values, bracketed scalar combinations, combinator composition and `on` application. Listings typeset the source token `<->` as $\leftrightarrow$ and pair delimiters `<`, `>` as $\langle$, $\rangle$.

A.2 Statics

Admission, elaboration and evaluation are defined together by induction over the ordered declarations and proper subprograms. In particular, evaluating a local call to fill a table column uses a proper subprogram or an already checked earlier definition. The grammar in Figure 11 describes definitions in an ordered environment Σ. A binding $f = \omega : a \leftrightarrow b$ is introduced after checking ω against earlier bindings. Write Σ ok when these declarations form a well-typed environment. Types are the finite trees of Figure 11. Both 0 and 1 are types, and $a + b$ and $a \otimes b$ are types whenever a and b are. Names may be chosen fresh by renaming. The rules for introducing them are

$$\frac{}{\emptyset \text{ ok}} \qquad \frac{\Sigma \text{ ok} \quad f \notin \mathrm{dom}(\Sigma) \quad \Sigma \vdash \omega : a \leftrightarrow b}{\Sigma, (f = \omega : a \leftrightarrow b) \text{ ok}}.$$

The declaration

$$\texttt{round}\ f : a + c = \{v_i \leftrightarrow e_i\}_{i \in I}$$

introduces the ordinary binding $f = \{v_i \leftrightarrow e_i\}_{i \in I} : (a + c) \leftrightarrow (a + c)$ after the literal clauses have passed the same check. The name denotes the open round. Closing it requires an explicit operation. Writing $\Sigma \vdash d \Rightarrow \Sigma'$ for declaration elaboration, its rule is

$$\frac{\Sigma \text{ ok} \quad f \notin \mathrm{dom}(\Sigma) \quad \Sigma \vdash \mathscr{T} : (a + c) \leftrightarrow (a + c)}{\Sigma \vdash \texttt{round}\ f : a + c = \mathscr{T} \Rightarrow \Sigma, (f = \mathscr{T} : (a + c) \leftrightarrow (a + c))}.$$

An ordinary iso signature still permits distinct external types. The binding $f = \omega : a \leftrightarrow b$ represents the surface declaration $f :: a \leftrightarrow b\ ;\ f = \omega$; listings may separate these parts by a newline. Write Δ for a finite set of distinct typed value variables. In Δ_1, Δ_2 the two domains must be disjoint. Each judgment below uses this convention, including inside a clause with an empty set of variables. The empty table presents the unique map on 0. Whenever we use a unit input or a nonzero operator line we assume the corresponding external type is inhabited.

Patterns and expressions. The finite basis is defined recursively:

$$\mathscr{B}(0) = \emptyset, \qquad \mathscr{B}(1) = \{*\},$$
$$\mathscr{B}(a + b) = \mathrm{left}\,\mathscr{B}(a) \sqcup \mathrm{right}\,\mathscr{B}(b), \quad \mathscr{B}(a \otimes b) = \mathscr{B}(a) \times \mathscr{B}(b).$$

An environment for Δ assigns each variable a basis value of its type. A variable matches any such value; a constructor matches its own tag, and a pair matches its two components. Write $\Delta \vdash v : a$ when a basis pattern is well typed. The rules are

$$\emptyset \vdash * : 1 \qquad x : a \vdash x : a \qquad \frac{\Delta \vdash v : a}{\Delta \vdash \mathrm{left}\ v : a + b} \qquad \frac{\Delta \vdash v : b}{\Delta \vdash \mathrm{right}\ v : a + b}$$

$$\frac{\Delta_1 \vdash v : a \qquad \Delta_2 \vdash w : b}{\Delta_1, \Delta_2 \vdash \langle v, w \rangle : a \otimes b}.$$

The basis assignments for a context Δ form $\mathscr{B}(\Delta) = \prod_{(x:a) \in \Delta} \mathscr{B}(a)$. The empty product contains the unique empty assignment. If any variable has an empty basis, the product is empty. Write $\mathscr{H}(\Delta) = \mathbb{C}^{\mathscr{B}(\Delta)}$. A product pattern p with $\Delta \vdash p : b$ gives a bijection $\gamma \mapsto p[\gamma]$ from $\mathscr{B}(\Delta)$ to

$\mathscr{B}(b)$. For unit it maps the unique empty assignment to $*$. For a variable it is the identity. For a pair, disjoint contexts let us combine the two component assignments uniquely. Induction on these three rules proves the bijection, including empty bases. Matching p is its inverse.

The unit, variable and pair rules also type the product patterns p, q in Figure 11. In the argument position q these rules assemble already bound values. In the result position p they bind fresh variables of the indicated types. A single variable in either position may stand for a value of any type. For expressions we write $\Sigma; \Delta \vdash e : a$ and use

$$\frac{\Delta \vdash v : a}{\Sigma; \Delta \vdash v : a} \qquad \frac{\Sigma; \Delta \vdash e_i : a \; (i \in I) \qquad \alpha_i \in \mathbb{C}}{\Sigma; \Delta \vdash \sum_{i \in I} \alpha_i e_i : a}$$

$$\frac{\Delta_1 \vdash q : a \qquad \Sigma \vdash \omega : a \leftrightarrow b \qquad \Delta_p \vdash p : b \qquad \Sigma; \Delta_2, \Delta_p \vdash e : d}{\Sigma; \Delta_1, \Delta_2 \vdash \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e : d}.$$

In the last rule the variables in p are fresh and pairwise distinct. A product pattern of type b binds every basis output, by the bijection above. The sum rule permits an empty index set in any stated context Δ; its denotation is the zero linear map from $\mathscr{H}(\Delta)$. Such an expression cannot supply an entire column of a nonempty unitary table. A finite first-order term is a closed expression, or an application of a closed iso to a term of its domain. Writing $\Sigma \vdash t : a$ for a closed term, its rules are

$$\frac{\Sigma; \emptyset \vdash e : a}{\Sigma \vdash e : a} \qquad \frac{\Sigma \vdash t : a \qquad \Sigma \vdash \omega : a \leftrightarrow b}{\Sigma \vdash \omega\, t : b}.$$

This suffices for feeding a superposition of ground values to a program.

For a matrix M with input columns s and output rows t, row s of $M^\dagger$ is the conjugate of column s of M. Consequently its Gram matrix has entries

$$(M^\dagger M)_{ss'} = \sum_{t \in \mathscr{B}(b)} \overline{M_{ts}} M_{ts'}.$$

Each product pairs amplitudes in the same output row. At $s = s'$ it contributes to the squared norm of one column; at $s \neq s'$ it contributes to the inner product between two columns.

Let M have one column for each $s \in \mathscr{B}(a)$, filled with the coefficients of the expression in the uniquely matching clause under its ground substitution. The following rule imposes a finite extensional check. Its orthogonality premise concerns fully expanded outputs, so it need not infer linear independence from surface syntax.

$$\frac{\begin{array}{c}\Delta_i \vdash v_i : a \quad \Sigma; \Delta_i \vdash e_i : b \quad (i \in I) \\ \big(v_i[\mathscr{B}(\Delta_i)]\big)_{i \in I} \text{ disjointly covers } \mathscr{B}(a) \qquad |\mathscr{B}(a)| = |\mathscr{B}(b)| \quad M^\dagger M = I\end{array}}{\Sigma \vdash \{v_i \leftrightarrow e_i\}_{i \in I} : a \leftrightarrow b}$$

Here $v_i[\mathscr{B}(\Delta_i)]$ is the set of all ground instances of pattern v_i. The coverage premise requires these sets to be disjoint and their union to be $\mathscr{B}(a)$. An empty instance set is permitted and contributes no column. This rule is exact over a specified coefficient domain with exact equality and arithmetic, for example rational complex numbers. With mathematical complex parameters it is a well-formedness predicate; exact equality of arbitrary descriptions of complex numbers is not assumed decidable. The finite semantics does not make floating-point tolerance an orthogonality proof.

Reversible combinators. For $M \in \mathbb{C}^{\mathscr{B}(b) \times \mathscr{B}(a)}$ the atom rule is

$$\frac{|\mathscr{B}(a)| = |\mathscr{B}(b)| \qquad M^\dagger M = I}{[M] : a \leftrightarrow b}.$$

Write u, v for reversible programs and omit the common Σ in the following rules. A bound name has the type at its declaration. For finite types we have the structural isos of Π as abbreviations for the corresponding permutation clauses, including identity, symmetries and distributivity. The reversible rules are

$$\frac{f : a \leftrightarrow b \in \Sigma}{f : a \leftrightarrow b} \qquad \frac{u : a \leftrightarrow b \qquad v : b \leftrightarrow d}{u\,;\, v : a \leftrightarrow d} \qquad \frac{u : a \leftrightarrow b}{u^\dagger : b \leftrightarrow a}$$

$$\frac{u : a \leftrightarrow b \qquad v : d \leftrightarrow f}{u \oplus v : a + d \leftrightarrow b + f} \qquad \frac{u : a \leftrightarrow b \qquad v : d \leftrightarrow f}{u \otimes v : a \otimes d \leftrightarrow b \otimes f} \qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathsf{trace}_c\, \beta : a \leftrightarrow b}.$$

A.3 Dynamics

Evaluating an expression. Let $\gamma \in \mathscr{B}(\Delta)$ assign basis values to the variables in a linear context Δ, and write $\gamma \vdash e \Downarrow \psi$ when evaluation returns the finite output vector ψ. For a value pattern v, $v[\gamma]$ is its ground instance. Values and sums evaluate by

$$\frac{}{\gamma \vdash v \Downarrow |v[\gamma]\rangle} \qquad \frac{\gamma \vdash e_i \Downarrow \psi_i \ (i \in I)}{\gamma \vdash \sum_i \alpha_i e_i \Downarrow \sum_i \alpha_i \psi_i}.$$

For let $p = \omega q$ in e, split the input assignment as $\gamma_1 \uplus \gamma_2$ according to the typing rule: q uses precisely γ_1, and e retains γ_2. Write $U|q[\gamma_1]\rangle = \sum_t a_t|t\rangle$, where $\omega \Downarrow U$. Matching the product pattern p with t gives the fresh assignment γ_t. The application rule is

$$\frac{\omega \Downarrow U \quad \gamma_2 \uplus \gamma_t \vdash e \Downarrow \psi_t \quad (t \in \mathscr{B}(b))}{\gamma_1 \uplus \gamma_2 \vdash \mathsf{let}\ p = \omega q\ \mathsf{in}\ e \Downarrow \sum_t a_t \psi_t}.$$

Here b is the output type of ω. Its admitted matrix supplies the coefficients a_t, and total matching of p supplies each γ_t. The same binding is used independently in the alternatives of a sum; it is split only where the linear typing rule splits it. For closed terms, $t \Downarrow \psi$ uses the empty variable assignment, with

$$\frac{\varnothing \vdash e \Downarrow \psi}{e \Downarrow \psi} \qquad \frac{\omega \Downarrow U \qquad t \Downarrow \psi}{\omega t \Downarrow U\psi}.$$

The fixed iso environment is omitted from these evaluation judgments. Named matrices are computed from earlier definitions; the reversible combinators act by their matrix operations, and a trace uses Proposition B.2. Thus expression evaluation is finite even when its matrix contains feedback with infinitely many nonzero exits. Induction on an expression's typing derivation shows that evaluation gives its linear interpretation: the value rule substitutes basis values, the sum rule is linearity, and the application rule composes the admitted iso with the continuation's linear map. These rules do not select a measurement outcome or normalize individual amplitude components. More explicitly, the local application's column at (γ_1, γ_2) is

$$E_{\mathsf{let}}(s; \gamma_1, \gamma_2) = \sum_{t \in \mathscr{B}(b)} U_{t, q[\gamma_1]}\, E_e(s; \gamma_2, \gamma_t).$$

Here $s \in \mathscr{B}(d)$ indexes the continuation output, and $t \in \mathscr{B}(b)$ indexes the intermediate output of the local iso. This is ordinary matrix multiplication: reassemble q's variables, apply U while carrying γ_2, match its output to the fresh product pattern, and apply the continuation matrix. The two pattern maps are the bijections just proved. Thus the rule acts linearly on arbitrary superpositions of assignments, including entangled ones. It neither duplicates an assignment nor measures the local result.

Elaboration and matrix evaluation. The matrix programs have grammar

$$\kappa ::= [M] \mid \kappa;\kappa \mid \kappa \oplus \kappa \mid \kappa \otimes \kappa \mid \kappa^\dagger \mid \mathsf{trace}_c\, \kappa.$$

The combinators κ of Section 2 have ground checked tables as atoms. Write $\mathscr{E}_\Sigma(\omega)$ for elaboration. If $f = \omega_f$ is a declaration in Σ, let Σ_f be its earlier prefix, the environment in which ω_f was checked. With M the evaluated columns of a table $\mathscr{T}$, the definition is

$$\begin{aligned}
\mathscr{E}_\Sigma(\mathscr{T}) &= [M], & \mathscr{E}_\Sigma(f) &= \mathscr{E}_{\Sigma_f}(\omega_f),\\
\mathscr{E}_\Sigma(\omega;\nu) &= \mathscr{E}_\Sigma(\omega);\mathscr{E}_\Sigma(\nu), & \mathscr{E}_\Sigma(\omega \oplus \nu) &= \mathscr{E}_\Sigma(\omega) \oplus \mathscr{E}_\Sigma(\nu),\\
\mathscr{E}_\Sigma(\omega \otimes \nu) &= \mathscr{E}_\Sigma(\omega) \otimes \mathscr{E}_\Sigma(\nu), & \mathscr{E}_\Sigma(\omega^\dagger) &= \mathscr{E}_\Sigma(\omega)^\dagger,\\
\mathscr{E}_\Sigma(\mathsf{trace}_c\, \omega) &= \mathsf{trace}_c\, \mathscr{E}_\Sigma(\omega).
\end{aligned}$$

These equations recurse into proper subprograms and, on lookup, earlier definitions. Computing a table uses this same evaluation on its local iso subprograms. This is a well-founded mutual definition with expression evaluation. Each clause of the definition preserves its source and target types. The declaration rule for `round` binds this open combinator and does not insert a trace.

If $\kappa \Downarrow M$ and $\lambda \Downarrow N$, the ordinary matrix rules, used with their typed premises, are

$$\begin{gathered}
[M] \Downarrow M, \qquad (\kappa;\lambda) \Downarrow NM,\\
(\kappa \oplus \lambda) \Downarrow M \oplus N, \quad (\kappa \otimes \lambda) \Downarrow M \otimes N, \quad \kappa^\dagger \Downarrow M^\dagger.
\end{gathered}$$

The feedback rule completes them:

$$\frac{\kappa \Downarrow U \quad U = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \quad F : X \to Y \quad (I_Y - D)F = C}{\mathsf{trace}_c\, \kappa \Downarrow A + BF}.$$

The premise solves a finite linear system. It neither inverts a possibly singular $I - D$ nor requires an execution to reach a last exit. Theorem B.1 gives a solution and proves that every solution gives the same output. Proposition B.2 computes that output by eliminating one return pattern at a time. Thus this relational evaluation has a result independent of the chosen wire solution, and exact elimination implements it. Write $\Sigma \vdash \omega \Downarrow M$ when its elaboration evaluates to M. Closed application then uses

$$\frac{\Sigma \vdash \omega \Downarrow M \qquad t \Downarrow \psi}{\omega t \Downarrow M\psi}.$$

For a well-typed expression and a basis assignment, induction on its syntax gives a unique output vector. In the local-binding case, the argument and result patterns give inverse assembly/matching bijections, and the finite coefficient sum is the matrix product displayed above. Expanding each clause therefore preserves all its columns. Induction through the combinators preserves evaluation and the linear denotation; the trace case uses the feedback theorem. Linearity then gives the same result on arbitrary superpositions, including assignments entangled with a retained factor.

The executable accepts unit and nested local patterns, empty types and tables, and the `dagger` combinator for adjoint. Its combinator spellings are ; for sequence, + for direct sum and * for tensor. Native numerical evaluation uses floating-point coefficients and may refuse a coupled near-unit pivot; it is not the exact field-arithmetic procedure of Proposition B.2.

A.4 Denotation

A checked table denotes the linear extension of its columns. Names, sequence, dagger, direct sum and tensor denote lookup, composition in execution order, adjoint, direct sum and tensor; trace denotes the exit sum (3.2). The evaluation identities above give these same maps, with the trace case supplied by Theorem B.1.

Control of an endo-iso u is $I \oplus u$ under the distributive iso for $(1+1)\otimes a$. Thus controlling a global phase can change a relative phase. The types above deliberately do not provide this operation on an arbitrary channel.

A.5 Effect extensions

Syntax. Write u for a reversible iso, v for an isometry program and h for a channel program. The two upper grammars are

$$v ::= \mathsf{arr}\, u \mid \mathsf{lift}\, u \mid v\,;\, v \mid v \oplus v \mid v \otimes v \mid \mathsf{count}_K\, u,$$
$$h ::= \mathsf{arr}\, v \mid \mathsf{hide}\, v \mid h\,;\, h \mid h \otimes h \mid \mathsf{while}_c\, u.$$

Statics. The type of the intermediate result must match the type expected by the next program. At the isometry layer this gives the following rules for sequence, tensor and direct sum.

$$\frac{v : a \rightarrowtail b \qquad w : b \rightarrowtail d}{v\,;w : a \rightarrowtail d} \qquad \frac{v : a \rightarrowtail b \qquad w : c \rightarrowtail d}{v \otimes w : a \otimes c \rightarrowtail b \otimes d} \qquad \frac{v : a \rightarrowtail b \qquad w : c \rightarrowtail d}{v \oplus w : a + c \rightarrowtail b + d}.$$

Channels also compose in sequence and by tensor product.

$$\frac{h : a \rightsquigarrow b \qquad k : b \rightsquigarrow d}{h\,;k : a \rightsquigarrow d} \qquad \frac{h : a \rightsquigarrow b \qquad k : c \rightsquigarrow d}{h \otimes k : a \otimes c \rightsquigarrow b \otimes d}.$$

These operations have their usual linear meanings, with sequence written in execution order. An identity on each layer is the embedded identity iso. The remaining rules change the kind of map or close a round.

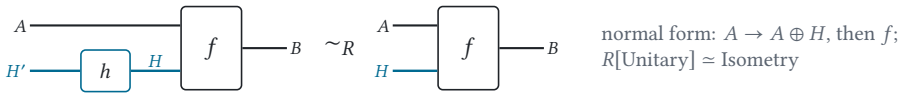
$$\frac{u : a \leftrightarrow b}{\mathsf{arr}\, u : a \rightarrowtail b} \qquad \frac{u : a + d \leftrightarrow b}{\mathsf{lift}\, u : a \rightarrowtail b} \qquad \frac{\beta : a + c \leftrightarrow b + c \qquad K \geq 1}{\mathsf{count}_K\, \beta : a \rightarrowtail [K] \otimes b}$$

$$\frac{v : a \rightarrowtail b}{\mathsf{arr}\, v : a \rightsquigarrow b} \qquad \frac{v : a \rightarrowtail b \otimes g \qquad |\mathscr{B}(g)| > 0}{\mathsf{hide}\, v : a \rightsquigarrow b} \qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathsf{while}_c\, \beta : a \rightsquigarrow b}.$$

Denotation. An R[Unitary] representative $U : X \oplus H \to X'$ exposes $u \mapsto U(u, 0)$, identifying changes confined to the unused summand. An L[Isometry] representative $V : X \to X' \otimes G$ exposes $\rho \mapsto \mathrm{tr}_G(V \rho V^\dagger)$, identifying isometric relabelings of the hidden factor and the equivalence relation they generate [18, Prop. 7, Thm. 22].

For $0 \leq j < K$, the allocation permutation $p_j : a + ([K-1] \otimes a) \leftrightarrow [K] \otimes a$ maps left x to $\langle j, x\rangle$, taking $[0] = 0$. Then $\mathsf{alloc}_j = \mathsf{lift}\, p_j$. The basis copier for inhabited b uses heap $[|\mathscr{B}(b)| - 1] \otimes b$, with $|\mathscr{B}(b)|^2 - |\mathscr{B}(b)|$ basis values, completing the diagonal injection to a permutation. An embedded unitary is the same map viewed as an isometry. An embedded isometry V is the channel $\mathscr{E}_V$, distinct from the syntactic elaboration $\mathscr{E}_\Sigma$. For lift and hide the denotations are restriction and partial trace as in Section 4. For count_K, tensor $I_{[K]}$ with β, distribute the tensor over the two tags and append $I_{[K]\otimes b} \oplus (S_K \otimes I_c)$ in the resulting order. Trace the summand $[K] \otimes c$, then precompose with $u \mapsto |0\rangle \otimes u$. This allocation is a lift of a permutation that places the exposed basis at counter value zero. The four blocks of the enlarged iso are $I \otimes A, I \otimes B, S_K \otimes C, S_K \otimes D$. Its kth exit is $S_K^k \otimes x_k$, proving (4.1). To hide the counter in the displayed syntax, compose with $\mathsf{arr}\, \sigma_{[K],b}$ before hide. Native `count K beta` returns data before counter, so its following `discard` hides the counter. Appending `swap` instead gives our displayed order. In the executable, `discard` is the derived first projection, $I_a \otimes \mathsf{discard}_b ; \mathsf{unit}_\otimes$, rather than the whole-register operation $\mathsf{discard}_b : b \rightsquigarrow 1$.

(a) $R[\mathscr{C}]$: a heap enters beside the input, and only access to it counts



(b) $L[\mathscr{C}]$: garbage leaves beside the output, and its contents do not count

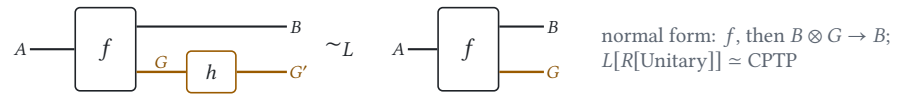


Fig. 12. Allocation exposes an input summand of a larger reversible map. Hiding exposes one output factor of an isometry. Preprocessing the unused summand or isometrically relabeling the hidden factor preserves the exposed map. Here $\mathscr{C}$ = Unitary in $R[\mathscr{C}]$ and $\mathscr{C}$ = Isometry in $L[\mathscr{C}]$.

For while, allocate a fresh bit at 0 and apply the ordinary iso

$$\omega_{\text{tag}} = \{ \begin{array}{l} \langle \text{exit } x, 0\rangle \leftrightarrow \langle \text{exit } x, 0\rangle, \\ \langle \text{exit } x, 1\rangle \leftrightarrow \langle \text{exit } x, 1\rangle, \\ \langle \text{back } y, 0\rangle \leftrightarrow \langle \text{back } y, 1\rangle, \\ \langle \text{back } y, 1\rangle \leftrightarrow \langle \text{back } y, 0\rangle \end{array} \} : (b + c) \otimes \texttt{Qubit} \leftrightarrow (b + c) \otimes \texttt{Qubit}.$$

The patterns partition the basis and merely permute its values, so the column test admits this iso. Its restriction to bit input 0 gives the isometry T. For output components $v \in [\![b]\!]$ and $y \in [\![c]\!]$, its linear action is

$$T(v, y) = (\iota_{\text{ext}} v) \otimes |0\rangle + (\iota_{\text{ret}} y) \otimes |1\rangle.$$

For the empty type, discard_0 is obtained by hiding the unit factor of the unique isometry $0 \rightarrowtail 1 \otimes 1$, constructed by lift from $0 + 1 \leftrightarrow 1 \otimes 1$. Its action is the unique map on the zero operator space. Set measure_0 to the embedded identity on 0. These supply the empty cases omitted from the inhabited-type formulas in the body.

Applying T after U and tracing out the copied bit gives the diagonal output blocks. Feedback now uses the classical sum. This construction works when the payload types differ. The notation while names this classical feedback operation on the translated body.

When $a = b = c$, let $\text{factor} : a + a \leftrightarrow a \otimes \texttt{Qubit}$ and $\text{dist} = \text{factor}^\dagger$, and put $\hat{\beta} = \text{dist}; \beta; \text{factor}$. Writing $E(\omega) = \text{arr}(\text{arr}\, \omega)$, the channel in Figure 9 is

$$E(\text{factor}); E(\hat{\beta}); (I_a \otimes \text{measure}_{\texttt{Qubit}}); E(\text{dist}).$$

The first factor; dist cancel, leaving the original body followed by exposure, measurement and restoration of its tag.

Soundness. Induct on the finite syntax in a well-founded order that puts prior definitions before uses. The unique match and the substitution clause give a column at each closed input. The admission premise makes these columns an orthonormal basis. Sequential, sum, tensor and dagger operations preserve the stipulated map classes. Theorem B.1 proves the trace case. Restriction of a unitary is an isometry, and a unitary after counter allocation is an isometry. Embedding an isometry and taking a partial trace after an isometry both give completely positive trace-preserving maps. The positive sum in (4.2) is completely positive, since each finite partial sum is and the completely

positive cone in finite dimension is closed; its trace preservation follows from $\sum_k x_k^\dagger x_k = I$. Thus every well-typed iso, isometry program and channel program denotes the indicated kind of map. A closed term $t : a$ instead denotes its evaluated vector in $\mathbb{C}^{\mathscr{B}(a)}$; its typing alone does not require that vector to have norm one. Conversely a given finite unitary V has as a definition the table $s \leftrightarrow \sum_t V_{ts} t$. Each column satisfies the typing rule since $V^\dagger V = I$, including the empty case.

B Convergence, probability and elimination

For a finite permutation, an external input follows a single path. It cannot revisit an internal basis value: unique predecessors would identify its original external input with an internal value. It therefore exits within $|\mathscr{B}(c)| + 1$ calls, while wholly internal cycles remain unreachable. The two external paths in Figure 13 combine to a swap, as in reversible flowchart semantics [14, Prop. 3.10][22, Prop. 5.3].

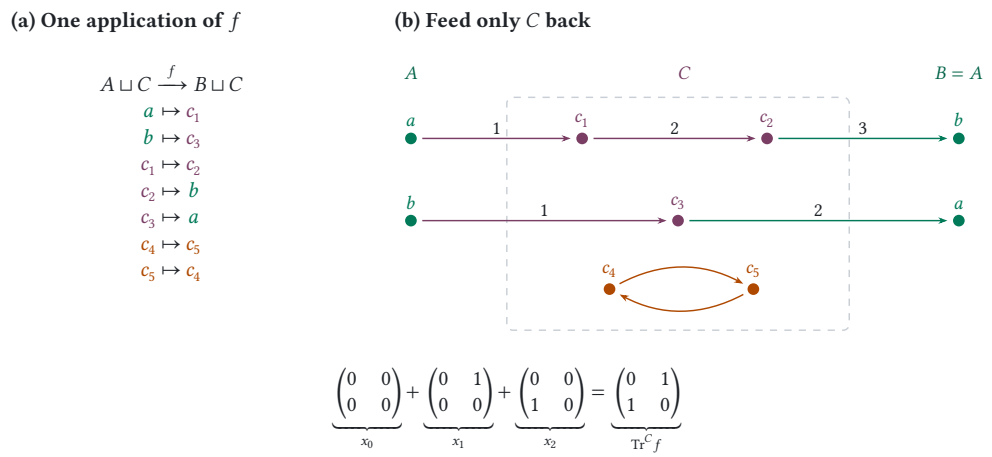
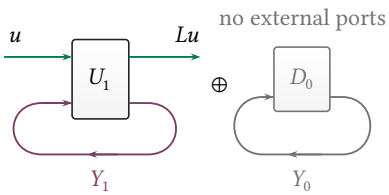


Fig. 13. A reversible loop with two external paths and an internal cycle. The two nonzero exit maps have disjoint domains and their sum is a swap.

Theorem B.1 (Finite feedback). *For a finite unitary $U : X \oplus Y \to X' \oplus Y$, the exits are absolutely summable in operator norm. Their sum $L : X \to X'$ is unitary, and $\sum_k x_k^\dagger x_k = I_X$. The wire equation $y = Cu + Dy$ has a solution for every u. Every solution gives the same output $Au + By = Lu$.*



$$\|u\|^2 = \|Au\|^2 + \|Cu\|^2,$$
$$\|Cu\|^2 = \|BCu\|^2 + \|DCu\|^2,$$
$$\|DCu\|^2 = \|BDCu\|^2 + \|D^2Cu\|^2.$$

Fig. 14. An internal basis change separates U_1 from the unreachable D_0. The three norm identities telescope because each return term supplies the next call.

Proof of Theorem B.1. From $U^\dagger U = I$ and $UU^\dagger = I$ we have

$$A^\dagger A + C^\dagger C = I, \qquad B^\dagger B + D^\dagger D = I, \qquad CC^\dagger + DD^\dagger = I.$$

The middle identity gives $\|Dy\|^2 + \|By\|^2 = \|y\|^2$. With $Y_0 = \bigcap_{j\ge 0} \ker BD^j$, it follows that $DY_0 \subseteq Y_0$ and $D|_{Y_0}$ is isometric. In finite dimension it maps Y_0 onto Y_0. Thus U maps $0_X \oplus Y_0$ onto $0_{X'} \oplus Y_0$. Unitarity maps their orthogonal complements $X \oplus Y_1$ and $X' \oplus Y_1$ onto one another. Hence $C(X) \subseteq Y_1$, $D(Y_1) \subseteq Y_1$ for $Y_1 = Y_0^{\perp}$, and B vanishes on Y_0.

Write $D_1 = D|_{Y_1}$ and $B_1 = B|_{Y_1}$. Put $m = \dim Y$. By Cayley–Hamilton a return vector is invisible exactly when its first m emissions vanish. If $Y_1 = 0$, all delayed exits vanish. Otherwise $m > 0$. For a unit $y \in Y_1$, conservation through m calls gives

$$1 - \|D_1^m y\|^2 = \sum_{j=0}^{m-1} \|BD^j y\|^2 > 0.$$

Equality would make every displayed emission zero and hence put y in Y_0. On the compact unit sphere of Y_1, the continuous function $y \mapsto \|D_1^m y\|$ therefore has a maximum $q < 1$. For $k = pm + r$, $0 \le r < m$, norm nonincrease on the remaining r calls gives $\|D_1^k\| \le \|D_1^m\|^p \|D_1^r\| \le q^p$. This geometric bound also makes every polynomially weighted exit series converge. Therefore $\sum \|x_k\|$ is finite and $F = (I - D_1)^{-1} C = \sum_j D_1^j C$ solves the wire equation. Every other solution differs by a vector in Y_0 fixed by D, which B kills. The norm identity for $U(u, Fu) = (Lu, Fu)$ proves that L is an isometry. Since $\dim X = \dim X'$ it is onto.

To account for the probability, let $T_j = C^\dagger D^{\dagger j} D^j C$. On a unit input u, $\langle u, T_j u\rangle$ is the probability still inside after call $j+1$. The first block identity gives $x_0^\dagger x_0 = I - T_0$. On later calls the weight that leaves is exactly the drop in the weight still inside, since

$$x_{j+1}^\dagger x_{j+1} = C^\dagger D^{\dagger j} \underbrace{B^\dagger B}_{I - D^\dagger D} D^j C = T_j - T_{j+1}.$$

Adding through exit k gives $\sum_{j=0}^{k} x_j^\dagger x_j = I_X - T_k$. The residual term tends to zero by the decay just proved, giving total exit probability one even for a body with infinitely many possible exits. □

For an internal state w, write $\mathcal{O}w = (BD^j w)_{j\ge 0}$ for the sequence of amplitudes it can still emit. Let $\mathcal{S}(z_0, z_1, \ldots) = (z_1, z_2, \ldots)$ be the left shift. Then $\mathcal{O}D_1 = \mathcal{S}\mathcal{O}$ on Y_1 says that advancing a live state drops its first emission from the sequence. The squared norms of the entries of $\mathcal{O}w$ sum to $\|w\|^2$ for $w \in Y_1$, by the same conservation argument as in the proof. Figure 15 illustrates this with $B = -4/5$ and $D = 3/5$ on the live direction, and with an independent dead direction.

B.1 Eliminating a pattern

Proposition B.2 (Elimination). *Given the expanded table, eliminating its returning basis values in any order produces a closed iso denoting L in exactly $|\mathcal{B}(c)|$ steps. If the original coefficients lie in a subfield of $\mathbb{C}$, the result has coefficients in that subfield.*

Proof of Proposition B.2. Place the selected return pattern in $Q = \mathbb{C}q$ and put $Y_{\text{rest}} = Y \ominus Q$, the orthogonal complement of Q in Y. The remaining input and output are $E_{\text{in}} = X \oplus Y_{\text{rest}}$ and $E_{\text{out}} = X' \oplus Y_{\text{rest}}$. Grouping those remaining ports together gives body blocks

$$\begin{pmatrix} M & b_q \\ c_q & d_q \end{pmatrix}, \qquad M : E_{\text{in}} \to E_{\text{out}}, \quad b_q : Q \to E_{\text{out}}, \quad c_q : E_{\text{in}} \to Q, \quad d_q \in \mathbb{C}.$$

Here $d_q = D_{qq}$ is the selected self-entry. For $w = (u, y_{\text{rest}})$, the row c_q gives $c_q w = (Cu)_q + \sum_{r \ne q} D_{qr} y_r$ from the q-coordinate of $y = Cu + Dy$, while b_q collects the coefficients of y_q in the other outputs. Closing Q asks for a scalar $\lambda = c_q w + d_q \lambda$ and returns $Mw + b_q \lambda$. When $d_q \ne 1$, solving gives $\lambda = c_q w/(1 - d_q)$ and output $(M + b_q c_q/(1 - d_q))w$. When $d_q = 1$, its self-entry has norm one, so

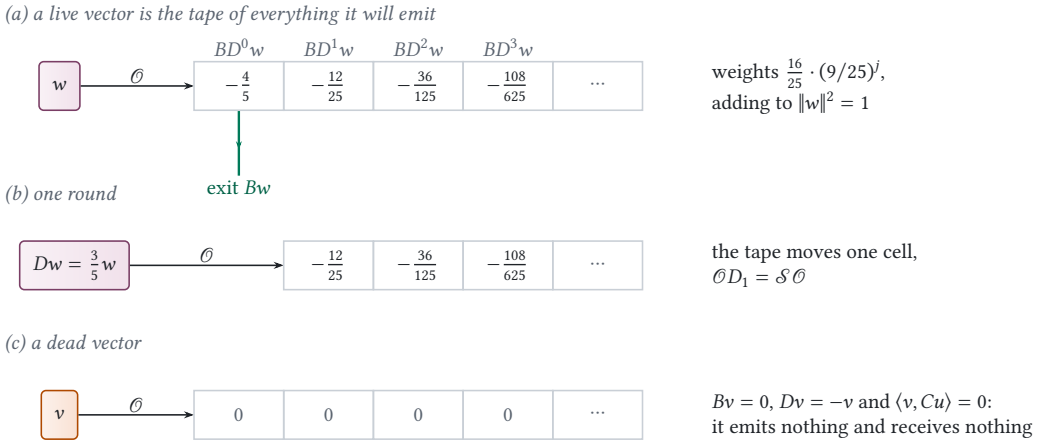


Fig. 15. The future emissions of a live state form its observation tape. Advancing that state shifts the tape by one position. A dead state has the zero tape. Here $Dv = -v$ is dead but not fixed; only the fixed subspace $\ker(I - D)$ obstructs the inverse at phase one.

every other entry in both its row and column is zero. Hence $b_q = c_q = 0$, any λ solves the wire equation, and the output is Mw. These cases are exhaustive. In particular $d_q = -1$ or $d_q = i$ uses the first branch, with $b_q = c_q = 0$; no divergent geometric series is invoked. This intermediate table is $\mathrm{Tr}^{Q}(U)$, with E_{in} and E_{out} as its external spaces. Theorem B.1 proves its unitarity in either case.

Figures 3 and 4 carry out these steps for `guardRound` and for the two return patterns of `flipRound`.

After one elimination, solving the remaining feedback equations gives exactly the output of solving all the original equations. Indeed a simultaneous solution restricts to one of the smaller system, and a solution of the smaller system can be extended by a solution for q at those incident values. The external value is independent of these choices by Theorem B.1. Induct on the number of return patterns. Each step removes one and uses only field operations with nonzero denominators. For a phase on the return wire apply the same argument to the modified unitary. No exchange in the order of a conditionally convergent series of individual walks is needed. □

Closed sums for the running rotation. For `guardRound`, both port spaces are $\mathbb{C}$, so the blocks are scalars: $A = D = t = 3/5$, $C = s = 4/5$ and $B = -s$. The general exit formula becomes $x_0 = t$ and $x_{k+1} = (-s)t^k s = -s^2 t^k$. To compare adding these amplitudes with adding their squared magnitudes, follow the first n calls on unit input. Write r_n for the amplitude still returning, ℓ_n for the sum emitted so far, and p_n for the weight of those exits when the guard is measured. The geometric sums give

$$r_n = st^{n-1},$$
$$\ell_n = t - s^2 \sum_{j=0}^{n-2} t^j = -1 + (1+t)t^{n-1},$$
$$p_n = t^2 + s^4 \sum_{j=0}^{n-2} t^{2j} = 1 - s^2 t^{2(n-1)}.$$

The sums are empty when $n = 1$. Table 1 follows the resulting emissions, residuals and sums. The squared length $|\ell_n|^2$ can fall and rise as later emissions interfere. The measured halting weight p_n only increases, because each call adds a nonnegative probability.

	one call		coherent feedback		guard measured
call n	emitted x_{n-1}	still returning r_n	sum emitted ℓ_n	squared length $\lvert\ell_n\rvert^2$	halted so far p_n
1	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{3}{5}$	0.3600	0.3600
2	$-\frac{16}{25}$	$\frac{12}{25}$	$-\frac{1}{25}$	0.0016	0.7696
3	$-\frac{48}{125}$	$\frac{36}{125}$	$-\frac{53}{125}$	0.1798	0.9171
4	$-\frac{144}{625}$	$\frac{108}{625}$	$-\frac{409}{625}$	0.4282	0.9701
5	$-\frac{432}{3125}$	$\frac{324}{3125}$	$-\frac{2477}{3125}$	0.6283	0.9893
$n \to \infty$	0	0	-1	1	1

Table 1. `guardRound` on $|*\rangle$. Coherent feedback adds emissions, while guard measurement adds their squared magnitudes. A record keeps emission x_{n-1} with label $n-1$. Only p_n is a halting probability; decimal entries are rounded to four places.

A table and an unknown gate. There is also no uniform approximation guarantee for an unknown body given only as a gate. For a scalar rotation with cosine t the trace is $t-(1-t^2)/(1-t) = -1$ for $-1 < t < 1$, but at the identity rotation it is 1. Put a bypass beside this scalar and the jump is a relative phase detectable by a Hadamard. A fixed finite circuit built by continuously composing the supplied body has a continuous response; eliminating a given clause table can branch on an exact equality [13, Remark B.1]. Appendix H follows the light from its first trip around an empty ring.

There is also a trace elimination result for diagrams built from polarizing beam splitters. Each path traverses a given feedback wire at most twice. For a diagram on at least two wires with invertible gates, Clément and Perdrix give an equivalent trace-free diagram [10, Prop. 26]. In `guardRound`, a single input contributes to infinitely many exits. Eliminating its one return value sums all of them.

C Records, measurement and phase

For a common payload $H = X = X' = Y$, the guard-first transport is $F_g = \mathrm{factor};\sigma_{H,\mathrm{Qubit}} : H + H \leftrightarrow \texttt{Qubit} \otimes H$. It gives $\widehat{U}_g = [\![F_g]\!]U[\![F_g]\!]^{-1}$. For a counter, $F_{K,g} = d_K^{-1};(I_{[K]} \otimes F_g)$ puts counter, guard and data in that order, with inverse transport at the output. These are the equal-payload forms of the general constructions in Section 4.

For general ports, distributivity d_K identifies $[K] \otimes (a + c)$ with $([K] \otimes a) + ([K] \otimes c)$. Write d'_K for the same distributivity at the output. The enlarged body of Figure 7 is

$$\beta_K = d_K^{-1}\,;\,(I_{[K]} \otimes \beta)\,;\,d'_K\,;\,(I_{[K]\otimes b} \oplus (S_K \otimes I_c)).$$

Its matrix and the resulting counting loop are

$$W_K = [\![\beta_K]\!] = \begin{pmatrix} I_K \otimes A & I_K \otimes B \\ S_K \otimes C & S_K \otimes D \end{pmatrix}. \qquad R_K u = (\mathrm{Tr}^{\mathbb{C}^K \otimes Y}\, W_K)(|0\rangle \otimes u).$$

To hide the counter we use $\mathsf{hide}(\mathsf{count}_K\beta; \mathsf{arr}\,\sigma)$ with $\sigma : [K] \otimes b \leftrightarrow b \otimes [K]$.

On the first call the input is $(|0\rangle \otimes u, 0)$. The ket is the prepared counter and the second zero means no returning amplitude. The two output blocks are $|0\rangle \otimes Au$ and $|1 \bmod K\rangle \otimes Cu$.

For `guardRound` on $|*\rangle$, keeping $|*\rangle$ as the data factor gives the exact record $\xi_{\mathrm{rot}} = t|0\rangle - s^2 \sum_{k\geq 1} t^{k-1}|k\rangle$. For `bypassRound`, $R|0\rangle = |0\rangle \otimes |0\rangle$ and $R|1\rangle = \xi_{\mathrm{rot}} \otimes |1\rangle$, so $R|+\rangle$ is entangled.

Use the Hilbert space $H_{\mathrm{rec}} = \ell^2(\mathbb{N})$ for an exact record, in the order counter then data. The squared norm of Ru is $\sum_k \|x_k u\|^2 = \|u\|^2$. For the isometric inclusion $J_K : \mathbb{C}^K \to H_{\mathrm{rec}}$ onto the first

K labels,

$$\|(J_K \otimes I)R_K - R\| \;\le\; \sum_{k\ge K} \big\|\,|k \bmod K\rangle - |k\rangle\big\|\,\|x_k\| \;\le\; 2\sum_{k\ge K}\|x_k\|.$$

For $k \ge K$, the two distinct orthogonal count labels have distance $\sqrt{2} \le 2$. The bound therefore tends to zero. This argument only compares the operators produced by finite bodies. The map that folds all square-summable sequences into residue classes would be unbounded, as would the map that sums their entries at phase one.

In Ying's free coin histories [35, Eq. (34)], let $J|k\rangle$ be the word $|1^k 0\rangle$ in the sector with $k+1$ coins. These words are orthonormal. Thus J is an isometry from the exact register onto their closed span. For matched exit maps the recorded state is $(J \otimes I)Ru$. Tracing out either record therefore gives the same data. This identifies the labels of completed histories. Ying's principal-system semantics also specifies an initial coin state and uses the symmetric semantics [35, Def. 9]. Assolini and Di Pierro keep a qubit from each test. Their unitary approximations can fail to converge while a branch keeps running. Restricting the output to completed branches gives a convergent sequence on each input [4, § 4.1, Thm. 5.9].

For an input of the form $|u\rangle\langle v|$, tracing the exact record over its orthonormal basis leaves $\sum_k x_k|u\rangle\langle v|x_k^\dagger$. Such operators span the input matrix algebra, giving (4.2). For a finite counter put $G_j = \sum_{k\equiv j \bmod K} x_k$. Discarding that counter instead gives $\sum_{j<K} G_j \rho G_j^\dagger$. As a result, unequal counts congruent modulo K retain cross terms. For the Fourier forms, the coefficient sum is absolutely convergent, and

$$G_j = \frac{1}{K}\sum_{z^K=1} \bar{z}^{\,j}\Theta(z), \qquad x_k = \frac{1}{2\pi}\int_0^{2\pi} e^{-ik\varphi}\Theta(e^{i\varphi})\,d\varphi.$$

Expanding $\Theta(z)\rho\Theta(z)^\dagger$ and integrating term by term removes all unequal indices, proving (6.1). For a uniform choice among the Kth roots, a pair of exits k, j acquires the factor $K^{-1}\sum_{z^K=1} z^{k-j}$. This geometric sum is one when $k \equiv j \pmod K$ and zero otherwise. The surviving terms are exactly $\sum_{r<K} G_r \rho G_r^\dagger$, the channel of the hidden K-counter. If $D^n = 0$, only counts $0, \dots, n$ can occur, so $n+1$ equally spaced phases give the measured loop exactly.

The table at the end of Section 4 gives these maps for `bypassRound`. For the classical trace, let $\theta(\rho,\sigma) = \operatorname{diag}(\rho,\sigma)$ insert the two operator blocks and let φ extract the diagonal output blocks. Then $\Phi_U = \varphi \circ \mathscr{E}_U \circ \theta$ is the branching map of Section 4. In the category of completely positive trace-nonincreasing maps on finite C^*-algebras, the classical-sum trace $\mathrm{Tr}_{\mathrm{cl}}^{\mathrm{End}(Y)}(\Phi_U)$ sends ρ to $\sum_k x_k \rho x_k^\dagger$ [3, Prop. 3.2.17].

Relation to monitored first returns. The published test-first loop measures before applying U [36, § 3]. To obtain our body-first convention when $X' = X$, apply U once before entering it. Let $\iota_X u = (u, 0)$ and $P = \iota_X \iota_X^\dagger$. For $n \ge 1$, the exit map on call n is $\iota_X^\dagger U((I-P)U)^{n-1}\iota_X = x_{n-1} : X \to X$. Thus the monitored-return count includes the initial call and is one more than our number of returns [7, § 2].

To see why this sum gives the least fixed point, put $Q = I - P$ and write $V = UQ$ for a return followed by another call. On a state σ just before the test, Ying's unfolding sends a map G to the map $\sigma \mapsto P\sigma P + G(V\sigma V^\dagger)$ [34, Prop. 5.1(6), Cor. 5.1]. Starting at the zero map and unfolding n times gives $\sum_{j=0}^{n-1} PV^j\sigma(V^j)^\dagger P$. Each term is positive for positive σ. Since $V^\dagger V = Q = I - P$, the trace of the sum is $\operatorname{tr}\sigma - \operatorname{tr}(V^n\sigma(V^n)^\dagger)$. The sums therefore increase and converge in finite dimension. Their limit solves the unfolding equation. Every positive solution bounds zero and, by unfolding,

bounds each partial sum. The limit is thus the least solution in the pointwise order on positive inputs. Starting it on $U\iota_X \rho \iota_X^\dagger U^\dagger$ and reading the external block gives $\sum_{k\geq 0} x_k \rho x_k^\dagger = \mathscr{M}(\rho)$.

D Agreement and replacement

Proof of Theorem 5.1. Assume $\mathscr{M} = \mathscr{E}_L$. For each unit input u, test the output on any w orthogonal to Lu. The sum of the nonnegative numbers $|\langle w, x_k u\rangle|^2$ is zero. Thus $x_k u$ is parallel to Lu for each k and all u. Put $V = L^\dagger x_k$. Since Vu lies along u, testing a basis vector e_i gives $Ve_i = \lambda_i e_i$. To see why the scalar cannot depend on i, test two distinct basis vectors together:

$$V(e_i + e_j) = \underbrace{\lambda_i e_i + \lambda_j e_j}_{\text{by linearity}} = \underbrace{\lambda_{ij}(e_i + e_j)}_{\text{parallel to the input}}.$$

Comparing the two coordinates gives $\lambda_i = \lambda_{ij} = \lambda_j$. Hence $V = b_k I$ and $x_k = b_k L$, including in dimension one where there is only one basis coefficient to begin with. By absolute convergence, $\sum_k b_k = 1$ since $\sum_k x_k = L \neq 0$. By $\sum_k x_k^\dagger x_k = I$, $\sum_k |b_k|^2 = 1$. The unit vector $\xi = \sum_k b_k |k\rangle$ therefore satisfies $Ru = \xi \otimes Lu$. Conversely that factorization implies $\mathscr{M} = \mathscr{E}_L$ after tracing out the count; testing each label shows $x_k = \langle k|\xi\rangle L$.

These conditions imply that $x_0, \dots, x_m$ span at most a line. If $m > 0$, Cayley–Hamilton writes D^m as a linear combination of $I, D, \dots, D^{m-1}$. Since $x_{k+1} = BD^k C$, multiplication by higher powers shows that every exit after x_m belongs to the span of $x_1, \dots, x_m$. If $m = 0$, all exits after x_0 vanish. Thus the span of the first $m+1$ exits is the span of all the exits. It cannot be zero because it contains their unitary sum on the nonzero input. If it is a line, the same sum L spans it, so $x_k = b_k L$ for each k. Completeness of the exits again gives the unit norm of ξ. This proves all implications. □

Channel contexts. Fix the hole type $a \rightsquigarrow b$ and abbreviate it by $\mathscr{H}$ to the left of a context judgment. For fixed closed channels h, h_j, the contexts are generated by

$$\mathscr{C} ::= [-] \mid h; \mathscr{C} \mid \mathscr{C}; h \mid h \otimes \mathscr{C} \mid \mathscr{C} \otimes h \mid \mathrm{case}\big(\Psi_i; \mathscr{C}, (\Psi_j; h_j)_{j\neq i}\big).$$

The rules for the hole, sequence and tensor are

$$\mathscr{H} \vdash [-] : a \rightsquigarrow b \qquad \frac{h : e \rightsquigarrow c \qquad \mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d}{\mathscr{H} \vdash h; \mathscr{C} : e \rightsquigarrow d} \qquad \frac{\mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d \qquad h : d \rightsquigarrow e}{\mathscr{H} \vdash \mathscr{C}; h : c \rightsquigarrow e}$$

$$\frac{h : e \rightsquigarrow f \qquad \mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d}{\mathscr{H} \vdash h \otimes \mathscr{C} : e \otimes c \rightsquigarrow f \otimes d} \qquad \frac{\mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d \qquad h : e \rightsquigarrow f}{\mathscr{H} \vdash \mathscr{C} \otimes h : c \otimes e \rightsquigarrow d \otimes f}.$$

For a finite instrument $(\Psi_j : c \to c_j)_{j\in J}$, meaning $\Psi_j : \mathrm{End}([\![c]\!]) \to \mathrm{End}([\![c_j]\!])$, the maps are completely positive and their output traces sum to the input trace. Its case rule is

$$\frac{i \in J \quad \mathscr{H} \vdash \mathscr{C} : c_i \rightsquigarrow d \quad h_j : c_j \rightsquigarrow d \ (j \neq i)}{\mathscr{H} \vdash \mathrm{case}\big(\Psi_i; \mathscr{C}, (\Psi_j; h_j)_{j\neq i}\big) : c \rightsquigarrow d}.$$

Plugging h into the hole interprets sequencing by channel composition and tensor by the tensor product of channels. The case denotes $[\![\mathscr{C}[h]]\!] \circ \Psi_i + \sum_{j\neq i} [\![h_j]\!] \circ \Psi_j$. Thus exactly one syntactic occurrence depends on the hole, and each branch continuation has the channel type stated by the rules. These contexts can use a hole on one instrument outcome and omit it on the others. They do not contain iso-level operations on that hole.

Proof of Corollary 5.2. If $X = 0$ then $X' = 0$ as well, and there is just one map on the zero operator space. Now suppose $X \neq 0$. On agreement the finite record has the form $R_K u = \xi_K \otimes Lu$, where $\xi_K = \sum_k b_k |k \bmod K\rangle$. Since R_K is an isometry and an input unit vector exists, $\|\xi_K\| = 1$, even if components with equal labels interfere. Discarding it gives $\mathscr{E}_L$. For the exact record the equality after discard needs no agreement condition. Channel equality on data entails equality after tensoring with identity on any finite reference, since elementary tensor operators span the joint matrix algebra. Composition with fixed channels, discarding and applying a fixed instrument on either program therefore preserve equality. For a measurement-conditioned continuation apply the same argument separately to each unnormalized outcome and then add if the outcome is forgotten. This proves the claim by induction over a context built with these constructors. Conversely, if the channels differ, some density operator distinguishes them since positive trace-one inputs span the matrix algebra over $\mathbb{C}$. The two resulting density operators are distinguished by a spectral projection of their Hermitian difference. This test is at the output of the empty context; finite exact isos and tag measurement can implement a measurement in the chosen eigenbasis.

For delayed discard, use the following separate semantic continuation calculus. Its ordinary data types are finite, while its opaque history names may carry countable semantic record spaces. Write $\Gamma \vdash K : a \Rightarrow b$ for a finite continuation that starts with ordinary data of type a and the fresh opaque history names in Γ, and returns data of type b after consuming all those names. Each name carries its designated record space. Each branch Ψ_j of a finite instrument sends operators on $[\![a]\!]$ to operators on $[\![a_j]\!]$ by a completely positive map. The traces of these outputs sum to one on each density input. An ordinary channel is its one-outcome case. For a binder, let $R : [\![a]\!] \to H \otimes [\![d]\!]$ be a recording isometry. Its history name g has space H. The caller rules are

$$\emptyset \vdash \mathsf{return} : a \Rightarrow a \qquad \frac{\Gamma \vdash K_j : a_j \Rightarrow b \quad (j \in J)}{\Gamma \vdash \mathsf{case}\,(\Psi_j; K_j)_{j\in J} : a \Rightarrow b}$$

$$\frac{\Gamma \vdash K : a \Rightarrow b \quad h \notin \Gamma}{\Gamma, h \vdash \mathsf{drop}\ h; K : a \Rightarrow b} \qquad \frac{\Gamma, g \vdash K : d \Rightarrow b \quad g \notin \Gamma}{\Gamma \vdash \mathsf{record}\ g = R\ \mathsf{in}\ K : a \Rightarrow b}.$$

Bound names are chosen fresh throughout the finite derivation by renaming. For an ordering of the live names, let H_Γ be their tensor product and I_Γ its identity. The semantics maps trace-class operators on $H_\Gamma \otimes [\![a]\!]$ to operators on $[\![b]\!]$. With canonical reassociations and permutations of named factors understood, its equations are

$$\begin{aligned}
[\![\mathsf{return}]\!] &= \mathrm{id},\\
[\![\mathsf{case}\,(\Psi_j; K_j)_{j\in J}]\!] &= \sum_{j\in J} [\![K_j]\!] \circ (\mathscr{E}_{I_\Gamma} \otimes \Psi_j),\\
[\![\mathsf{drop}\ h; K]\!] &= [\![K]\!] \circ \mathrm{tr}_h,\\
[\![\mathsf{record}\ g = R\ \mathsf{in}\ K]\!] &= [\![K]\!] \circ \mathscr{E}_{I_\Gamma \otimes R}.
\end{aligned}$$

The instrument selects a continuation by its classical outcome and passes all live names to that continuation. A binder adds its new record, and a drop traces out just the named factor. Only the empty history context can return, so each returning branch drops every live name exactly once. For K_h starting with a distinguished h, fresh for every internal record binder, delete its drops to obtain K, leaving all other binders and drops in place. The rules show that K is well typed with h removed. On each returning branch and for any joint trace-class input τ,

$$[\![K_h]\!](\tau) = [\![K]\!](\mathrm{tr}_h\, \tau).$$

For a drop it is the definition. For an ordinary operation Ψ before the drop, use $\mathrm{tr}_h((\mathscr{E}_{I_h} \otimes \Psi)(\tau)) = \Psi(\mathrm{tr}_h\, \tau)$, by taking the partial trace in a basis of h. This holds for each subnormalized instrument outcome, and tensoring with a reference leaves it valid. Nested recording is an isometry on the ordinary data and the new record, with the older name carried by the identity. Partial traces over distinct names commute at their eliminations. Induction over sequencing, measurement branches and nested bindings proves the formula. Consequently a recording atom R followed by K_h may be replaced by its marginal channel $\rho \mapsto \mathrm{tr}_H(R\rho R^\dagger)$ followed by K. When R is the exact exit record of our body β, this marginal is $\mathsf{while}_c\, \beta$ by (4.2). Under agreement it may also be replaced by the embedded coherent iso. A finite clock has its own marginal and does not in general admit the unconditional exact-record replacement. If a caller is a limit of increasing finite executions with subnormalized outputs, the same pathwise identities preserve its lost trace under that limit. This last observation assumes such a semantics; general recursive callers are outside the finite grammar. □

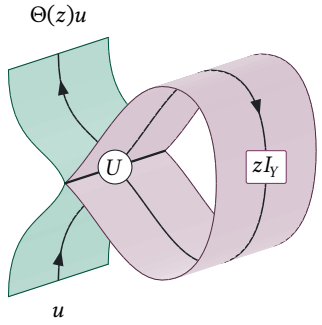
Why one finite clock does not suffice. The output interface matters. If a caller reads the count, even a fixed count independent of the data is an observable output. Conversely, a finite clock may hide a failure of agreement. Use the permutation that fixes `exit 0` and cycles through `exit` $1 \mapsto$ `back` $q_1 \mapsto \cdots \mapsto$ `back` $q_K \mapsto$ `exit` 1, with K distinct return values $q_1, \dots, q_K$. Its only nonzero exits are $x_0 = |0\rangle\langle 0|$ and $x_K = |1\rangle\langle 1|$. The K-clock assigns label 0 to both exits. Ignoring it preserves $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, whereas a measurement on every return dephases $|+\rangle$ in that basis. Checking just this clock would accept the replacement incorrectly.

E The derivative and its average

Define

$$Y_0 = \{y \in Y \mid BD^j y = 0 \text{ for every } j \geq 0\}, \qquad Y_1 = Y_0^\perp.$$

By definition, $D(Y_0) \subseteq Y_0$ and $B|_{Y_0} = 0$. Unitarity gives $\|Dy\|^2 + \|By\|^2 = \|y\|^2$, so D preserves norm on Y_0. In finite dimension this restriction is onto. Unitarity then maps the orthogonal complements of these internal components onto one another, giving $C(X) \subseteq Y_1$ and $D(Y_1) \subseteq Y_1$. Put $B_1 = B|_{Y_1}$ and $D_1 = D|_{Y_1}$, regarding C as $X \to Y_1$. The geometric decay from Section 3 makes $I - zD_1$ invertible for $|z| \leq 1$, even when $I - zD$ on all of Y is singular.



	$j=0$	$j=1$	$j=2$
$k=0$	T_{00}	$z^{-1}T_{01}$	$z^{-2}T_{02}$
$k=1$	zT_{10}	T_{11}	$z^{-1}T_{12}$
$k=2$	z^2T_{20}	zT_{21}	T_{22}

$$T_{kj} = x_k \rho x_j^\dagger$$

$$\Theta_{\mathrm{rot}}(z) = \frac{t - z}{1 - tz},$$

$$\Theta_{\mathrm{bypass}}(z) = \mathrm{diag}(1, \Theta_{\mathrm{rot}}(z)).$$

$$\Theta(z) = \sum_{k \geq 0} z^k x_k = A + zB_1(I - zD_1)^{-1}C, \qquad L = \Theta(1).$$

Fig. 16. Each return adds a factor z at left. In the channel at right, exit pair k, j carries $z^{k-j}T_{kj}$. Averaging over the circle retains only the orange diagonal. The same cancellation applies to later exits. The displayed families Θ_{rot} and Θ_{bypass} belong to `guardRound` and `bypassRound`.

Proof of Theorem 6.1. Work on Y_1. The geometric bound in Theorem B.1 gives an open disk of radius greater than one on which F_z and $\Theta(z)$ are analytic. Empty live space gives $F_z = 0$. The identity $F_z = C + zD_1F_z$ says

$$U_1(u, zF_zu) = (\Theta(z)u, F_zu).$$

Use the wire equation twice, first at (v, w) and then at (u, z). Because $U_1 : X \oplus Y_1 \to X' \oplus Y_1$ preserves inner products, the inner product of the inputs equals that of the outputs. With the inner product linear in its second argument, the two sides are

$$\underbrace{\langle v, u\rangle + \bar{w}z\langle F_wv, F_zu\rangle}_{\text{external and returning inputs}} = \underbrace{\langle \Theta(w)v, \Theta(z)u\rangle + \langle F_wv, F_zu\rangle}_{\text{external and returning outputs}}.$$

Collecting terms gives $I - \Theta(w)^\dagger\Theta(z) = (1-\bar{w}z)F_w^\dagger F_z$. Keep w fixed on the unit circle and differentiate with respect to z. At $z = w$ the factor $1 - \bar{w}w$ is zero, so the term containing F_z' vanishes. The remaining equality is $\Theta(w)^\dagger\Theta'(w) = \bar{w}F_w^\dagger F_w$. Now put $w = e^{i\varphi}$. The angular derivative multiplies $\Theta'(w)$ by iw, giving $Q(w) = F_w^\dagger F_w$ as claimed.

By the telescope in Appendix B, $\sum_{k=1}^{p} kx_k^\dagger x_k = \sum_{j=0}^{p-1} T_j - pT_p$. Geometric decay makes $pT_p \to 0$. Thus $N = \sum_j T_j$. In $F_z^\dagger F_z$ the terms at m, n have factor z^{n-m} on the unit circle. As in Figure 16, integration keeps only the cells where $m = n$. Uniform absolute convergence permits this termwise integration, leaving $\sum_j T_j$. This gives the phase average. For the trace of N, cyclicity changes the summand $\operatorname{tr} T_j$ into $\operatorname{tr}(D_1^j CC^\dagger D_1^{\dagger j})$. From the Y_1 output block of $U_1U_1^\dagger = I_{X'\oplus Y_1}$ we have $CC^\dagger = I - D_1D_1^\dagger$. The partial sum once again telescopes,

$$\sum_{j=0}^{p-1} D_1^j CC^\dagger D_1^{\dagger j} = I_{Y_1} - D_1^p D_1^{\dagger p} \longrightarrow I_{Y_1}.$$

Thus $\operatorname{tr} N = \dim Y_1$. On each unit input u the measured probability of exit k is $\|x_ku\|^2$, so its mean count is $\sum_k k\|x_ku\|^2 = \langle u, Nu\rangle$. □

For the monitored first-return convention the generating function is $z\Theta(z)$, whose relative derivative is $I + Q(z)$. Its mean number of calls is therefore $1 + \langle u, Nu\rangle$ [7, § 3, Eq. (8)]. A phase on the return tag is an ordinary iso, so forming this family requires no counter or guard measurement.

Phase invariance and agreement. On a nonempty external type, the phase family also characterizes agreement. If every exit is $x_k = b_kL$, then $\Theta(z) = f(z)L$ with $|f(z)| = 1$ on the circle, so every marking phase gives the same data channel. Conversely, if those channels are all equal, they equal $\mathscr{E}_L$ because $\Theta(1) = L$. Their average is $\mathscr{M}$ by (6.1), so $\mathscr{M} = \mathscr{E}_L$. Thus agreement says exactly when changing the return phase leaves the data channel unchanged.

F Clause programs and their observations

Many small loop bodies can be written as two clauses with a variable for the payload. We give their definitions here so that one can reproduce each diagram directly from an iso. Each program closes its returning summand with `trace`. We abbreviate `exit` and `back` by E and B in this section's source equations. In these equations B is an injection tag; a matrix block appears only in a denotational equation. In executable types, `x` spells tensor product, and a coefficient `(a :+ b)` denotes $a + ib$ for real a, b.

Two scalar loops in sequence. Use two copies of the scalar guard rotation, with $t = 3/5$ and $s = 4/5$. Their separate returns each receive the phase z, and their external ports connect in sequence. The responses therefore multiply to $f(z)^2$, where $f(z) = (t - z)/(1 - tz)$. Figure 17 gives both this cascade and its single-round matrix W on $\mathbb{C} \oplus (\mathbb{C} \oplus \mathbb{C})$. Multiplying its blocks, or solving its two return equations, gives the same $f(z)^2$. The columns of W give the following three clauses; its two returning basis values are `back 0` and `back 1`.

```
round twoSections : Unit + Qubit = {
  exit unit ↔ [9/25 . exit unit + 4/5 . back 0 + 12/25 . back 1];
  back 0 ↔ [-12/25 . exit unit + 3/5 . back 0 + -16/25 . back 1];
  back 1 ↔ [-4/5 . exit unit + 3/5 . back 1]
}

trace twoSections on unit
```

The command returns $|*\rangle$, since $f(1)^2 = 1$. A period-two clock also returns label 0: both $f(1)^2$ and $f(-1)^2$ are one, so the odd residue sum is zero.

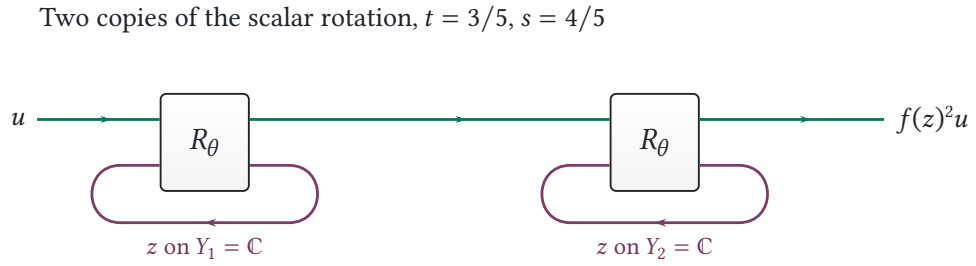


Fig. 17. Two specified scalar sections give response $f(z)^2$. The numerical matrix is the open body of this cascade in external-first order.

Now carry data u unchanged through both sections and apply a gate L once on the final external output. As Figure 18 shows, this amounts to tensoring the internal evolution with the identity on the data. The response is $f(z)^2 L$. Every exit coefficient multiplies the same L, so dropping the count preserves the data channel by Theorem 5.1. On the unit circle $|f| = 1$, hence the relative derivative of the squared scalar response is twice that of one section. Its phase average gives mean count two, as expected from the mean of one for each rotation. Changing coordinates on the internal space or adjoining an inaccessible summand leaves the exits unchanged.

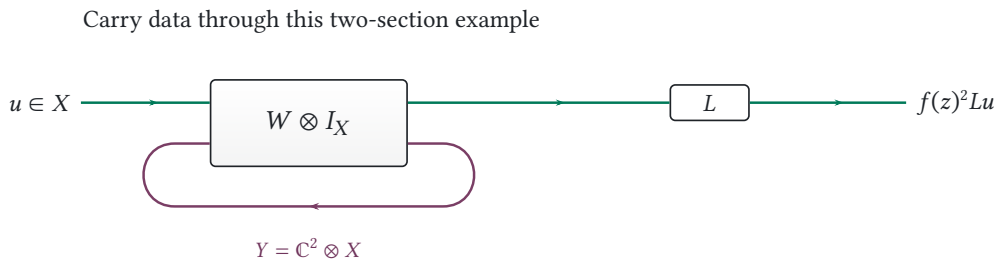


Fig. 18. Tensor the specified two-section body with the data identity and apply L on its external output. This constructed body passes agreement and has mean count two.

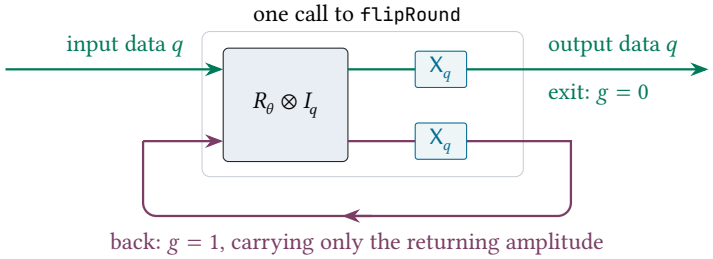
F.1 A flip on every call

For a loop that also transforms a payload on every call, take a guard and a data qubit, apply R_θ to the guard and the data flip $J = \mathsf{X}$ to the data, and close the returning summand. This is more structure than the scalar loop of the main text needs, but it gives a second example where measured and coherent outputs differ. Each clause binds the data and applies its gate to that value.

```
X :: Qubit ↔ Qubit
X = { 0 ↔ 1; 1 ↔ 0 }

round flipRound : Qubit + Qubit = {
  exit x ↔ let y = X x in [ 3/5 . exit y + 4/5 . back y];
  back x ↔ let y = X x in [-4/5 . exit y + 3/5 . back y]
}
```

The first definition names the data gate X inside the box in Figure 19. On input `exit 0` the first clause binds $x = 0$, so `let y = X x` binds $y = 1$. Each dot scales a tagged output, and the brackets form their superposition. The coefficients $3/5$ and $4/5$ are the first column of R_θ, since `exit` means that the guard enters at 0. Following `back 1` into the second clause flips the data once more and uses the other column, with coefficients $-4/5$ and $3/5$. This is where the second call acquired its minus sign. With the guard first and the data second, Figure 20 is the matrix of $R_\theta \otimes \mathsf{X}$.



Call	Component entering this call	Leaves the loop	Continues to the next call
1	$\|00\rangle$	$\frac{3}{5}\|01\rangle$	$\frac{4}{5}\|11\rangle$
2	$\frac{4}{5}\|11\rangle$	$-\frac{16}{25}\|00\rangle$	$\frac{12}{25}\|10\rangle$
3	$\frac{12}{25}\|10\rangle$	$-\frac{48}{125}\|01\rangle$	$\frac{36}{125}\|11\rangle$

Fig. 19. The coherent loop connects the body's returning output to its returning input. The two ports are alternative guard values, each carrying the same data qubit q. Both alternatives apply X_q; this data gate commutes with the guard rotation. The table starts at $|00\rangle$ and writes joint states $|gq\rangle$, with guard first. Emitted components do not enter a later row; only the last column does. The guard stays coherent throughout.

```
exit 0 <-> [3/5 . exit 1 + 4/5 . back 1]
exit 1 <-> [3/5 . exit 0 + 4/5 . back 0]
back 0 <-> [-4/5 . exit 1 + 3/5 . back 1]
back 1 <-> [-4/5 . exit 0 + 3/5 . back 0]
```

one column for each ground input

input basis patterns

outputs	exit 0	exit 1	back 0	back 1
exit 0	A 0	$\frac{3}{5}$	0	$-\frac{4}{5}$ B
exit 1	$\frac{3}{5}$	0	$-\frac{4}{5}$	0
back 0	0	$\frac{4}{5}$	0	$\frac{3}{5}$
back 1	C $\frac{4}{5}$	0	$\frac{3}{5}$	0 D

Compare inputs `exit 0` and `back 0`

output row	column product
`exit 1`	$(3/5)(-4/5)$
`back 1`	$(4/5)(3/5)$
inner product	0

The other rows contribute zero.

Each squared norm is $9/25 + 16/25 = 1$.

Fig. 20. The four columns of the payload-flipping body, in external-first order. This is $R_\theta \otimes \mathrm{X}$.

The exit maps of `flipRound`. Put $t = 3/5$ and $s = 4/5$. The two clauses of `flipRound` give $A = t\mathrm{X}$, $C = s\mathrm{X}$ on entry and $B = -s\mathrm{X}$, $D = t\mathrm{X}$ on return. Hence $x_0 = t\mathrm{X}$ and $x_1 = BC = -s^2 I$. The data flips twice before that second exit. In general $x_{k+1} = -s^2 t^k \mathrm{X}^{k+2}$. Grouping equal data gates gives two ordinary geometric series,

$$\underbrace{t - s^2 t \sum_{j\geq 0} t^{2j}}_{\text{coefficient of X}} = 0, \qquad \underbrace{-s^2 \sum_{j\geq 0} t^{2j}}_{\text{coefficient of } I} = -1, \qquad s^2 = 1 - t^2.$$

Thus the whole coherent loop returns $-u$. In particular it can never return data 1 on input 0. With an exact count the exits that return 0 on this input have odd counts, with amplitudes $-s^2, -s^2t^2, \ldots$.

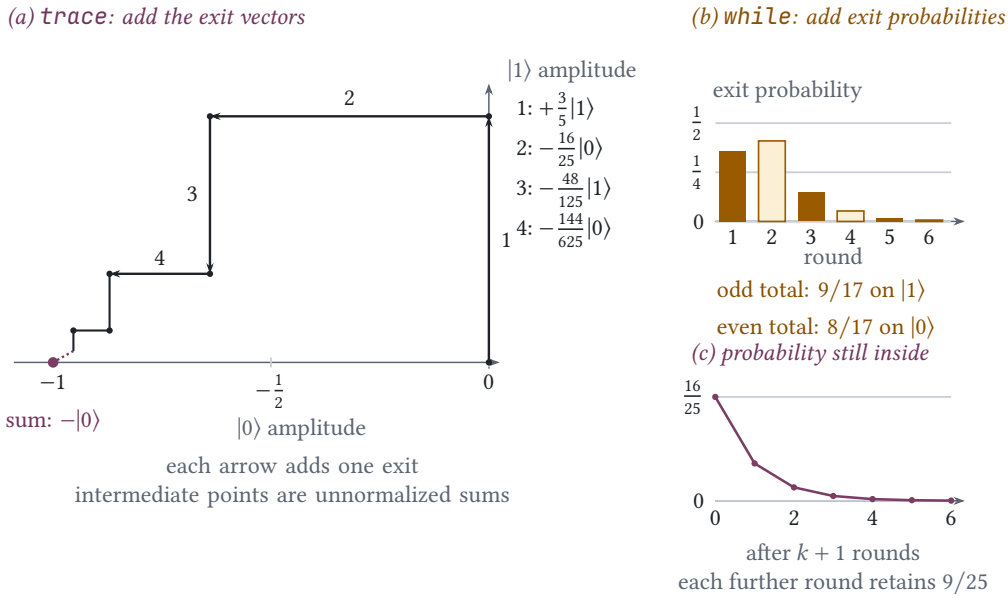


Fig. 21. On input $|0\rangle$, the left panel adds exit amplitudes in the data space. The right panel keeps the exit rounds distinct and plots their probabilities. Calls are numbered from one.

Their squared magnitudes sum to $s^4/(1-t^4) = 8/17$. The complementary probability $9/17$ goes to 1. These are the two observations in Figure 21.

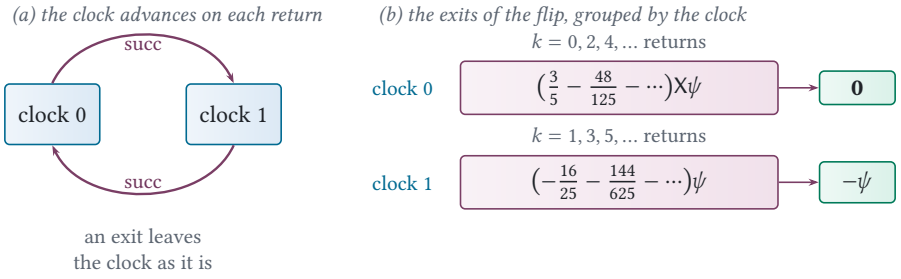


Fig. 22. For the flip, folding by parity cancels all even counts as amplitudes. Measurement of an exact count has positive probability at every count.

The numerical elimination also gives a finite calculation of this loop. Unlike `guardRound`, it has two returning basis patterns. Figure 4 shows the clauses after each step.

The first substitution in Figure 4 removes `back 0` and leaves coefficient $(3/5)^2 = 9/25$ on the remaining return amplitude y_1. It also changes the coefficient of y_1 in external output v_1 to $(3/5)(-4/5) = -12/25$. On external basis input 0, an amplitude $4/5$ enters this remaining return pattern. Solving its scalar wire equation gives the two entries in that output column:

$$\text{on data } 0 : \quad \frac{(4/5)(-4/5)}{1 - 9/25} = -1,$$

$$\text{on data } 1 : \quad \frac{3}{5} + \frac{(4/5)(-12/25)}{1 - 9/25} = 0.$$

Interchanging the data bits gives the other column. Thus the finite rewrite produces the same $-I$ that we obtained by adding the exits round by round.

Moving the flip to the exit. A variant that passes the agreement test has the circuit in Figure 23. In the ground clauses, the bit after **`exit`** is the flipped input bit and the bit after **`back`** is the input itself.

```
round flipOnExit : Qubit + Qubit = {
  exit 0 ↔ [ 3/5 . exit 1 + 4/5 . back 0];
  exit 1 ↔ [ 3/5 . exit 0 + 4/5 . back 1];
  back 0 ↔ [-4/5 . exit 1 + 3/5 . back 0];
  back 1 ↔ [-4/5 . exit 0 + 3/5 . back 1]
}
```

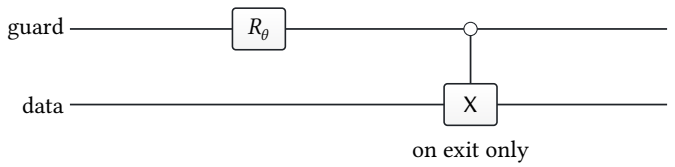


Fig. 23. The same rotation as in `flipRound`, with the flip controlled on the outgoing guard being 0. A returning component keeps its data. Whichever round exits, that component of the data is flipped exactly once.

Running `trace flipOnExit on 0` gives $-|1\rangle$. The measured command `while flipOnExit on 0` gives $|1\rangle\langle 1|$. Its exits are $(3/5)\mathsf{X}$ at count zero and $-(16/25)(3/5)^{k-1}\mathsf{X}$ at counts $k \geq 1$, using the scalar coefficients calculated in Section 3. Every exit is proportional to X, so agreement holds on every input, including one entangled with a reference. The coherent completion is $-\mathsf{X}$.

Rotation with a data iso. For $c = \cos\theta$, $s = \sin\theta$ and an iso $M : a \leftrightarrow a$, define $\mathrm{rotbody}(\theta, M) : a + a \leftrightarrow a + a$ by the clauses

$$E\,x \leftrightarrow \text{let } y = M\,x \text{ in } c\,E\,y + s\,B\,y,$$

$$B\,x \leftrightarrow \text{let } y = M\,x \text{ in } -s\,E\,y + c\,B\,y.$$

The notation is a schema for named first-order isos, not a higher-order value. Putting $M = \mathsf{X}$ gives the four ground instances of the flip. For $|c| < 1$ its exits are cM and $-s^2c^kM^{k+2}$. Summing them gives

$$f_\theta(M) = M(cI - M)(I - cM)^{-1}.$$

On an eigenvector of M with eigenvalue w, the equation $f_\theta(w) = 1$ reduces to $w^2 - 2cw + 1 = 0$, while $f_\theta(w) = -1$ reduces to $w^2 = 1$. The eigenvalues of $R_{-\theta}$ are $c \mp is$, the roots of the first polynomial, so $f_\theta(R_{-\theta}) = I$. When $|c| = 1$, the guard never enters and the immediate exit gives the same conclusion for that rotation. The eigenvalues of X are ± 1, so $f_\theta(\mathsf{X}) = -I$ for $|c| < 1$.

Here is the inverse-rotation instance at $c = 3/5$, including its invocation. The first declaration supplies $M = R_{-\theta}$; the second applies it locally on each call.

```
inverseRotation :: Qubit ↔ Qubit
inverseRotation = {
  0 ↔ [3/5 . 0 + -4/5 . 1];
  1 ↔ [4/5 . 0 + 3/5 . 1]
}

round undoRound : Qubit + Qubit = {
  exit x ↔ let y = inverseRotation x in
    [3/5 . exit y + 4/5 . back y];
  back x ↔ let y = inverseRotation x in
    [-4/5 . exit y + 3/5 . back y]
}

trace undoRound on 0
```

The command returns $|0\rangle$. On an arbitrary vector the completed gate is I, as the eigenvalue calculation shows. The measured loop need not agree: on $|+\rangle$ it gives $\frac{1}{650}\begin{pmatrix} 349 & -107 \\ -107 & 301 \end{pmatrix}$ rather than $|+\rangle\langle+|$. The cancellation giving the coherent identity takes place between exits on different rounds.

A guard whose two payloads differ. A tag test need not be a measurement of a qubit register beside an unchanged data type. Take the two external patterns 0, 1 and just one return pattern $*$. The body below is the three-cycle of monitored recurrence, with its basis relabeled [7, Example 3.1]. It binds no history variable.

```
cycle3 :: (Qubit + Unit) ↔ (Qubit + Unit)
cycle3 = {
  left (left unit)  ↔ right unit;
  left (right unit) ↔ left (left unit);
  right unit        ↔ left (right unit)
}
```

Input 1 exits immediately as 0, whereas input 0 returns once and exits as 1. Thus $x_0 = |0\rangle\langle 1|$, $x_1 = |1\rangle\langle 0|$, and all later exits vanish. Calling `trace` on this iso gives the flip X, so $|+\rangle$ returns unchanged. Calling `count` with period two gives $(|0\rangle|0\rangle + |1\rangle|1\rangle)/\sqrt{2}$, with counter first. Calling `while` on the very same iso returns $I/2$ on that input. The two exit vectors on $|+\rangle$ have zero inner product, but they are values of the data we retain. Coherent feedback still preserves the terms $|0\rangle\langle 1|$ and $|1\rangle\langle 0|$ between them. A separate ignored count removes these terms because it gives the two data vectors different orthogonal labels. Thus distinguishable data values alone do not make recording the round harmless.

A cycle, a reflection and a phase. The cycle3 program just defined gives the data flip after closing its one return pattern. One may add a dead pattern with its own identity clause, obtaining the same flip on a two-pattern return type. A body with no data can instead be the reflection $E * \leftrightarrow t\, E * + s\, B *$, $B * \leftrightarrow s\, E * - t\, B *$ at $t = 3/5, s = 4/5$. Its immediate exit is t, and its delayed sum is $s^2/(1+t)$. They add to 1.

```
round reflection : Unit + Unit = {
  exit unit ↔ [3/5 . exit unit + 4/5 . back unit];
  back unit ↔ [4/5 . exit unit + -3/5 . back unit]
}

trace reflection on unit
```

This command returns $|*\rangle$. The measured loop also returns the sole external value with probability one. Its parity record is $|1\rangle \otimes |*\rangle$: the even exits cancel and the odd exits add to one. This reflection and `guardRound` therefore have the same measured data output but opposite coherent phases.

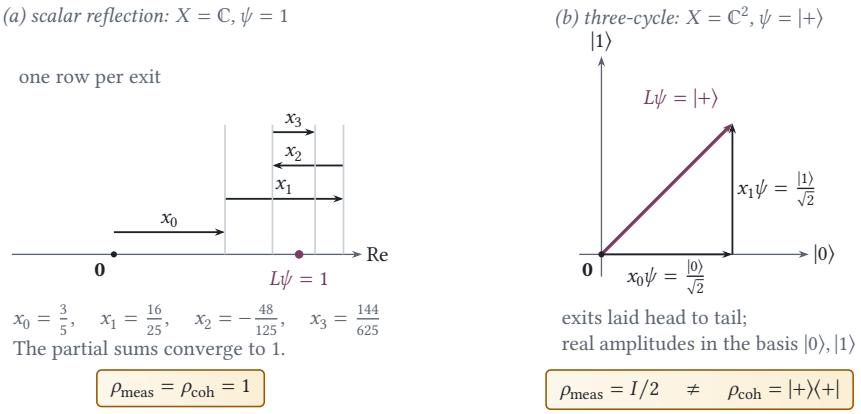


Fig. 24. The scalar reflection gives the same data under measurement or coherent feedback. For the three-cycle, measuring yields $I/2$ while coherent feedback returns the pure state $|+\rangle$.

For another example, take the four permutation clauses

$$E0 \mapsto E0, \quad E1 \mapsto B1, \quad B0 \mapsto B0, \quad B1 \mapsto E1.$$

Multiply every output by g with $|g| = 1$. Input 0 receives one factor of g and input 1 receives two, giving $\mathrm{diag}(g, g^2)$. The one-call channel is unchanged by this common phase but the loop channel in general changes. Placing a phase z on the *returning* outputs alone instead gives $\mathrm{diag}(1, z)$. The markings occur in different places, but at $z = g$ their completions differ only by the global factor g. They therefore give the same data channel for this permutation. That channel can still differ from the unmarked identity channel.

The following two programs instantiate $g = z = i$, written `(0 :+ 1)` in executable syntax. Both use the exact input $3|0\rangle/5 + 4|1\rangle/5$.

```
round unmarked : Qubit + Qubit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ back 1;
  back 0 ↔ back 0;
  back 1 ↔ exit 1
}

round wholePhase : Qubit + Qubit = {
  exit 0 ↔ (0 :+ 1) . exit 0;
  exit 1 ↔ (0 :+ 1) . back 1;
  back 0 ↔ (0 :+ 1) . back 0;
  back 1 ↔ (0 :+ 1) . exit 1
}

trace wholePhase on 3/5 . 0 + 4/5 . 1
```

This invocation returns $3i|0\rangle/5 - 4|1\rangle/5$.

```
round returnPhase : Qubit + Qubit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ (0 :+ 1) . back 1;
  back 0 ↔ (0 :+ 1) . back 0;
  back 1 ↔ exit 1
}

trace returnPhase on 3/5 . 0 + 4/5 . 1
```

The second returns $3|0\rangle/5 + 4i|1\rangle/5$. The first output differs from this one by the phase i, so both give the same density operator.

Kickback with a returned ancilla. Let $J = \mathsf{X}$, H be Hadamard, $G = HJ$, and $\Pi_\pm = (I \pm J)/2$. Use rotbody$(\pi/4, M)$ with ancilla and data as payload, where $M = G \otimes \Pi_+ + JG \otimes \Pi_-$. In iso syntax this M is an operation controlled in the J basis of the data: change that basis with H, apply G or JG to the ancilla according to the resulting tag, then undo the basis change. In the first sector $G^2 = \sqrt{2}G - I$, so $f_{\pi/4}(G) = I$. In the second $(JG)^2 = I$, so $f_{\pi/4}(JG) = -I$. The completed loop is $I \otimes (\Pi_+ - \Pi_-) = I \otimes J$. It therefore acts as the identity on the ancilla for every joint input.

In the basis of the two payload bits, the projector sum gives

$$M|0, d\rangle = \frac{|0, 1-d\rangle - |1, 1-d\rangle}{\sqrt{2}}, \qquad M|1, d\rangle = \frac{|0, d\rangle + |1, d\rangle}{\sqrt{2}}.$$

The data bit flips in the first case and stays in the second. Expanding the guard rotation now gives the clauses below. Each nonzero coefficient of M is $\pm 1/\sqrt{2}$, and the guard contributes another $1/\sqrt{2}$, leaving only coefficients $\pm 1/2$. Thus no rounded Hadamard coefficient is needed in this source.

```
round kickbackExact : (Qubit x Qubit) + (Qubit x Qubit) = {
  exit ⟨0, 0⟩ ↔ [1/2 . exit ⟨0, 1⟩ + -1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + -1/2 . back ⟨1, 1⟩];
  exit ⟨0, 1⟩ ↔ [1/2 . exit ⟨0, 0⟩ + -1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + -1/2 . back ⟨1, 0⟩];
  exit ⟨1, 0⟩ ↔ [1/2 . exit ⟨0, 0⟩ + 1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + 1/2 . back ⟨1, 0⟩];
  exit ⟨1, 1⟩ ↔ [1/2 . exit ⟨0, 1⟩ + 1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + 1/2 . back ⟨1, 1⟩];
  back ⟨0, 0⟩ ↔ [-1/2 . exit ⟨0, 1⟩ + 1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + -1/2 . back ⟨1, 1⟩];
  back ⟨0, 1⟩ ↔ [-1/2 . exit ⟨0, 0⟩ + 1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + -1/2 . back ⟨1, 0⟩];
  back ⟨1, 0⟩ ↔ [-1/2 . exit ⟨0, 0⟩ + -1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + 1/2 . back ⟨1, 0⟩];
  back ⟨1, 1⟩ ↔ [-1/2 . exit ⟨0, 1⟩ + -1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + 1/2 . back ⟨1, 1⟩]
}

trace kickbackExact on 3/5 . ⟨0, 0⟩ + 4/5 . ⟨1, 1⟩
```

The input is entangled between ancilla and data. Its output is $3|0,1\rangle/5 + 4|1,0\rangle/5$, exactly the result of applying J to the second component. The same identity on arbitrary joint inputs follows from the two-sector calculation above.

F.2 Other retained histories

In the map example of *From Symmetric Pattern-Matching to Quantum Control*, the iso argument $g : a \leftrightarrow b$ is applied to a head and the recursive iso f to the tail. In the source's list notation the two clauses read [30, § 4]

$$[] \leftrightarrow [],$$
$$h : t \leftrightarrow \text{ let } x = g\,h \text{ in let } y = f\,t \text{ in } x : y.$$

If g is Hadamard and the input has length two, every output of the recursion still has length two. A superposition of input lengths two and three likewise produces two components that can be distinguished by length alone. This explains how a reversible program can retain the depth of its recursion without ever measuring it. The cyclic register in (4.1) lets us make the same distinction, up to its period, on a body whose input need not contain a list.

An exact record may also be carried by orthogonal time slots: one tick of delay on the return wire puts $x_0u, x_1u, \ldots$ in consecutive output slots [3, § 3.4]. In the common-clock halting construction, a common post-halt ancilla sequence likewise distinguishes halt times at a fixed observation time. The original model also allows different ancilla sequences across branches, which can restore interference later [24]. Our exact record requires distinct halt times to remain orthogonal at observation. Appendix H gives the corresponding short-pulse and single-frequency experiments.

A case inside or outside feedback. Let θ have $\cos\theta = t = 3/5$ and rotate the guard alone. A phase z on the returning output gives $\theta_0(z) = (t - z)/(1 - tz)$ by summation of the scalar exits. The case $I \oplus \mathrm{trace}(\mathrm{rotbody}(\theta, I); (I \oplus zI))$ on a reference qubit therefore denotes $\mathrm{diag}(1, \theta_0(z))$. If instead the data bit selects the phase on a return *inside* the guard rotation, the two data sectors get $\theta_0(1)$ and $\theta_0(z)$. The result is $\mathrm{diag}(-1, \theta_0(z))$. The bypass in the first program makes no visit to the loop,

whereas both sectors in the second visit it. At $z = i$, the two open bodies are the following literal programs. A factor `(0 :+ 1)` contributes i only to a returning output.

```
round phaseOutside : Qubit + Unit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ [3/5 . exit 1 + (0 :+ 1) . 4/5 . back unit];
  back unit ↔ [-4/5 . exit 1 + (0 :+ 1) . 3/5 . back unit]
}

trace phaseOutside on 3/5 . 0 + 4/5 . 1

round phaseInside : Qubit + Qubit = {
  exit 0 ↔ [3/5 . exit 0 + 4/5 . back 0];
  exit 1 ↔ [3/5 . exit 1 + (0 :+ 1) . 4/5 . back 1];
  back 0 ↔ [-4/5 . exit 0 + 3/5 . back 0];
  back 1 ↔ [-4/5 . exit 1 + (0 :+ 1) . 3/5 . back 1]
}

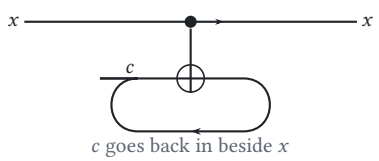
trace phaseInside on 3/5 . 0 + 4/5 . 1
```

Here $\theta_0(i) = (15 - 8i)/17$. On the stated input the first returns $3|0\rangle/5 + (12/17 - 32i/85)|1\rangle$, while the second negates only the $|0\rangle$ coefficient. Thus the position of the case changes the relative phase between the two data alternatives. On $|+\rangle$ followed by H, their probabilities of outcome 1 are $1/17$ and $16/17$, respectively.

G Reading the feedback diagrams

The sheet diagrams read from bottom to top. Each summand has a separate surface. Looking along the depth of one surface reveals the wires of its tensor factors, so two registers on one sheet can be used together, while different sheets carry alternative components. Where the surfaces meet, the dark seam labelled U acts on both summands, even where the returning sheet hides part of the external wire. Closing a summand bends its entire surface back to the input. This last bend denotes the trace proved here. The underlying sheet calculus describes sum, tensor and composition [11, Defs. 16, 18–20]. A trace in the sense of this paper closes an entire summand. For instance the ordinary partial matrix trace of controlled-X over the target gives $2|0\rangle\langle 0|$. For the summand trace in Figure 25, reorder the tensor factors to group by the target c, which carries the exit or return tag. The control x remains the external data. Control 0 leaves c unchanged and exits immediately. Control 1 flips c twice and exits after one return. The exit operators are $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$, whose sum is the identity on the control. These computations even have different types of interpretation.

(a) trace for ⊗: a wire fed back beside the others

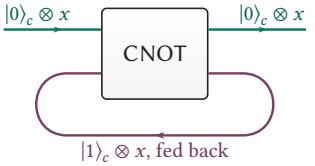


$\sum_k (I \otimes \langle k|)\,\mathrm{CNOT}\,(I \otimes |k\rangle)$
$= |0\rangle\langle 0|\,\mathrm{tr}(I) + |1\rangle\langle 1|\,\mathrm{tr}(X) = 2|0\rangle\langle 0|$

nothing splits and nothing passes twice;
not unitary

$\mathrm{Tr}^{\mathbb{C}^2_c}_{\otimes}(\mathrm{CNOT}) : \mathbb{C}^2_x \longrightarrow \mathbb{C}^2_x$

(b) trace for ⊕: the guard splits the state



$A = |0\rangle\langle 0|,\ B = C = |1\rangle\langle 1|,\ D = |0\rangle\langle 0|$ on x
$\mathrm{Tr}^{\oplus}\,\mathrm{CNOT} = A + BC = I$

$x = |1\rangle$ goes round once and leaves;
$\mathrm{Tr}^{Y}_{\oplus}(U) : X \longrightarrow X$ unitary

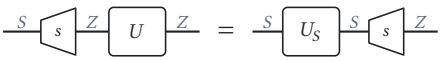
one controlled flip, x controls c; the guard decomposition reorders the tensor factors

Fig. 25. Two traces of controlled-X. Closing the target as a tensor factor gives twice a projection. Closing it as a return summand gives the identity.

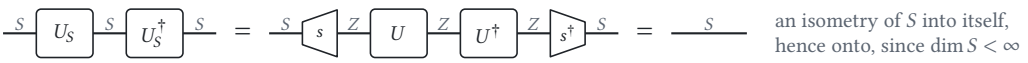
G.1 The part of the return space that an input can reach

Figure 26 separates the return states that never emit from the rest of the finite unitary body.

(a) U maps S into S: $Us = sU_S$

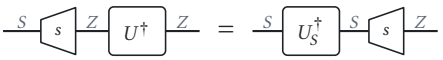


(b) step 1: $U_S^\dagger U_S = s^\dagger U^\dagger U s = s^\dagger s = I_S$

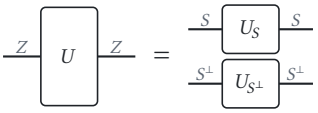


an isometry of S into itself, hence onto, since $\dim S < \infty$

(c) step 2: $U^\dagger s = sU_S^\dagger$, so $U^\dagger$ maps S into S



(d) step 3 and the conclusion: $U = U_S \oplus U_{S^\perp}$



$Z = S \oplus S^\perp$, and the two wires never meet

Fig. 26. For a unitary $U : Z \to Z$, let $s : S \rightarrowtail Z$ be the inclusion of a finite invariant subspace. Norm preservation gives an isometry on S, which is onto in finite dimension; hence S reduces U.

To find that subspace without diagonalizing D, follow the emissions in Figure 27. Set $K_0 = Y$ and $K_{j+1} = \ker B \cap D^{-1}(K_j)$. Then K_j is the set of vectors with their first j emissions zero. An equality $K_j = K_{j+1}$ makes this space invariant and fixes all subsequent terms; strictly decreasing steps can occur at most $\dim Y$ times. Consequently $K_m = Y_0$ for $m = \dim Y$. When $m > 0$ and $Y_1 \neq 0$, the norm

of D_1^m is strictly less than one: a unit vector achieving norm one would have emitted nothing in the first m rounds and would lie in Y_0. Compactness supplies a maximum, hence a uniform geometric bound. This gives another proof of the decay used in Appendix B.

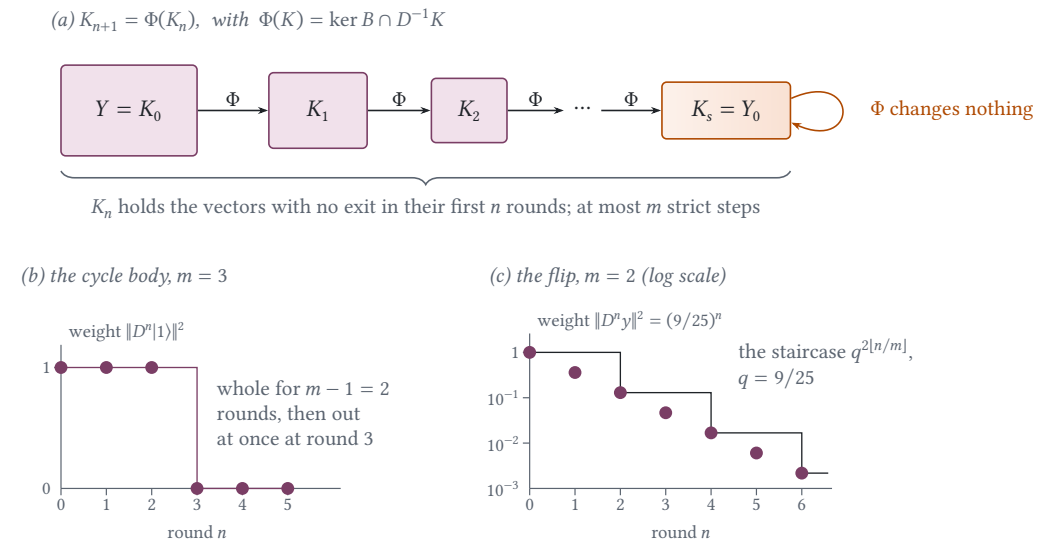


Fig. 27. The no-emission spaces stabilize within $\dim Y$ steps. Panel (b) is the four-state cycle $0 \mapsto 1 \mapsto 2 \mapsto 3 \mapsto 0$, with external state 0. Its three-dimensional return block has $D|1\rangle = |2\rangle$, $D|2\rangle = |3\rangle$ and $D|3\rangle = 0$. In panel (c), $\|D_1\| = 3/5$ and $m = 2$, so $q = \|D_1^2\| = 9/25$. The exact residual weight is $(9/25)^n$, and the staircase bounds it by $q^{2\lfloor n/m \rfloor}$.

G.2 Effects and trace laws

Figure 30 places the three readings in the Π family. Figure 28 adds the partial-injection model and the compact completion. The finite counter is obtained by extending the body before tracing. Figure 29 shows why coherent sum does not descend to channels. The trace on finite isometries agrees on embedded unitaries with the original coherent trace [3, Cor. 3.3.16], so allocation alone does not turn coherent feedback into measurement. The exact countable record is a semantic limit of the finite recording programs. Measuring the tag instead uses the classical sum of operator blocks. All three constructions start from the same iso body.

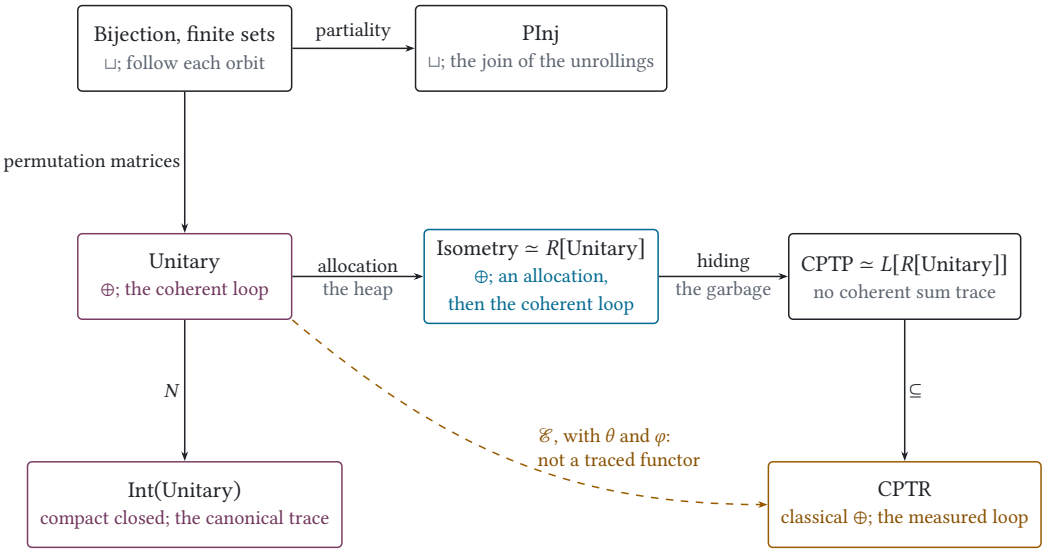


Fig. 28. The measured loop uses a classical sum of operator algebras. The dashed body translation is $U \mapsto \Phi_U = \varphi \circ \mathscr{E}_U \circ \theta$. The direct-sum trace of an embedded unitary remains coherent on the isometry layer.

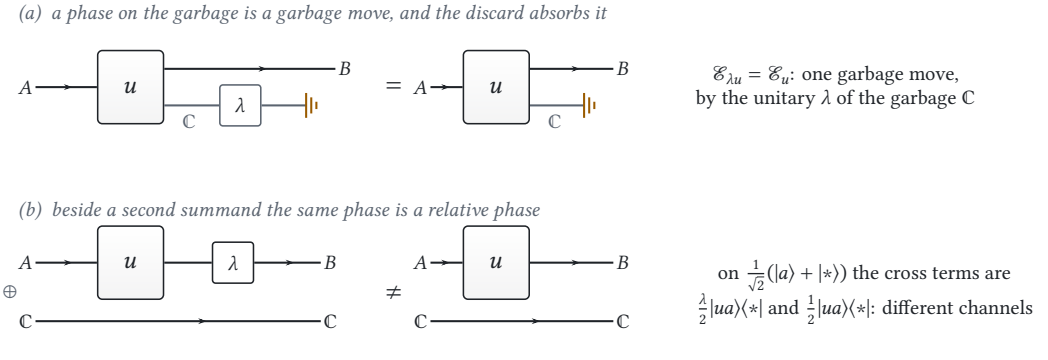


Fig. 29. Here $u : A \rightarrowtail B$ is an isometry, $A \neq 0$, and $|\lambda| = 1$ with $\lambda \neq 1$. A one-dimensional garbage factor absorbs this phase. Putting u in one arm of a coherent sum exposes it instead.

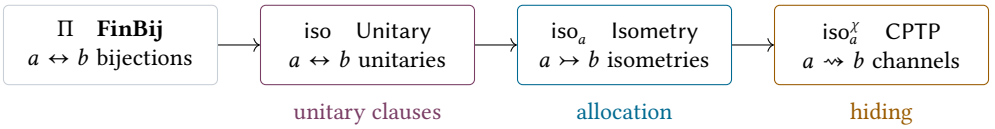


Fig. 30. The finite source extensions in the Π family. Allocation gives $R[\text{Unitary}] \simeq \text{Isometry}$ and hiding gives $L[\text{Isometry}] \simeq \text{CPTP}$ [18]. The next subsection draws the coherent trace laws [5, 21].

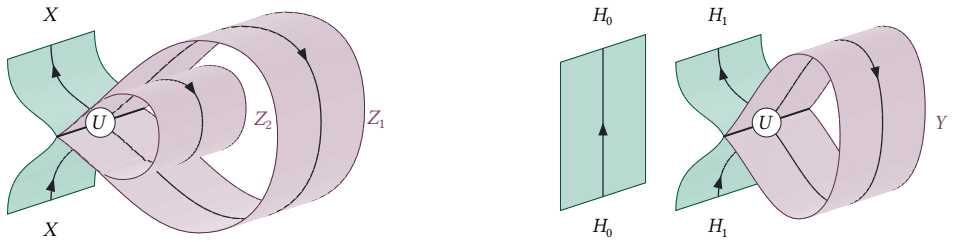
G.3 Composition and the backward wire

The summand matters. In a traced monoidal category the trace closes one component of a morphism $X \odot Y \to X' \odot Y$, for the category's chosen product $\odot$ [21]. Here it is $\oplus$. The usual tensor trace

on finite linear maps is a different operation and need not send a unitary to a unitary: tracing the factor $\mathbb{C}^2$ out of $I_X \otimes I_{\mathbb{C}^2}$ gives $2I_X$.

Bartha's finite unitary trace satisfies the trace laws for every body, including ones with a trapped return [5, Thms. 7–8] [3, Cor. 3.3.16]. For a cartesian product the trace also gives a Conway fixed-point operator [17, Thm. 3.1]. For the direct sum the diagonal $u \mapsto (u, u)$ doubles the squared norm of every nonzero u, so it cannot be a unitary in our language.

Consider the iso swapping external and internal copies of a (Figure 34). Its input makes one return and comes out unchanged, so its coherent trace is the identity: the law called *yanking*. A K-clock with $K > 1$ distinguishes this path from an immediate exit by labels 1 and 0. Naturality and sliding move external and internal operations across a trace, as drawn in Figure 35 [5, 21]. The two sheets below show how the trace acts on a nested loop and on a loop beside a bypass. Both equations use canonical reassociations of the summands.



$$\mathrm{Tr}^{Z_1}(\mathrm{Tr}^{Z_2} U) = \mathrm{Tr}^{Z_1 \oplus Z_2} U \qquad I \oplus \mathrm{Tr}^{Y} U = \mathrm{Tr}^{Y}(I \oplus U)$$

Fig. 31. At left, both return ports of U are closed. At right, only the input on H_1 enters the loop.

Writing the guard as a tensor factor and applying distributivity gives exactly the two injection tags in a clause (Figure 32). Naturalities of the additive trace apply to this representation. Closing the body $I \oplus U$ over U's return port gives $I \oplus L$: the immediate exit is $I \oplus x_0$ and the later exits are $0 \oplus x_k$. Closing disjoint internal ports successively gives the simultaneous solution by the argument of Appendix B.1, as drawn in Figure 33. For two loops in sequence, at a common phase z their responses multiply as $\Theta_2(z)\Theta_1(z)$, because the external output of the first enters the second and their return ports are separate. Expanding the product gives the convolution $\sum_{j=0}^{k} x_{k-j}^{(2)} x_j^{(1)}$ at total count k. The adjoint of a coherent loop is the loop of the adjoint body: its blocks give exactly $x_k^\dagger$ as exits and the series is absolutely convergent (Figure 36).

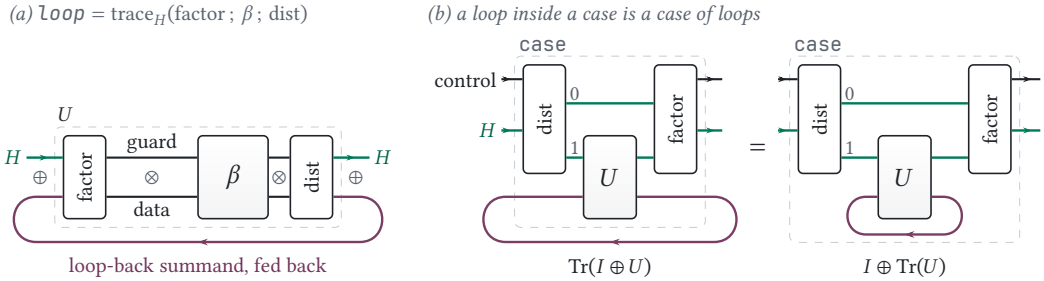


Fig. 32. Distributivity makes the guard a summand. A case with a separate untouched arm distributes over the trace of the other arm.

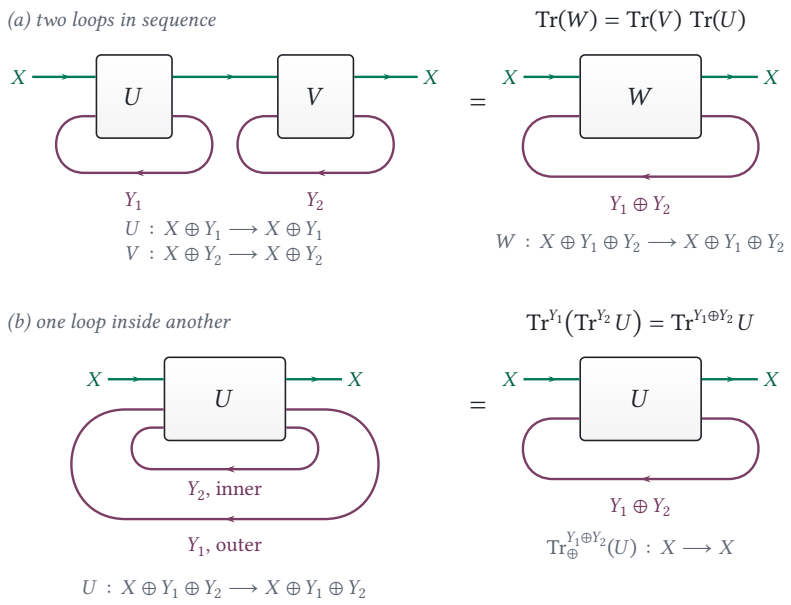


Fig. 33. In sequence the exit operators convolve. Nested traces solve the two return equations together.

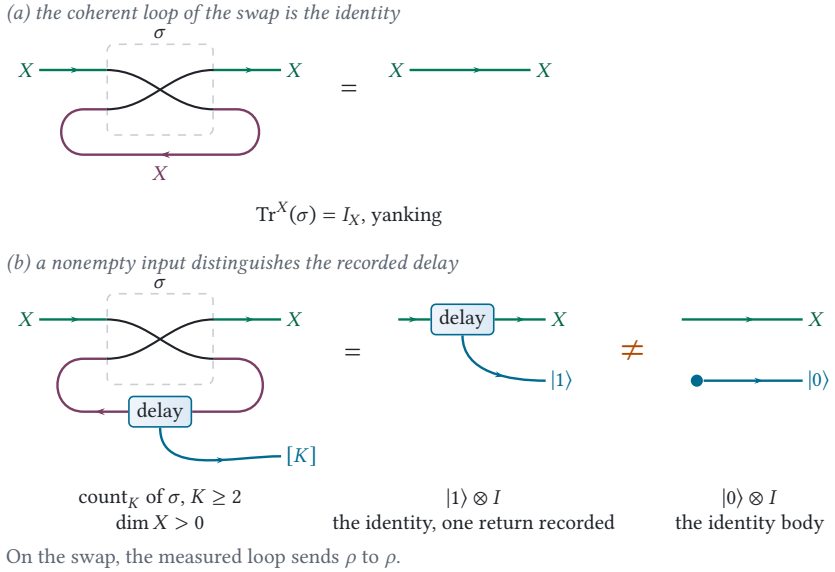


Fig. 34. The swap is between two copies of X. Its coherent trace returns the input. For $\dim X > 0$ and $K \geq 2$, a recording loop distinguishes one return from an immediate exit.

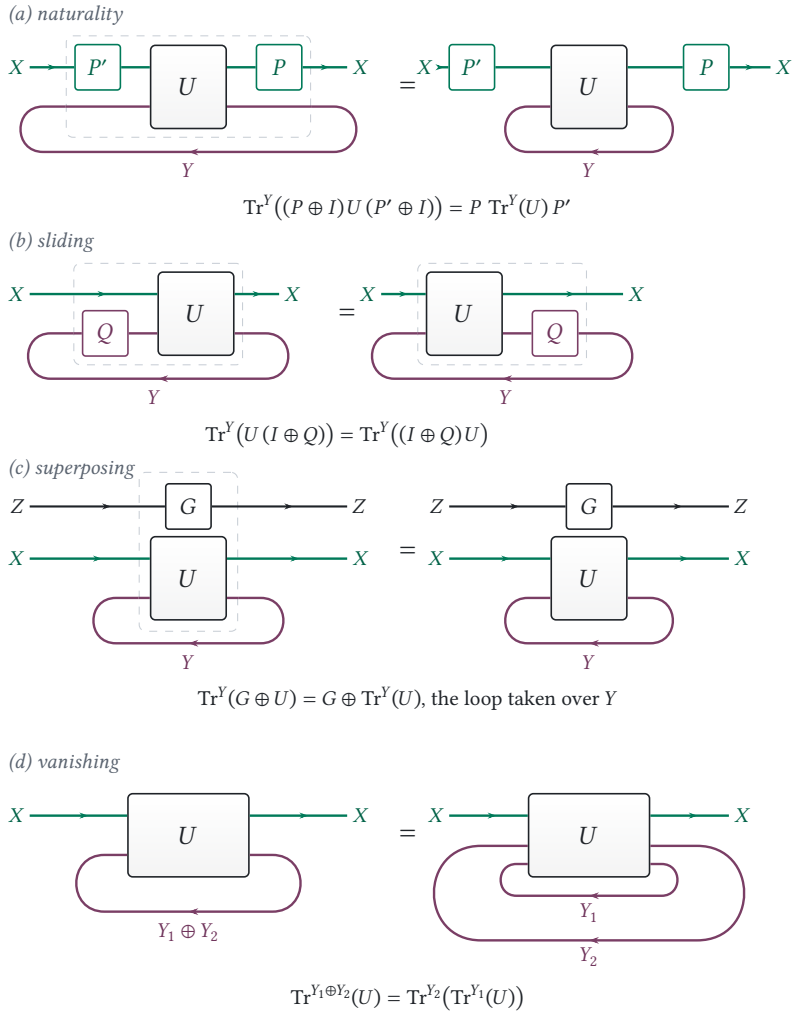


Fig. 35. Naturality, sliding, superposing and vanishing are typed equations for coherent loops. In panel (a), $P, P' : X \to X$.

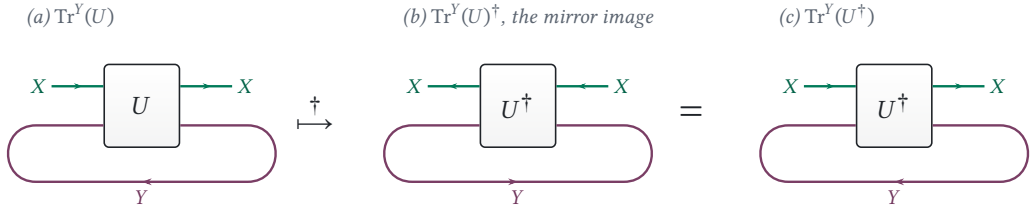


Fig. 36. Reversing a coherent body exchanges entry and exit and takes the adjoint of each exit amplitude.

In the loop diagram a wire runs backwards from the returning output to the next input. To make this drawing into a typed composition, the Int construction pairs each forward type P^+ with a

backward type P^-. A map from P to Q can then consume P^+ from its left and Q^- from its right, so its underlying unitary has type $P^+ \oplus Q^- \to Q^+ \oplus P^-$. Figure 37 draws these ports.

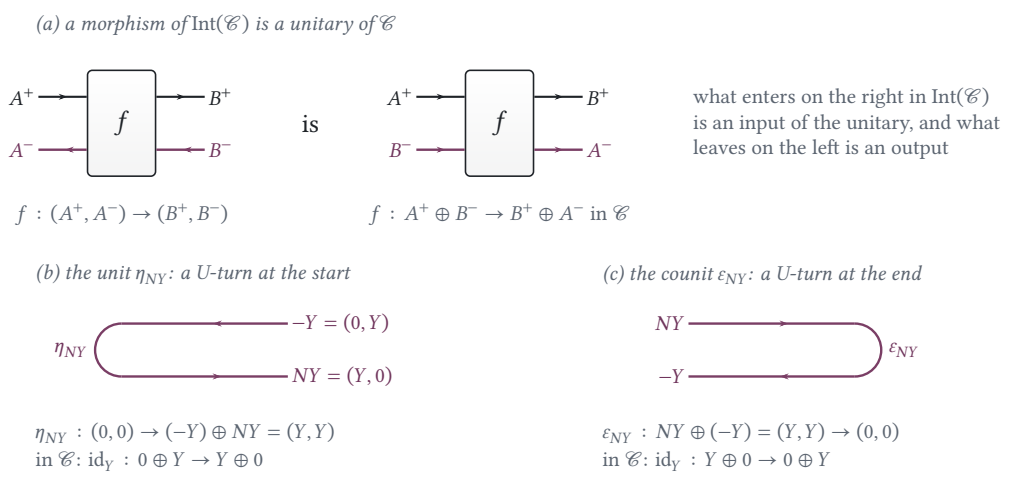


Fig. 37. Objects in the compact completion have forward and backward components. Cups and caps bend a port within that completion.

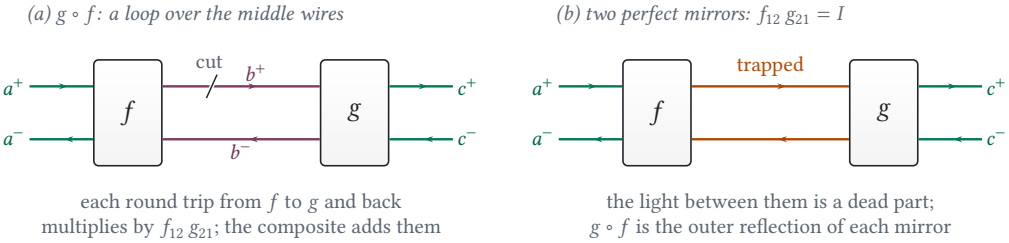


Fig. 38. Composition closes the middle ports. Between two perfect mirrors, a round trip returns the light unchanged and never reaches an outer port.

In Figure 38, f sends b^+ to g and g sends b^- back to f. On external inputs a^+ and c^- the two middle amplitudes satisfy

$$b^+ = f_{11}a^+ + f_{12}b^-, \qquad b^- = g_{21}b^+ + g_{22}c^-.$$

These are also the equations for light scattering between two components. The wave leaving f enters g, and the part reflected by g enters f again. When the internal inverse exists, eliminating these two amplitudes gives the scattering matrix of the pair, the Redheffer star product [15, Figs. 10–11]. Starting at the cut on the upper wire, a return passes through g and then f, so its coefficient is $D = f_{12}g_{21}$. Closing the cut identifies its outgoing and incoming amplitudes. Substituting the second equation into the first gives the wire equation of Section 3 with $y = b^+$. Its external outputs are c^+ and a^-.

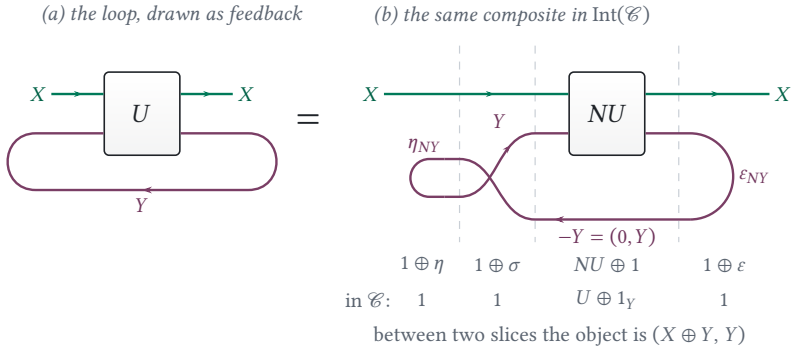


Fig. 39. Both diagrams close Y in $U : X \oplus Y \to X \oplus Y$. The cap takes the return onto the backward wire $(0, Y)$ and the cup brings it to the input.

The fully faithful embedding $N(X) = (X, 0)$ puts an ordinary space in this construction with no backward input. Bending Y around an embedded body identifies the amplitude leaving it, $Cu + Dy$, with the amplitude y supplied on the same wire. The compact closed trace in Figure 39 therefore returns exactly $N(L)$, by the very wire equation used in Theorem B.1 [21] [25, Thms. 5.15–5.16]. Inserting a successor changes what comes back along that wire. Measuring the tag changes what the wire carries from an amplitude to a classical alternative.

The sum in Int reverses the order of negative components:

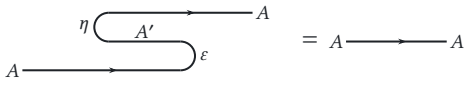
$$(P^+, P^-) \oplus (Q^+, Q^-) = (P^+ \oplus Q^+, Q^- \oplus P^-).$$

The dual of $(Y, 0)$ is $(0, Y)$, and the unit and counit are identity wires on Y with their ports rearranged. Composing two maps plugs the forward and backward middle ports together and takes the original trace. With $\eta_{NY} : 0 \to (-Y) \oplus NY$ and $\varepsilon_{NY} : NY \oplus (-Y) \to 0$, the typed closure is

$$N(\mathrm{Tr}^Y U) = (I_{NX'} \oplus \varepsilon_{NY})(NU \oplus I_{-Y})(I_{NX} \oplus \sigma_{-Y,NY})(I_{NX} \oplus \eta_{NY}).$$

The swap puts the cup's two outputs in the order required by $NU \oplus I_{-Y}$. Its underlying map is $I_{Y \oplus Y}$, since both source and target pairs are (Y, Y). With the external identity this is $I_{X \oplus Y \oplus Y}$. From left to right, the four regions separated by dashed lines in the right panel of Figure 39 show these four factors in execution order. If two perfect reflectors enclose a mode, that mode need not have a unique internal amplitude, but it has a unique external response by Theorem B.1. A finite clock bends the enlarged internal type $[K] \otimes Y$ and advances its label on each passage. These statements require no tensor-product duals of unitary gates themselves. Figure 40 separates these constructions. Under direct sum a cup from 0 to $Y \oplus Y^*$ cannot be unitary unless $Y = 0$. Under tensor, a Hilbert-space cup and cap obeying the snake identity on an n-dimensional object have product of norms at least n: closing to a scalar gives n and Cauchy–Schwarz bounds it by the product. Thus for $n \geq 2$ they cannot both be contractions. The compact completion uses the sum, where its objects are pairs.

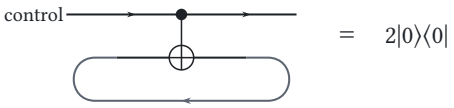
(a) the tensor product: a dual would make the loop of the identity the dimension



the snake equation defines the dual

so $\varepsilon\,\sigma\,\eta = \dim A$, while $|\varepsilon\,\sigma\,\eta| \le \|\varepsilon\|\,\|\eta\| \le 1$ among contractions: no dual once $n \ge 2$

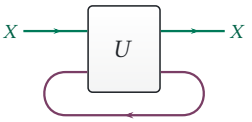
(b) the trace of a factor adds the blocks of one application



tr over the target, a factor

$(I \otimes \langle 0|)\,\mathrm{CNOT}\,(I \otimes |0\rangle) + (I \otimes \langle 1|)\,\mathrm{CNOT}\,(I \otimes |1\rangle)$
$= |0\rangle\langle 0| + |0\rangle\langle 0|$, not a contraction

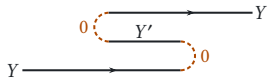
(c) the trace of a summand adds the exits of the rounds



the loop-back summand Y

$y = Cu + Dy$ implies
$U(u, y) = (\mathrm{Tr}^Y(U)u, y)$,
so $\|\,\mathrm{Tr}^Y(U)u\| = \|u\|$

(d) the direct sum: the unit and counit through the zero space are zero maps

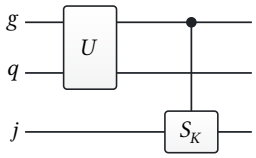


not the identity of Y: every map into or out of the zero space is zero, and a unitary into or out of it exists only when $Y = 0$

Fig. 40. The compact completion supplies the negative type. The original tensor product on contractions does not supply a dual of dimension two or greater.

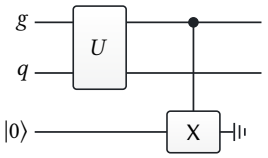
Figure 41 puts the counter increment and the tag-copying step beside the same body.

Keep the count modulo K



$S_K|j\rangle = |j + 1 \bmod K\rangle$.

Measure only the guard tag



Discard the fresh bit after this call.

Fig. 41. One call to the same body U, followed by an operation controlled on the outgoing guard $g = 1$. A filled dot denotes a coherent control. The left circuit carries j into the next round. In the right circuit the discarded bit distinguishes the tags without distinguishing states of the payload q. Crossing wires interact only at a marked gate.

H A ring realization of the same exits

In Figure 6, light stays in its waveguide with amplitude $t = 3/5$ and crosses to the other waveguide with amplitude $is = 4i/5$. This coupler is the rotation from `guardRound` in a different basis on the return wire, $(1 \oplus i)R_\theta(1 \oplus (-i))$. Light that enters the ring, stays for k further trips and leaves has amplitude $(is)t^k(is) = -s^2t^k$. Hence the coupler and the rotation have the same exit on every round and the same response $\Theta_{\mathrm{ring}}(z) = (t - z)/(1 - tz)$ at round-trip phase z. The two clauses give the following iso program.

```
round ringCoupler : Unit + Unit = {
  exit unit ↔ [3/5 . exit unit + (0 :+ 1) . 4/5 . back unit];
  back unit ↔ [(0 :+ 1) . 4/5 . exit unit + 3/5 . back unit]
}

trace ringCoupler on unit
```

With no added phase the command returns $-|*\rangle$.

The same exit amplitudes give both outputs in Figure 8. Write $u(T)$ for the incident field, with time measured relative to the direct path. The copy that makes k trips arrives $k\tau_0$ later with the amplitude of exit k. The field at the marked port is therefore

$$v(T) = t\,u(T) - s^2 \sum_{k\geq 1} t^{k-1} u(T - k\tau_0).$$

For a pulse shorter than τ_0, the copies have disjoint support. For a steady wave at round-trip phase 1, all the delayed copies equal $u(T)$. Their coefficients sum to $-s^2/(1-t) = -8/5$. Adding the direct $3/5$ gives the sign returned by the program [2, Eqs. (6)–(7)] [28, Eqs. (30)–(33)].

To realize `bypassRound`, match the bypass to the straight-through path in phase and travel time. Prepare the external qubit in $|+\rangle$, run the two paths, and apply H to compare them. At a frequency for which one trip around the ring brings the wave back in phase with itself, the round-trip factor is $z = 1$. This is resonance. The ring response is then -1 and the difference port has probability $|1-(-1)|^2/4 = 1$. For a short pulse the first time bin contributes $(1-t)^2/4$ to this port, and the later bins contribute their probabilities separately:

$$\Pr(b=1) = \frac{1}{4}\left((1-t)^2 + \sum_{k\geq 1} s^4 t^{2(k-1)}\right) = \frac{1}{4}\big((1-t)^2 + 1 - t^2\big) = \frac{1}{5}.$$

These are the coherent and measured predictions from Section 4. The scalar ring passes agreement, but the two-arm body fails: its immediate exit contains the bypass and its later exits do not. Forming that bypass uses a coherent sum of isos before channelization, an operation unavailable to a channel context. For a finite pulse, the interference depends on how much the delayed pulses overlap in the observed output mode.

Switching on the light. Fairbanks and Selinger vary the angle of this same rotation. Its trace jumps from $+1$ at $\theta = 0$ to -1 for $0 < \theta < 2\pi$ [13, Remark B.1]. To see the jump in the ring, start with the ring empty and switch on a wave of unit amplitude at resonance. Write $t = \cos\theta$ and $s = \sin\theta$. Before light has had time to go around once, the output is t. After n round trips the first n delayed exits have reached the output. For $0 < \theta < \pi/2$ they give

$$t - s^2(1 + t + \cdots + t^{n-1}) = -1 + (1+t)t^n \qquad (n \geq 1).$$

At any fixed n the output approaches $+1$ as θ approaches zero. At any fixed angle in this interval the output approaches -1 as more trips arrive. Thus the jump appears when we let the ring settle at each angle before taking the coupling to zero. At any fixed observation time the output varies continuously with the coupling. At zero coupling no light enters the ring and the output stays at $+1$.

For the steady response at resonance, the first crossing contributes is on the return wire and each further trip multiplies it by t. On unit input the field there is $is/(1-t)$. Its squared magnitude is the weight from Section 6, $Q(1) = s^2/(1-t)^2 = \cot^2(\theta/2)$ for $0 < \theta < \pi/2$. It grows without bound as the coupling tends to zero. If each trip takes time τ_0 and the coupler is independent of frequency,

changing the angular frequency ω changes the return phase by $d\varphi/d\omega = \tau_0$. The Wigner–Smith delay at resonance is therefore $-i\Theta^\dagger d\Theta/d\omega = \tau_0 Q(1)$ [31].

More than one return port. For a given finite unitary body $U : X \oplus Y \to X' \oplus Y$, use one optical mode for each basis value. Reck et al. give a construction of the open U from beam splitters and phase shifters [29]. Connect each return output to its matching input with an ideal lossless delay line. Choose the lengths so that each line returns light with phase 1 at the working frequency. If u enters the external ports and y comes back along the lines, the amplitudes at the coupler network obey

$$y = Cu + Dy, \qquad v = Au + By.$$

These are the evaluator's wire equations. Even when a return mode can circulate in isolation, every solution gives the same external output $v = Lu$ (Section 3). Thus the steady response of this network at the chosen frequency realizes the direct-sum trace of the given body. The construction uses its matrix to set the optical elements. The rotation above shows why no fixed number of trips suffices for all couplings.

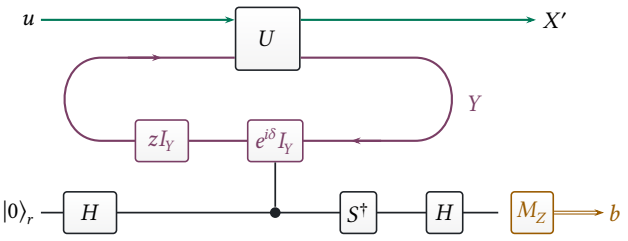
Czachor uses the ring as a model of looped quantum evolution in a study of time travel [12, Appendix]. Gough and James obtain the same block formula for scattering in the zero-delay limit when the internal inverse exists [16, Thm. 12, Eq. (12)]. Here the program fixes what each trip contributes. Its exit after $k+1$ returns has amplitude BD^kCu, exactly the amplitude that enters through C, circulates k more times and leaves through B.

I Observing the response

For a fixed $|z| = 1$, the phase iso $P_z^c = \{s \leftrightarrow z \cdot s\}_{s \in \mathcal{B}(c)} : c \leftrightarrow c$ gives the marked source body $\beta;(I_b \oplus P_z^c)$. The bypass caller is

$$H; \mathsf{trace}_1\big(\texttt{bypassRound}; (I_{\texttt{Qubit}} \oplus P_z^1)\big); H.$$

Thus the phase variation used in the body of the paper is an ordinary finite iso construction.



$U : X \oplus Y \to X' \oplus Y$

guardRound, $u = 1$

z	F_z	$Q(z)$
1	$s/(1-t) = 2$	4
-1	$s/(1+t) = 1/2$	1/4

$S^\dagger = \mathrm{diag}(1, -i)$

$$\Pr(0) - \Pr(1) = \mathrm{Im}\langle \Theta(z)u, \Theta(ze^{i\delta})u\rangle = \delta\langle u, Q(z)u\rangle + O(\delta^2).$$

Fig. 42. The reference chooses return phase z or $ze^{i\delta}$ and accompanies both summands unchanged through U. Its final measurement reads the relative derivative. At right, $t = 3/5$, $s = 4/5$, and F_z is the scalar return amplitude before the phase gate.

The bypass in `bypassRound` makes no return, giving $N_{\text{bypass}} = \mathrm{diag}(0, 1)$. Its measured mean on $|+\rangle$ is therefore $1/2$, the equal-weight average of bypass count zero and scalar-loop mean one.

Putting return phase $z = e^{i\varphi}$ into the original Hadamard caller gives these observable probabilities:

$$\Pr(b = 1) = |1 - \Theta_{\text{rot}}(z)|^2/4$$

φ	0	$\pi/2$	π
$\Theta_{\text{rot}}(e^{i\varphi})$	-1	$(15-8i)/17$	1
$\Pr(b=1)$	1	$1/17$	0

Averaging this test over the phase circle gives the measured probability $1/5$ by (6.1).

The controlled body. For $|z| = 1$ and $\delta \in \mathbb{R}$, let the reference space be $H_r = \mathbb{C}^2$, and put $P_\delta = \mathrm{diag}(1, e^{i\delta})$. Use distributivity to identify $H_r \otimes (X \oplus Y)$ with $(H_r \otimes X) \oplus (H_r \otimes Y)$. The body in Figure 42 is the finite unitary

$$W_{z,\delta} = \big(I_{H_r \otimes X'} \oplus (P_\delta \otimes zI_Y)\big)(I_{H_r} \otimes U).$$

It has external type $H_r \otimes X \to H_r \otimes X'$ and returning summand $H_r \otimes Y$. For reference basis value $r = 0$ its returning phase is z, and for $r = 1$ it is $ze^{i\delta}$. It never changes that basis value. Its kth exit on the two sectors is therefore respectively $z^k x_k$ and $(ze^{i\delta})^k x_k$. The closed gate is

$$|0\rangle\langle 0|_r \otimes \Theta(z) + |1\rangle\langle 1|_r \otimes \Theta(ze^{i\delta}).$$

For reference input $|+\rangle$ and unit data u, the closed gate returns

$$\frac{|0\rangle_r \otimes \Theta(z)u + |1\rangle_r \otimes \Theta(ze^{i\delta})u}{\sqrt{2}}.$$

Applying $S^\dagger$ and H gives data vectors on reference outcomes $0, 1$: $(\Theta(z)u - i\Theta(ze^{i\delta})u)/2$ and $(\Theta(z)u + i\Theta(ze^{i\delta})u)/2$. Their squared-norm difference is $\mathrm{Im}\langle\Theta(z)u, \Theta(ze^{i\delta})u\rangle$. Taylor expansion at $\delta = 0$ gives the displayed derivative readout in Figure 42.

Thus the straight reference line in the drawing denotes a tensor factor carried on both alternative sheets. Its controlled phase acts on each return. No coherent control of a channel hole is used.

At $z = 1$, this construction compares $\Theta(1)u$ and $\Theta(e^{i\delta})u$. For the scalar response $(3 - 5z)/(5 - 3z)$, choose $\delta = \pi/2$. The imaginary overlap is $8/17$, so the two outcome probabilities are $25/34$ and $9/34$. This finite-phase readout is the exact overlap, whereas the derivative formula is its first-order expansion at $\delta = 0$. This specializes Figure 42 to the controlled body $W_{1,\delta}$.